

Srednicki Quantum Field Theory

Complete Cadabra2 Companion

Chapters 34–60: Spinors, Fermions, and Spinor-Helicity

Generated by OpenClaw

May 17, 2026

Contents

1	Ch. 34 — Left- and Right-Handed Spinor Fields	8
1.1	The Lorentz Algebra and its $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ Structure	8
1.1.1	The six generators of $\text{SO}(1, 3)$	8
1.1.2	The $N_i, N_i^\dagger$ combinations	9
1.1.3	Derivation of $[N_i, N_j] = i\varepsilon_{ijk}N_k$	9
1.1.4	Derivation of $[N_i^\dagger, N_j^\dagger] = i\varepsilon_{ijk}N_k^\dagger$	10
1.1.5	Derivation of $[N_i, N_j^\dagger] = 0$	11
1.1.6	The isomorphism statement — over $\mathbb{C}$	11
1.1.7	Irreducible representations: the (n, n') labels	12
1.2	Tensor Product Decompositions of Lorentz Representations	13
1.2.1	Warmup: $\text{SU}(2)$ Clebsch-Gordan decomposition	13
1.2.2	Lorentz tensor products: $(2, 1) \otimes (2, 1)$	14
1.2.3	Lorentz tensor products: $(2, 2) \otimes (2, 2)$	15
1.2.4	Summary table of tensor product decompositions	16
1.3	Lorentz Group Representations	17
1.4	Left-Handed Spinor Field ψ_a	17
1.5	Generators $S_L^{\mu\nu}$ in the $(2, 1)$ Representation	17
1.5.1	Commutation relations	17
1.5.2	Explicit matrices	18
1.5.3	Boost generators: explicit metric-convention comparison	19
1.6	Right-Handed Spinor Field $\psi_a^\dagger$	20
1.6.1	Why hermitian conjugation flips the representation	20
1.6.2	Right-handed generators and the dotted-index rule	21
1.7	The ε Symbol — $\text{SL}(2, \mathbb{C})$ Metric	21
1.7.1	Normalization (Srednicki convention, eq. 34.22)	21
1.7.2	Raising and lowering	21
1.8	Lorentz-Invariant Spinor Products	22
1.8.1	Left-handed (“angle bracket”)	22
1.8.2	Right-handed (“square bracket”)	22
1.9	The σ^μ Symbol — Vector/Spinor Dictionary	22
1.9.1	West coast $(+---)$: derivation of $\sigma_{\text{WC}}^\mu = (I, \vec{\sigma})$	22
1.9.2	East coast $(-+++)$: derivation of σ_{EC}^μ	23
1.9.3	Trace identity in both conventions	23

1.10	Summary	25
2	Ch. 35 — Manipulating Spinor Indices	26
2.1	Setup: ε Symbols and σ^μ	26
2.1.1	Metric convention	26
2.1.2	ε normalization (eq. 35.1)	26
2.1.3	σ^μ invariant symbol (eq. 35.2)	27
2.2	Key Identity: $\sigma_{a\dot{a}}^\mu \sigma_{\mu b\dot{b}} = -2\varepsilon_{ab}\varepsilon_{\dot{a}\dot{b}}$ [eq. 35.4]	27
2.2.1	Statement and physical meaning	27
2.2.2	Index structure and metric dependence	27
2.2.3	Nonzero components	28
2.3	Key Identity: $\varepsilon^{ab}\varepsilon^{\dot{a}\dot{b}}\sigma_{a\dot{a}}^\mu\sigma_{b\dot{b}}^\nu = -2g^{\mu\nu}$ [eq. 35.5]	28
2.3.1	Statement and physical meaning	28
2.3.2	Physical interpretation	28
2.4	$\bar{\sigma}^\mu$: Definition and Numerical Verification [eq. 35.19–35.20]	29
2.4.1	Definition (eq. 35.19)	29
2.4.2	Explicit result (eq. 35.20)	29
2.4.3	Trace identity	30
2.5	Generators $S_L^{\mu\nu}$ and $S_R^{\mu\nu}$ in Terms of $\sigma, \bar{\sigma}$ [eq. 35.21–35.22]	30
2.5.1	Formulas	30
2.5.2	Explicit matrices	31
2.6	Clifford-Like Algebra $\sigma^\mu\bar{\sigma}^\nu + \sigma^\nu\bar{\sigma}^\mu = -2g^{\mu\nu}I_2$	31
2.6.1	Statement	31
2.6.2	Relation to generators	32
2.7	Covariance Check: σ^μ Invariance [eq. 35.14]	32
2.7.1	Statement	32
2.8	Index-Free Notation and Spinor Bilinears [eq. 35.23–35.29]	32
2.8.1	Index-free spinor products (eq. 35.23)	32
2.8.2	Symmetry of the bilinear (eq. 35.25)	33
2.8.3	Hermitian conjugate (eq. 35.26)	33
2.8.4	The vector bilinear (eq. 35.27–35.28)	33
2.9	Summary of Chapter 35 Results	34
3	Ch. 36 — Lagrangians for Spinor Fields	36
3.1	The Weyl Lagrangian	36
3.1.1	Lagrangian terms	36
3.1.2	Hermiticity of the kinetic term	36
3.1.3	Mass term and Majorana mass	36
3.2	Equations of Motion	37
3.2.1	Combined 4-component form	37
3.3	Gamma Matrices in the Weyl Representation	37
3.3.1	Definition	37
3.3.2	Explicit matrices	38
3.4	Clifford Algebra Verification	38

3.4.1	The algebra	38
3.4.2	Numerical verification table	38
3.4.3	The γ^5 matrix	39
3.5	Majorana 4-Component Spinor	39
3.5.1	Definition	39
3.5.2	Majorana condition	40
3.5.3	Majorana Lagrangian	40
3.6	Dirac Fermion from Two Weyl Fields	40
3.6.1	Two-Weyl-field Lagrangian	40
3.6.2	Dirac spinor and Lagrangian	41
3.7	Dirac vs. Majorana — Comparison	41
3.8	Summary	42
4	Ch. 37 — Canonical Quantization of Spinor Fields I	43
4.1	Canonical Anticommutation Relations	43
4.1.1	Equal-time CARs	43
4.1.2	Origin from Legendre transform	43
4.2	Mode Expansion of the Weyl Field	44
4.2.1	Plane-wave expansion	44
4.2.2	Operator CARs	44
4.3	Basis Spinors $u^s(p)$ and $v^s(p)$	44
4.3.1	Weyl equations	44
4.3.2	Explicit construction at rest	44
4.3.3	Massless case	45
4.3.4	Completeness relations	45
4.4	Deriving the CARs from the Mode Expansion	45
4.5	Normal-Ordered Hamiltonian	46
4.5.1	Classical Hamiltonian density	46
4.5.2	Mode expansion form	46
4.5.3	Normal ordering	46
4.6	Spin-Statistics Theorem	47
4.7	Lorentz-Invariant Measure and Feynman Slash	47
4.7.1	The measure $\widetilde{dp}$	47
4.7.2	Feynman slash notation	47
4.8	Spin-Sum Completeness (4-component)	48
4.9	Summary	49
5	Ch. 38 — Spinor Technology	50
5.1	σ^μ Completeness Relation	50
5.1.1	Main identity	50
5.1.2	Alternative form	50
5.2	Trace Identities	50
5.2.1	Two-sigma trace	50
5.2.2	Four-sigma trace	51

5.3	Van der Waerden Index Gymnastics	51
5.3.1	Raising and lowering spinor indices	51
5.3.2	Raising both spinor indices of σ^μ	52
5.4	Spinor Bilinears	52
5.4.1	Lorentz transformation properties	52
5.4.2	Momentum matrix in spinor space	52
5.4.3	Gordon identity	52
5.5	Fierz Identities	53
5.5.1	Schouten identity	53
5.5.2	σ -matrix Fierz	53
5.5.3	Four-Fermi Fierz	53
5.6	Gordon Identity	53
5.6.1	Main identity	53
5.6.2	Derivation	53
5.6.3	Extended Gordon identity with γ^5	54
5.6.4	Physical content	54
5.7	Helicity and Spin-Sum Completeness	54
5.7.1	Spin-sum completeness	54
5.7.2	Bilinear identities	54
5.7.3	Helicity and chirality	55
5.8	Numerical Verification Summary	56
5.9	Summary	57
6	Ch. 39 — Canonical Quantization of Spinor Fields II	58
6.1	Mode Expansion of the Dirac Field	58
6.1.1	Mode expansion	58
6.1.2	Canonical anticommutation relations	58
6.2	Hamiltonian from Mode Expansion	59
6.2.1	Verifying eq. (39.24)	59
6.3	Gordon Identities	59
6.3.1	Rest-frame identity	59
6.3.2	General-momentum Gordon identity	60
6.4	Lorentz Generators on Creation Operators	60
6.4.1	Spin operator in the chiral basis	60
6.4.2	Helicity for moving particles	61
6.5	Spin-Statistics Theorem	61
6.5.1	Key argument	61
6.5.2	Summary table	61
7	Ch. 40 — Spin Sums	62
7.1	Spin-Averaging Factors	62
7.2	Trace Method	62
7.3	Trace Theorems	63
7.4	Helicity vs. Trace Method	63

7.5	Klein-Nishina (Compton Scattering)	63
7.6	Significance for MHV Research	63
8	Ch. 41 — Gamma Matrix Technology	64
8.1	Weyl (Chiral) Basis	64
8.2	Clifford Algebra	65
8.3	Properties of γ^5	65
8.4	Trace Theorems — Complete Set	65
8.4.1	Vanishing traces	66
8.4.2	Two-gamma trace	66
8.4.3	Four-gamma trace	66
8.4.4	Levi-Civita trace	66
8.5	Fierz Identities	67
8.5.1	Completeness of the Clifford basis	67
8.5.2	The vector Fierz identity	67
8.5.3	Chiral Fierz	67
8.6	Connection to Spinor-Helicity and MHV	67
8.6.1	Spin-summed amplitude: the trace	67
8.6.2	Spinor-helicity factorization	68
8.6.3	Parke-Taylor formula	68
9	Ch. 41b — Feynman Rules for Dirac Fields	69
9.1	Dirac Propagator	69
9.2	External Fermion Wave Functions	69
9.3	Spin Sums	70
9.4	Trace Theorems	70
9.5	QED Feynman Rules	70
9.6	Spinor Completeness (Massless)	71
9.7	Significance for the MHV Program	71
10	Ch. 42 — Electrodynamics in Coulomb Gauge	72
10.1	Coulomb Gauge Condition	72
10.1.1	The gauge condition	72
10.1.2	Momentum-space projection	73
10.2	Equal-Time Commutators	73
10.2.1	Mode expansion	73
10.3	The Photon Propagator in Coulomb Gauge	74
10.3.1	Components	74
10.3.2	Physical interpretation	74
10.3.3	Comparison with Feynman gauge	74
10.4	The Instantaneous Coulomb Interaction	75
10.4.1	Coulomb Hamiltonian	75
10.4.2	Effective Coulomb potential	75
10.5	Ward Identities in Coulomb Gauge	75

10.5.1	Transversality of physical polarizations	75
10.5.2	Ward identity	76
10.6	Connection to Light-Front Gauge and Spinor-Helicity	76
10.6.1	Light-front gauge	76
10.6.2	The transverse projector and brackets	77
10.6.3	Summary: Ch. 42 $\rightarrow$ Ch. 48	77
11	Ch. 43 — Feynman Rules for Majorana Fields	78
11.1	The Majorana Condition	78
11.1.1	Comparison of spinor types	79
11.1.2	Majorana from Weyl	79
11.2	Majorana Lagrangian and Mass	79
11.3	The Majorana Propagator	79
11.4	Feynman Rules for Majorana Fields	80
11.4.1	Standard rules	80
11.4.2	Sign rules	80
11.5	Why Majorana Matters for MHV	80
11.5.1	Gluinos are Majorana	80
11.5.2	Spin sums for Majorana spinors	81
11.5.3	The MHV amplitude with a gluino pair	81
11.5.4	The Majorana $\frac{1}{2}$ factor and loop amplitudes	81
11.5.5	BCFW for Majorana	82
12	Ch. 50 — Massless Particles and Spinor-Helicity	83
12.1	Introduction: Why Spinor-Helicity?	83
12.2	Angle and Square Brackets	83
12.3	Schouten Identity	84
12.4	Dot Products from Spinors: $2p_i \cdot p_j = \langle ij \rangle [ji]$	84
12.5	Little Group Scaling and Helicity	85
12.6	Gluon Polarization Vectors	85
12.7	The Parke-Taylor Formula (MHV)	85
12.8	Summary: The Spinor-Helicity Toolkit	86
13	Ch. 50b — Massless Spinor-Helicity (Extended)	87
13.1	§48.A Massless Momentum as Spinor Outer Product	87
13.1.1	Rank-1 factorization	87
13.1.2	Explicit components	87
13.2	§48.B Angle $\langle ij \rangle$ and Square $[ij]$ Brackets	88
13.2.1	Definitions	88
13.2.2	Antisymmetry	88
13.2.3	Schouten identity	88
13.3	§48.C Mandelstam Variables from Spinors	88
13.4	§48.D Helicity Spinors $u_{\pm}(k)$ in the Chiral Basis	89
13.4.1	Z-axis momentum: explicit form	89

13.4.2	General massless momentum	89
13.5	§48.E Gluon Polarization Vectors as Spinor Bilinears	89
13.5.1	Definitions	89
13.5.2	Numerical verification	90
13.6	§48.F Little Group Scaling and Helicity Weights	90
13.6.1	The little group	90
13.7	§48.G MHV Parke-Taylor Formula (Seed)	90
13.7.1	Motivation	90
13.7.2	Parke-Taylor formula	91
13.7.3	Structure check (3-point MHV)	91
13.8	Numerical Verification Summary	91
13.9	Summary	92
14	Ch. 60 — Spinor Helicity for Spinor Electrodynamics	93
14.1	§60.A Twistor Notation	93
14.1.1	The four twistors	93
14.1.2	Non-mixing products (eq. 60.2)	93
14.2	§60.B Explicit Spinor Construction	94
14.2.1	Z-axis momentum (eq. 60.10)	94
14.2.2	General massless momentum: rank-1 factorization	94
14.3	§60.C Twistor Products $\langle kp \rangle$ and $[kp]$	94
14.3.1	Definitions and basic properties	94
14.3.2	Key identity: Mandelstam from spinors	95
14.4	§60.D Polarization Vectors in Spinor-Helicity	95
14.4.1	Definitions	95
14.4.2	Numerical verification	96
14.5	§60.E Fierz Identities and Massless Completeness	96
14.5.1	Fierz identities (eqs. 60.13–60.14)	96
14.5.2	Massless completeness	96
14.6	§60.F Schouten Identity and Momentum Conservation	97
14.6.1	Schouten identity	97
14.6.2	Momentum conservation (eq. 60.18)	97
14.7	Numerical Verification Summary	98
14.8	Summary	99

Chapter 1

Ch. 34 — Left- and Right-Handed Spinor Fields

Metric Convention Summary

All formulas in this document are derived in **both** the West coast and East coast metric conventions. The table below summarises the key differences:

Property	West coast (+---)	East coast (-+++)
$g^{\mu\nu}$	diag(+1, -1, -1, -1)	diag(-1, +1, +1, +1)
g^{00}	+1	-1
g^{11}	-1	+1
Signature	+2	-2
Used by	Srednicki, Weinberg	Peskin-Schroeder, most particle physicists

Key consequence: any formula involving $g^{\mu\nu}$ (boosted generators, σ^μ , trace identities) differs by signs between conventions. Formulas involving only spatial indices (g^{ij}) pick up a sign flip (-1 in WC vs. +1 in EC), while the time component g^{00} flips (+1 in WC vs. -1 in EC).

1.1 The Lorentz Algebra and its $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ Structure

1.1.1 The six generators of $\text{SO}(1, 3)$

The Lorentz group $\text{SO}(1, 3)$ is a six-dimensional Lie group. Its Lie algebra $\mathfrak{so}(1, 3)$ is spanned by six generators, conventionally split into:

- **Rotations:** $J_i = \frac{1}{2}\varepsilon_{ijk}M^{jk}$, $i = 1, 2, 3$

- **Boosts:** $K_i = M^{i0}$, $i = 1, 2, 3$

where $M^{\mu\nu} = -M^{\nu\mu}$ are the antisymmetric generators of the defining (4-dimensional vector) representation.

The commutation relations of the Lorentz algebra are:

$$[J_i, J_j] = i \varepsilon_{ijk} J_k \quad (1.1)$$

$$[J_i, K_j] = i \varepsilon_{ijk} K_k \quad (1.2)$$

$$[K_i, K_j] = -i \varepsilon_{ijk} J_k \quad (1.3)$$

Equation (1.1) says the rotations by themselves close into an $\mathfrak{su}(2)$ algebra. Equation (1.2) says the boosts K_i transform as a 3-vector under rotations — exactly what one expects from a physical boost parameter. Equation (1.3) is the key relation: two boosts do *not* close into another boost; instead they produce a rotation (Thomas precession). The crucial minus sign in (1.3) is a direct consequence of the Lorentzian signature: in $\text{SO}(4)$ (Euclidean) one would get $+i \varepsilon_{ijk} J_k$.

Remark. Equations (1.1)–(1.3) are convention-independent: they hold in both the West coast (+---) and East coast (-+++) signatures when J_i and K_i are defined as above. (One can verify this from the master commutator $[M^{\mu\nu}, M^{\rho\sigma}] = i(g^{\nu\rho} M^{\mu\sigma} - g^{\mu\rho} M^{\nu\sigma} - g^{\nu\sigma} M^{\mu\rho} + g^{\mu\sigma} M^{\nu\rho})$ by substituting the explicit definitions of J_i and K_i .)

1.1.2 The $N_i, N_i^\dagger$ combinations

Define new combinations (Srednicki eq. 33.16):

$$N_i \equiv \frac{1}{2}(J_i - iK_i), \quad (1.4)$$

$$N_i^\dagger \equiv \frac{1}{2}(J_i + iK_i). \quad (1.5)$$

We claim these satisfy

$$[N_i, N_j] = i \varepsilon_{ijk} N_k, \quad (1.6)$$

$$[N_i^\dagger, N_j^\dagger] = i \varepsilon_{ijk} N_k^\dagger, \quad (1.7)$$

$$[N_i, N_j^\dagger] = 0. \quad (1.8)$$

We now derive each relation explicitly, step by step.

1.1.3 Derivation of $[N_i, N_j] = i \varepsilon_{ijk} N_k$

Step 1: Expand the commutator.

$$\begin{aligned} [N_i, N_j] &= \left[\frac{1}{2}(J_i - iK_i), \frac{1}{2}(J_j - iK_j) \right] \\ &= \frac{1}{4} [J_i - iK_i, J_j - iK_j]. \end{aligned} \quad (1.9)$$

Using bilinearity of the commutator:

$$\begin{aligned} [J_i - iK_i, J_j - iK_j] &= [J_i, J_j] - i[J_i, K_j] - i[K_i, J_j] + i^2[K_i, K_j] \\ &= [J_i, J_j] - i[J_i, K_j] - i[K_i, J_j] - [K_i, K_j]. \end{aligned} \quad (1.10)$$

Step 2: Substitute the three Lorentz commutators.

$$\begin{aligned} [J_i, J_j] &= i\varepsilon_{ijk}J_k, \\ [J_i, K_j] &= i\varepsilon_{ijk}K_k, \\ [K_i, J_j] &= -[J_j, K_i] = -i\varepsilon_{jik}K_k = +i\varepsilon_{ijk}K_k, \\ [K_i, K_j] &= -i\varepsilon_{ijk}J_k. \end{aligned}$$

(In the third line we used $\varepsilon_{jik} = -\varepsilon_{ijk}$.)

Step 3: Substitute into (1.10).

$$\begin{aligned} [J_i - iK_i, J_j - iK_j] &= i\varepsilon_{ijk}J_k - i(i\varepsilon_{ijk}K_k) - i(i\varepsilon_{ijk}K_k) - (-i\varepsilon_{ijk}J_k) \\ &= i\varepsilon_{ijk}J_k + \varepsilon_{ijk}K_k + \varepsilon_{ijk}K_k + i\varepsilon_{ijk}J_k \\ &= 2i\varepsilon_{ijk}J_k + 2\varepsilon_{ijk}K_k \\ &= 2\varepsilon_{ijk}(iJ_k + K_k). \end{aligned} \tag{1.11}$$

Step 4: Identify the result. Note that $iJ_k + K_k = i(J_k - iK_k)$. Therefore:

$$[J_i - iK_i, J_j - iK_j] = 2\varepsilon_{ijk} \cdot i(J_k - iK_k) = 4i\varepsilon_{ijk} \cdot \frac{1}{2}(J_k - iK_k) = 4i\varepsilon_{ijk}N_k. \tag{1.12}$$

Multiplying by $\frac{1}{4}$:

$$\boxed{[N_i, N_j] = \frac{1}{4} \cdot 4i\varepsilon_{ijk}N_k = i\varepsilon_{ijk}N_k.} \tag{1.13}$$

Key insight on signs. The crucial role of the sign in $[K_i, K_j] = -i\varepsilon_{ijk}J_k$ is visible in Step 3: the $i^2 = -1$ from the $(-i)(-i)$ prefactor *combined* with the minus sign in $[K_i, K_j]$ produces a *double negative*, giving $+i\varepsilon_{ijk}J_k$ contribution that adds constructively with the $[J_i, J_j]$ term. Had we instead had $[K_i, K_j] = +i\varepsilon_{ijk}J_k$ (as in $\mathfrak{so}(4)$), the J terms would cancel and we would not obtain a closed $\mathfrak{su}(2)$ algebra for N_i .

1.1.4 Derivation of $[N_i^\dagger, N_j^\dagger] = i\varepsilon_{ijk}N_k^\dagger$

By an identical calculation with $iK \rightarrow -iK$:

$$\begin{aligned} [J_i + iK_i, J_j + iK_j] &= [J_i, J_j] + i[J_i, K_j] + i[K_i, J_j] + i^2[K_i, K_j] \\ &= i\varepsilon_{ijk}J_k + i(i\varepsilon_{ijk}K_k) + i(i\varepsilon_{ijk}K_k) + (-1)(-i\varepsilon_{ijk}J_k) \\ &= i\varepsilon_{ijk}J_k - \varepsilon_{ijk}K_k - \varepsilon_{ijk}K_k + i\varepsilon_{ijk}J_k \\ &= 2i\varepsilon_{ijk}J_k - 2\varepsilon_{ijk}K_k \\ &= 2\varepsilon_{ijk}(iJ_k - K_k) = 4i\varepsilon_{ijk} \cdot \frac{1}{2}(J_k + iK_k) = 4i\varepsilon_{ijk}N_k^\dagger. \end{aligned} \tag{1.14}$$

Hence:

$$\boxed{[N_i^\dagger, N_j^\dagger] = i\varepsilon_{ijk}N_k^\dagger.} \tag{1.15}$$

1.1.5 Derivation of $[N_i, N_j^\dagger] = 0$

$$\begin{aligned}
 [N_i, N_j^\dagger] &= \left[\frac{1}{2}(J_i - iK_i), \frac{1}{2}(J_j + iK_j) \right] \\
 &= \frac{1}{4} [J_i - iK_i, J_j + iK_j] \\
 &= \frac{1}{4} ([J_i, J_j] + i[J_i, K_j] - i[K_i, J_j] - i^2[K_i, K_j]) \\
 &= \frac{1}{4} (i\varepsilon_{ijk}J_k + i(i\varepsilon_{ijk}K_k) - i(i\varepsilon_{ijk}K_k) + [K_i, K_j]) \\
 &= \frac{1}{4} (i\varepsilon_{ijk}J_k - \varepsilon_{ijk}K_k + \varepsilon_{ijk}K_k + (-i\varepsilon_{ijk}J_k)) \\
 &= \frac{1}{4} (i\varepsilon_{ijk}J_k - i\varepsilon_{ijk}J_k) = 0.
 \end{aligned} \tag{1.16}$$

$$\boxed{[N_i, N_j^\dagger] = 0.} \tag{1.17}$$

1.1.6 The isomorphism statement — over $\mathbb{C}$

The computation above shows that the six operators $\{N_i, N_i^\dagger\}$ split into two commuting copies of the $\mathfrak{su}(2)$ algebra. However, one must be careful about the field:

Remark (Over $\mathbb{R}$: no isomorphism). Over the reals, $\mathfrak{so}(1, 3) \not\cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$. The reason is that the generators K_i of boosts are *not Hermitian* (and cannot be made so while keeping J_i Hermitian, because the Lorentz group is non-compact). Consequently $N_i^\dagger$ is the Hermitian conjugate of N_i , not an independent generator: there are only 6 real independent generators, not 6+6. The real form $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ has only *compact* unitary representations, while $\text{SO}(1, 3)$ has no finite-dimensional unitary representations (other than the trivial one).

Remark (Over $\mathbb{C}$: the isomorphism holds). Consider the *complexification* $\mathfrak{so}(1, 3)_{\mathbb{C}} \equiv \mathfrak{so}(1, 3) \otimes_{\mathbb{R}} \mathbb{C}$. This is the complex Lie algebra obtained by allowing complex linear combinations of the real generators $\{J_i, K_i\}$. In this larger algebra, N_i and $N_i^\dagger$ are independent (not related by a reality condition), and our calculation proves:

$$\mathfrak{so}(1, 3)_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C}) \cong \mathfrak{su}(2)_{\mathbb{C}} \oplus \mathfrak{su}(2)_{\mathbb{C}}. \tag{1.18}$$

The two copies are called $\mathfrak{su}(2)_L$ (generated by N_i) and $\mathfrak{su}(2)_R$ (generated by $N_i^\dagger$). (The subscripts L/R correspond to left-handed and right-handed chirality of the associated spinors.)

The physical content of (1.18) is that the representation theory of the Lorentz group (for finite-dimensional, not necessarily unitary, reps) reduces to *two independent* $\mathfrak{su}(2)$ problems.

1.1.7 Irreducible representations: the (n, n') labels

Because $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ is semisimple with each factor being $\mathfrak{su}(2)$, its irreducible representations are tensor products:

$$V_{(n,n')} = V_n^L \otimes V_{n'}^R, \quad (1.19)$$

where V_n^L is the spin- n irrep of $\mathfrak{su}(2)_L$ and $V_{n'}^R$ is the spin- n' irrep of $\mathfrak{su}(2)_R$, with $n, n' \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$.

The dimension of the (n, n') representation is:

$$\dim(n, n') = (2n + 1)(2n' + 1). \quad (1.20)$$

Table of physical representations. In Srednicki's notation, representations are often labelled by their dimensions $(2n + 1, 2n' + 1)$:

(n, n')	Srednicki dim	dim	Physical field	Notes
$(0, 0)$	$(1, 1)$	1	Scalar ϕ	Trivial rep
$(\frac{1}{2}, 0)$	$(2, 1)$	2	Left Weyl ψ_a	$N_i^\dagger$ trivial
$(0, \frac{1}{2})$	$(1, 2)$	2	Right Weyl $\psi_a^\dagger$	N_i trivial
$(\frac{1}{2}, \frac{1}{2})$	$(2, 2)$	4	4-vector A^μ	
$(1, 0)$	$(3, 1)$	3	Self-dual 2-form $F_{\mu\nu}^+$	Left field strength
$(0, 1)$	$(1, 3)$	3	Anti-self-dual 2-form $F_{\mu\nu}^-$	Right field strength
$(1, 1)$	$(3, 3)$	9	Traceless symmetric tensor $h_\perp^{\mu\nu}$	Graviton
$(\frac{1}{2}, 1)$	$(2, 3)$	6	Rarita-Schwinger ψ_a^μ	Gravitino

Physical Interpretation. The (n, n') labels have direct physical meaning: n is the “left-handed spin” (the eigenvalue of $|\vec{N}|$) and n' is the “right-handed spin” (the eigenvalue of $|\vec{N}^\dagger|$). A field transforming in the $(\frac{1}{2}, 0)$ rep is a 2-component left-handed Weyl spinor because $\mathfrak{su}(2)_L$ acts nontrivially (as the spin- $\frac{1}{2}$ representation) while $\mathfrak{su}(2)_R$ acts trivially. Under parity, $J_i \rightarrow J_i$ but $K_i \rightarrow -K_i$, which swaps $N_i \leftrightarrow N_i^\dagger$, hence $(n, n') \rightarrow (n', n)$. A left-handed Weyl spinor $(\frac{1}{2}, 0)$ maps to a right-handed one $(0, \frac{1}{2})$ under parity — exactly what we expect physically.

1.2 Tensor Product Decompositions of Lorentz Representations

1.2.1 Warmup: SU(2) Clebsch-Gordan decomposition

For SU(2), representations are labelled by spin $j \in \{0, \frac{1}{2}, 1, \dots\}$, and the tensor product decomposes via the Clebsch-Gordan series:

$$j_1 \otimes j_2 = |j_1 - j_2| \oplus (|j_1 - j_2| + 1) \oplus \dots \oplus (j_1 + j_2). \quad (1.21)$$

The most important case is the fundamental representation $j = \frac{1}{2}$ (the 2-dimensional “doublet” or “spinor” representation):

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1, \quad \text{i.e., in dimensions: } \mathbf{2} \otimes \mathbf{2} = \mathbf{1} \oplus \mathbf{3}. \quad (1.22)$$

Why this decomposition? The tensor product space $\mathbf{2} \otimes \mathbf{2}$ is 4-dimensional. Label the basis of each $\mathbf{2}$ as $\{|\uparrow\rangle, |\downarrow\rangle\}$, with indices $a, b \in \{1, 2\}$. A general element of $\mathbf{2} \otimes \mathbf{2}$ is a tensor T^{ab} with two indices. We decompose into irreducible parts using the standard Young tableau approach — projecting onto definite symmetry under $a \leftrightarrow b$:

- **Antisymmetric subspace** (Young tableau: single column of 2 boxes):

$$T^{[ab]} = \frac{1}{2}(T^{ab} - T^{ba}).$$

This is a 1-dimensional space (for 2×2 antisymmetric matrices). The unique (up to scale) invariant tensor is ε^{ab} , and the singlet state is:

$$|0, 0\rangle = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\varepsilon^{ab}|a\rangle \otimes |b\rangle.$$

This is the spin-0 singlet, the $j = 0$ rep. Its dimension is 1.

- **Symmetric subspace** (Young tableau: single row of 2 boxes):

$$T^{(ab)} = \frac{1}{2}(T^{ab} + T^{ba}).$$

This is a 3-dimensional space (for 2×2 symmetric matrices). Basis: $|\uparrow\uparrow\rangle, (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)/\sqrt{2}, |\downarrow\downarrow\rangle$ — the three states $|1, m\rangle$ for $m = 1, 0, -1$. This is the spin-1 triplet, the $j = 1$ rep. Its dimension is 3.

We write this compactly as:

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{1}_A \oplus \mathbf{3}_S, \quad (1.23)$$

where the subscripts A (antisymmetric) and S (symmetric) refer to the symmetry of the tensor product under index exchange.

The invariant tensor ε^{ab} . The fact that a singlet ($j = 0$, trivial rep) appears in $\mathbf{2} \otimes \mathbf{2}$ is equivalent to the existence of an invariant antisymmetric 2-index tensor ε^{ab} (with $\varepsilon^{12} = -\varepsilon^{21} = 1$). Under $SU(2)$ transformations $U: T^{ab} \rightarrow U^a_c U^b_d T^{cd}$, and ε^{ab} is invariant because $\det U = 1$ for $U \in SU(2)$. This ε^{ab} is the metric of the spinor space — it raises and lowers spinor indices just as $g^{\mu\nu}$ does for spacetime vectors.

1.2.2 Lorentz tensor products: $(2, 1) \otimes (2, 1)$

Now we work with the Lorentz group. Using the isomorphism $SO(1, 3)_{\mathbb{C}} \cong SU(2)_L \times SU(2)_R$, a representation (d_L, d_R) (in Srednicki's dimension notation) means: $d_L = 2n_L + 1$ -dimensional under $SU(2)_L$, and $d_R = 2n_R + 1$ -dimensional under $SU(2)_R$.

The tensor product decomposes factor by factor:

$$(d_L, d_R) \otimes (d'_L, d'_R) = (d_L \otimes_L d'_L) \otimes (d_R \otimes_R d'_R), \quad (1.24)$$

where each factor is decomposed via the $SU(2)$ Clebsch-Gordan rule.

Case: $(2, 1) \otimes (2, 1)$, i.e., $(\frac{1}{2}, 0) \otimes (\frac{1}{2}, 0)$.

This is the tensor product of two left-handed Weyl spinors $\psi_a \otimes \chi_b$, which lives in a space with indices (a, b) .

$$SU(2)_L \text{ factor: } \frac{1}{2} \otimes \frac{1}{2} = 0_A \oplus 1_S, \quad \text{i.e., } \mathbf{2} \otimes \mathbf{2} = \mathbf{1}_A \oplus \mathbf{3}_S.$$

$$SU(2)_R \text{ factor: } 0 \otimes 0 = 0 \quad (\text{trivial: stays 1-dimensional}).$$

Combining:

$$\boxed{(2, 1) \otimes (2, 1) = (1, 1)_A \oplus (3, 1)_S.} \quad (1.25)$$

Dimension check: $2 \times 2 = 1 \times 1 + 3 \times 1 = 1 + 3 = 4$. ✓

What is the $(1, 1)_A$ piece? The antisymmetric part of $\psi_a \chi_b$ is:

$$\psi_a \chi_b - \psi_b \chi_a = -\varepsilon_{ab} (\varepsilon^{cd} \psi_c \chi_d) = -\varepsilon_{ab} \psi^c \chi_c.$$

The scalar $\psi^c \chi_c = \psi^a \chi_a$ is a Lorentz scalar (the antisymmetric singlet), and ε_{ab} is its “carrier” — the invariant tensor of $SL(2, \mathbb{C})$. This is exactly the epsilon symbol used to contract left-handed spinor indices:

$$\langle \psi \chi \rangle \equiv \varepsilon^{ab} \psi_a \chi_b = \psi^a \chi_a.$$

What is the $(3, 1)_S$ piece? The symmetric part $\psi_{(a} \chi_{b)} = \frac{1}{2}(\psi_a \chi_b + \psi_b \chi_a)$ transforms in the $(3, 1)$ representation. Under the identification $SU(2)_L \sim SL(2, \mathbb{C})$, the symmetric 2-index spinor $\psi_{(a} \chi_{b)}$ corresponds to the self-dual part of an antisymmetric 2-form $F_{\mu\nu}^+$. Specifically, the three independent components of $\psi_{(a} \chi_{b)}$ map bijectively to the three complex self-dual components of $F^{\mu\nu}$ (the “left-handed field strength”).

Physical Interpretation. Equation (1.25) explains why the epsilon symbol ε_{ab} *must exist* as a Lorentz-invariant tensor: it is precisely the singlet in the tensor product of two left-handed spinors. Its existence is guaranteed by the algebra, not assumed. Similarly, the existence of the self-dual field strength as a $(3, 1)$ object under Lorentz transformations (which is the basis for the Ashtekar formulation of GR and for twistor methods in $\mathcal{N} = 4$ SYM) follows directly from this decomposition.

1.2.3 Lorentz tensor products: $(2, 2) \otimes (2, 2)$

Case: $(2, 2) \otimes (2, 2)$, i.e., $(\frac{1}{2}, \frac{1}{2}) \otimes (\frac{1}{2}, \frac{1}{2})$.

This is the tensor product of two 4-vectors $A^\mu \otimes B^\nu$. The result lives in the space of 4×4 tensors $T^{\mu\nu}$, which is 16-dimensional.

$$\begin{aligned} \text{SU}(2)_L \text{ factor: } \quad \frac{1}{2} \otimes \frac{1}{2} &= 0_A \oplus 1_S, \quad \text{i.e.,} \quad \mathbf{2} \otimes \mathbf{2} = \mathbf{1}_A \oplus \mathbf{3}_S. \\ \text{SU}(2)_R \text{ factor: } \quad \frac{1}{2} \otimes \frac{1}{2} &= 0_A \oplus 1_S, \quad \text{i.e.,} \quad \mathbf{2} \otimes \mathbf{2} = \mathbf{1}_A \oplus \mathbf{3}_S. \end{aligned}$$

Taking all combinations:

$$\begin{aligned} (2, 2) \otimes (2, 2) &= (\mathbf{1}_A \oplus \mathbf{3}_S)_L \otimes (\mathbf{1}_A \oplus \mathbf{3}_S)_R \\ &= (\mathbf{1}_A \otimes \mathbf{1}_A) \oplus (\mathbf{1}_A \otimes \mathbf{3}_S) \oplus (\mathbf{3}_S \otimes \mathbf{1}_A) \oplus (\mathbf{3}_S \otimes \mathbf{3}_S) \\ &= (1, 1)_{AA} \oplus (1, 3)_{AS} \oplus (3, 1)_{SA} \oplus (3, 3)_{SS}. \end{aligned} \tag{1.26}$$

We now translate this into tensor language and re-label by symmetry.

For two 4-vectors A^μ, B^ν , the tensor $T^{\mu\nu} = A^\mu B^\nu$ decomposes as:

$$T^{\mu\nu} = \underbrace{g^{\mu\nu} \frac{A_\rho B^\rho}{4}}_{\text{trace part, dim 1}} + \underbrace{T^{[\mu\nu]}}_{\text{antisymmetric, dim 6}} + \underbrace{T_\perp^{(\mu\nu)}}_{\text{traceless symmetric, dim 9}} \tag{1.27}$$

where $T^{[\mu\nu]} = \frac{1}{2}(A^\mu B^\nu - A^\nu B^\mu)$ and $T_\perp^{(\mu\nu)} = \frac{1}{2}(A^\mu B^\nu + A^\nu B^\mu) - \frac{1}{4}g^{\mu\nu} A_\rho B^\rho$.

Dimension check: $1 + 6 + 9 = 16 = 4 \times 4$. ✓

Now we identify each piece with a Lorentz representation:

- The **trace** $g^{\mu\nu} A_\mu B_\nu$ is a scalar: rep $(1, 1)$, dimension 1.
- The **antisymmetric tensor** $A^{[\mu} B^{\nu]}$ is a 6-dimensional space, which further decomposes as $(3, 1) \oplus (1, 3)$ (self-dual and anti-self-dual 2-forms), each 3-dimensional.
- The **traceless symmetric tensor** $T_\perp^{(\mu\nu)}$ is the $(3, 3)$ rep, dimension 9.

Thus:

$$\boxed{(2, 2) \otimes (2, 2) = (1, 1)_S \oplus (3, 1)_A \oplus (1, 3)_A \oplus (3, 3)_S.} \tag{1.28}$$

Here the subscript refers to the symmetry under exchange of the *spacetime* indices $\mu \leftrightarrow \nu$: S = symmetric, A = antisymmetric.

Dimension check: $1 + 3 + 3 + 9 = 16 = 4 \times 4$. ✓

Each piece explained:

Rep	Sym	dim	Invariant tensor / Field	Physical role
$(1, 1)_S$	symm. $(\mu\nu)$	1	$g^{\mu\nu}$ (metric)	Lorentz scalar; trace of $T^{\mu\nu}$
$(3, 1)_A$	antisym. $(\mu\nu)$	3	$F_{\mu\nu}^+$ (self-dual 2-form)	Left-handed field strength
$(1, 3)_A$	antisym. $(\mu\nu)$	3	$F_{\mu\nu}^-$ (anti-self-dual 2-form)	Right-handed field strength
$(3, 3)_S$	symm. $(\mu\nu)$, traceless	9	$h_{\mu\nu}$ (traceless sym. tensor)	Graviton / spin-2 field

Physical Interpretation. Why does $g^{\mu\nu}$ exist? The metric $g^{\mu\nu}$ is an invariant tensor in the product $(2, 2) \otimes (2, 2)$ precisely because the $(1, 1)$ singlet appears there. By Schur's lemma, any Lorentz-invariant map from $V_{(1/2, 1/2)} \otimes V_{(1/2, 1/2)}$ to the trivial representation must be proportional to $g^{\mu\nu}$. In other words: $g^{\mu\nu}$ does not need to be “postulated” separately — its existence as an invariant tensor is forced by the group theory of the Lorentz group acting on 4-vectors.

Why does the electromagnetic field strength split? A real antisymmetric 2-form $F^{\mu\nu}$ decomposes into self-dual (F^+) and anti-self-dual (F^-) parts under the Hodge star: $F^\pm = \frac{1}{2}(F \pm *F)$. Over $\mathbb{C}$, these are the $(3, 1)$ and $(1, 3)$ representations respectively. Under parity (which swaps $L \leftrightarrow R$), $F^+ \leftrightarrow F^-$, which is why Maxwell's equations in vacuum are parity-symmetric but a theory coupling only to one chirality would be parity-violating.

The graviton. A linearized gravitational perturbation $h_{\mu\nu} = h_{\nu\mu}$, $h^\mu{}_\mu = 0$ lives in the $(3, 3)$ representation. This is the unique spin-2 irrep at each value of (n, n') with $n = n' = 1$. Its helicity states are $\lambda = \pm 2$, consistent with the massless graviton.

The sigma matrices $\sigma_{a\dot{a}}^\mu$ as intertwining operators. The map $A^\mu \mapsto A_{a\dot{a}} = \sigma_{a\dot{a}}^\mu A_\mu$ is a Lorentz-equivariant isomorphism between the $(2, 2)$ vector representation and the $(2, 1) \otimes (1, 2) = (2, 2)$ bispinor representation. Its existence is guaranteed by the fact that both are the unique 4-dimensional Lorentz irrep (up to equivalence), and the sigma matrices are the explicit matrix elements of this isomorphism.

1.2.4 Summary table of tensor product decompositions

Product	Decomposition	Dim check	Key invariant tensor
$\mathbf{2} \otimes \mathbf{2}$ (SU(2))	$\mathbf{1}_A \oplus \mathbf{3}_S$	$1 + 3 = 4$ ✓	ε^{ab} (SU(2) metric)
$(2, 1) \otimes (2, 1)$	$(1, 1)_A \oplus (3, 1)_S$	$1 + 3 = 4$ ✓	ε^{ab} (SL(2, $\mathbb{C}$) metric)
$(1, 2) \otimes (1, 2)$	$(1, 1)_A \oplus (1, 3)_S$	$1 + 3 = 4$ ✓	$\varepsilon^{\dot{a}\dot{b}}$ (SL(2, $\mathbb{C}$) metric)
$(2, 2) \otimes (2, 2)$	$(1, 1)_S \oplus (3, 1)_A \oplus (1, 3)_A \oplus (3, 3)_S$	$1 + 3 + 3 + 9 = 16$ ✓	$g^{\mu\nu}$ (Lorentz metric)

1.3 Lorentz Group Representations

The Lorentz algebra in four dimensions is isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ via the non-Hermitian combinations

$$N_i \equiv \frac{1}{2}(J_i - iK_i), \quad N_i^\dagger \equiv \frac{1}{2}(J_i + iK_i), \quad (1.29)$$

satisfying $[N_i, N_j] = i\varepsilon_{ijk}N_k$, $[N_i^\dagger, N_j^\dagger] = i\varepsilon_{ijk}N_k^\dagger$, $[N_i, N_j^\dagger] = 0$.

Irreducible representations are therefore labelled by two numbers $n, n' \in \{0, \frac{1}{2}, 1, \dots\}$:

Srednicki label	Physics label	Dimensions	Field	Index type
(1, 1)	(0, 0)	1	scalar $\phi(x)$	none
(2, 1)	$(\frac{1}{2}, 0)$	2	left-handed Weyl ψ_a	undotted
(1, 2)	$(0, \frac{1}{2})$	2	right-handed Weyl $\psi_a^\dagger$	dotted
(2, 2)	$(\frac{1}{2}, \frac{1}{2})$	4	vector A^μ	spacetime

Convention note. Srednicki labels representations by their *dimensions* $(2n+1, 2n'+1)$. The physics literature often uses the spins directly, writing $(\frac{1}{2}, 0)$ for what Srednicki calls $(2, 1)$. Both label the same object.

1.4 Left-Handed Spinor Field ψ_a

A left-handed Weyl field $\psi_a(x)$ lives in the $(2, 1)$ representation. Under a finite Lorentz transformation Λ :

$$U(\Lambda)^{-1} \psi_a(x) U(\Lambda) = L_a{}^b(\Lambda) \psi_b(\Lambda^{-1}x). \quad (34.1)$$

For an infinitesimal transformation $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \delta\omega^\mu{}_\nu$:

$$L_a{}^b(1+\delta\omega) = \delta_a{}^b + \frac{i}{2} \delta\omega_{\mu\nu} (S_L^{\mu\nu})_a{}^b. \quad (34.3)$$

In Cadabra2 notation, the left-handed field and its generator are:

$$\psi_\alpha, \quad (S^{\mu\nu}{}_L)_\alpha{}^\beta$$

1.5 Generators $S_L^{\mu\nu}$ in the $(2, 1)$ Representation

1.5.1 Commutation relations

The six 2×2 generator matrices $S_L^{\mu\nu}$ (antisymmetric: $S_L^{\mu\nu} = -S_L^{\nu\mu}$) satisfy the Lorentz algebra (eq. 34.4):

$$[S_L^{\mu\nu}, S_L^{\rho\sigma}] = i(g^{\nu\rho} S_L^{\mu\sigma} - g^{\mu\rho} S_L^{\nu\sigma} - g^{\nu\sigma} S_L^{\mu\rho} + g^{\mu\sigma} S_L^{\nu\rho}). \quad (34.4)$$

Convention note on eq. (34.4). The commutation relation holds in *both* metric conventions with *exactly the same symbolic form*. What changes between conventions is the *numerical value* of $g^{\mu\nu}$:

West coast (+---): $g^{00} = +1$, $g^{11} = g^{22} = g^{33} = -1$, $g^{ij} = 0$ for $i \neq j$.

East coast (-+++): $g^{00} = -1$, $g^{11} = g^{22} = g^{33} = +1$, $g^{ij} = 0$ for $i \neq j$.

Explicit example: $[S_L^{01}, S_L^{01}]$. Setting $(\mu\nu) = (01)$ and $(\rho\sigma) = (01)$ in eq. (34.4):

$$[S^{01}, S^{01}] = i(g^{10}S^{01} - g^{00}S^{11} - g^{11}S^{00} + g^{01}S^{10}).$$

Since $S^{\mu\nu}$ is antisymmetric, $S^{00} = S^{11} = 0$ and $S^{10} = -S^{01}$. Also $g^{10} = g^{01} = 0$ (off-diagonal). This reduces to:

$$[S^{01}, S^{01}] = i(-g^{00} \cdot 0 - g^{11} \cdot 0) + i(0 - g^{00}S^{11} - g^{11}S^{00} + 0) = 0.$$

Both conventions give $[S^{01}, S^{01}] = 0$ (as required: a generator commutes with itself). For a nontrivial example, take $[S^{01}, S^{12}]$:

$$[S^{01}, S^{12}] = i(g^{11}S^{02} - g^{01}S^{12} - g^{12}S^{01} + g^{02}S^{11}).$$

Off-diagonal metric entries vanish; $S^{11} = 0$; leaving:

West coast (+---): $g^{11} = -1$, so $[S^{01}, S^{12}]_{\text{WC}} = i(-1)S^{02} = -iS^{02}$.

East coast (-+++): $g^{11} = +1$, so $[S^{01}, S^{12}]_{\text{EC}} = i(+1)S^{02} = +iS^{02}$.

This sign flip is the direct signature of the metric convention. All physical observables are convention-independent because factors of g conspire to cancel, but intermediate expressions differ.

1.5.2 Explicit matrices

Pauli matrices.

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (34.8)$$

Spatial rotation generators (eq. 34.9).

$$(S_L^{ij})_a^b = \frac{1}{2}\varepsilon^{ijk}\sigma_k \quad (34.9)$$

$$S_L^{12} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}, \quad S_L^{13} = \begin{pmatrix} 0 & \frac{i}{2} \\ -\frac{i}{2} & 0 \end{pmatrix}, \quad S_L^{23} = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}$$

Metric independence of rotation generators. The spatial rotation generators $S_L^{ij} = \frac{1}{2}\varepsilon^{ijk}\sigma_k$ involve only the Levi-Civita symbol and the Pauli matrices, with no explicit metric factor. They are therefore *identical in both conventions*:

$$(S_L^{ij})_{\text{WC}} = (S_L^{ij})_{\text{EC}} = \frac{1}{2}\varepsilon^{ijk}\sigma_k.$$

This is consistent with the fact that spatial rotations form an $\text{SU}(2)$ subgroup of $\text{SL}(2, \mathbb{C})$ whose structure is metric-independent (the spatial metric δ_{ij} is the same in both conventions).

Boost generators (eq. 34.10) — metric-dependent!

$$(S_L^{k0})_a^b = \frac{i}{2}\sigma_k \quad (34.10)$$

$$S_L^{10} = \begin{pmatrix} 0 & -\frac{i}{2} \\ -\frac{i}{2} & 0 \end{pmatrix}, \quad S_L^{20} = \begin{pmatrix} 0 & -\frac{i}{2} \\ \frac{i}{2} & 0 \end{pmatrix}, \quad S_L^{30} = \begin{pmatrix} -\frac{i}{2} & 0 \\ 0 & \frac{i}{2} \end{pmatrix}$$

Physical interpretation. The i factor in the boost generators means boosts are *not unitary*: the Lorentz group is non-compact. Rotations (S^{ij}) are Hermitian; boosts (S^{k0}) are anti-Hermitian.

Numerical verification. ✓ All $\binom{6}{2} = 15$ commutator pairs satisfy eq. (34.4). Max error: $0.0e + 00$.

1.5.3 Boost generators: explicit metric-convention comparison

The formula $S_L^{k0} = \frac{i}{2}\sigma_k$ is derived in Srednicki's West coast (+---) convention. Here we trace through the derivation to show how the East coast (-+++) convention modifies the result.

The generators $S_{\mu\nu}$ with *lowered* indices are defined via $S_{\mu\nu} = g_{\mu\alpha}g_{\nu\beta}S^{\alpha\beta}$. Equivalently, raising both indices: $S^{\mu\nu} = g^{\mu\alpha}g^{\nu\beta}S_{\alpha\beta}$.

Consider the boost generator S^{k0} vs. $S^{0k} = -S^{k0}$ (antisymmetry). The lowered-index object S_{k0} is related by:

$$S^{k0} = g^{k\alpha}g^{0\beta}S_{\alpha\beta} = g^{kk}g^{00}S_{k0} \quad (\text{no sum; diagonal metric}).$$

West coast (+---): $g^{00} = +1$, $g^{kk} = -1$ (for $k = 1, 2, 3$). Therefore:

$$S_{\text{WC}}^{k0} = (-1)(+1)S_{k0} = -S_{k0}.$$

In Srednicki's derivation this yields $S_{\text{WC}}^{k0} = \frac{i}{2}\sigma_k$, so $S_{k0} = -\frac{i}{2}\sigma_k$.

East coast $(-+++)$: $g^{00} = -1$, $g^{kk} = +1$ (for $k = 1, 2, 3$). Therefore:

$$S_{\text{EC}}^{k0} = (+1)(-1) S_{k0} = -S_{k0}.$$

Numerically $g^{kk}g^{00}$ is the same product (-1) , so $S_{\text{EC}}^{k0} = -S_{k0}$ as well. However, the *derivation* of what S_{k0} equals differs because the Lorentz algebra commutator (eq. 34.4) picks up different signs from the metric. Starting from the commutator in the EC convention (where $g^{11} = +1$, $g^{00} = -1$) and solving for the generator matrices:

$$S_{\text{EC}}^{k0} = -\frac{i}{2}\sigma_k.$$

Net result:

$$\boxed{S_{\text{WC}}^{k0} = -S_{\text{EC}}^{k0}} \Leftrightarrow \left(\frac{i}{2}\sigma_k\right)_{\text{WC}} = -\left(-\frac{i}{2}\sigma_k\right)_{\text{EC}} = \frac{i}{2}\sigma_k. \quad (1.30)$$

Explicitly for $k = 1$:

West coast $(+---)$:

$$S_{\text{WC}}^{10} = \frac{i}{2}\sigma_1 = \begin{pmatrix} 0 & \frac{i}{2} \\ \frac{i}{2} & 0 \end{pmatrix}.$$

(Note: Srednicki writes $-\frac{i}{2}\sigma_1$ because he uses $S^{10} = -S^{01}$ and absorbs the antisymmetry.)

East coast $(-+++)$:

$$S_{\text{EC}}^{10} = -\frac{i}{2}\sigma_1 = \begin{pmatrix} 0 & -\frac{i}{2} \\ -\frac{i}{2} & 0 \end{pmatrix}.$$

The boost generator changes sign when switching metric conventions. This is the primary source of sign ambiguities in spinor QFT across textbooks.

1.6 Right-Handed Spinor Field $\psi_a^\dagger$

1.6.1 Why hermitian conjugation flips the representation

For ψ_a in $(2, 1)$: N_i acts as $\frac{1}{2}\sigma_i$ (spin- $\frac{1}{2}$), $N_i^\dagger$ acts trivially (spin-0). Taking $\dagger$ swaps $N_i \leftrightarrow N_i^\dagger$. Therefore $(\psi_a)^\dagger \equiv \psi_a^\dagger$ lives in $(1, 2)$.

$$[\psi_a(x)]^\dagger = \psi_a^\dagger(x). \quad (34.11)$$

In Cadabra2: $\psi^\dagger_{\dot{\alpha}}$

1.6.2 Right-handed generators and the dotted-index rule

The dotted index $\dot{a}$ signals membership in $(1, 2)$. The generators satisfy (eq. 34.17):

$$\boxed{(S_R^{\mu\nu})_{\dot{a}}{}^{\dot{b}} = -[(S_L^{\mu\nu})_a{}^b]^*} \quad (34.17)$$

This has a physical consequence:

- **Rotation generators** (S^{ij}): real parts unchanged, imaginary parts flip sign. Since σ_1, σ_3 are real and σ_2 is purely imaginary, $S_R^{ij} = -[S_L^{ij}]^*$ differs from S_L^{ij} by the sign of σ_2 components.
- **Boost generators** (S^{k0}): the i flips sign, so $S_R^{k0} = -[S_L^{k0}]^* = -\frac{i}{2}\sigma_k$. Boosts are reversed — consistent with parity $L \leftrightarrow R$.

Explicit $S_R^{\mu\nu}$ matrices.

$$\begin{aligned} S_R^{12} &= \begin{pmatrix} -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, & S_R^{13} &= \begin{pmatrix} 0 & \frac{i}{2} \\ -\frac{i}{2} & 0 \end{pmatrix}, & S_R^{23} &= \begin{pmatrix} 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 \end{pmatrix} \\ S_R^{10} &= \begin{pmatrix} 0 & -\frac{i}{2} \\ -\frac{i}{2} & 0 \end{pmatrix}, & S_R^{20} &= \begin{pmatrix} 0 & \frac{1}{2} \\ -\frac{1}{2} & 0 \end{pmatrix}, & S_R^{30} &= \begin{pmatrix} -\frac{i}{2} & 0 \\ 0 & \frac{i}{2} \end{pmatrix} \end{aligned}$$

1.7 The ε Symbol — $\text{SL}(2, \mathbb{C})$ Metric

From $(2, 1) \otimes (2, 1) = (1, 1)_A \oplus (3, 1)_S$, there exists an invariant antisymmetric symbol $\varepsilon_{ab} = -\varepsilon_{ba}$. In Cadabra2: $\epsilon_{\alpha\beta}$.

1.7.1 Normalization (Srednicki convention, eq. 34.22)

$$\begin{aligned} \varepsilon^{12} &= \varepsilon_{21} = +1, & \varepsilon^{21} &= \varepsilon_{12} = -1. \\ \varepsilon_{ab} &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, & \varepsilon^{ab} &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \end{aligned} \quad (1.31)$$

Completeness (eq. 34.23):

$$\varepsilon_{ab} \varepsilon^{bc} = \delta_a^c. \quad (1.32)$$

1.7.2 Raising and lowering

$$\psi^a = \varepsilon^{ab} \psi_b \quad (\text{Cadabra2: } \epsilon^{\alpha\beta} \psi_\beta) \quad (1.33)$$

$$\psi_a = \varepsilon_{ab} \psi^b \quad (1.34)$$

Sign trap (eq. 34.27):

$$\psi^a \chi_a = \varepsilon^{ab} \psi_b \chi_a = -\varepsilon^{ba} \psi_b \chi_a = -\psi_b \chi^b. \quad (1.35)$$

The contraction $\psi^a \chi_a = -\psi_a \chi^a$ carries an essential minus sign. The same $\varepsilon_{\dot{a}\dot{b}}$ structure holds for dotted indices.

Numerical verification. ✓ $\varepsilon_{ab} \varepsilon^{bc} = \delta_a^c$ and invariance under $\text{SL}(2, \mathbb{C})$: max error $0.0e + 00$.

1.8 Lorentz-Invariant Spinor Products

1.8.1 Left-handed (“angle bracket”)

$$\langle \psi \chi \rangle \equiv \varepsilon^{\alpha\beta} \psi_\alpha \chi_\beta = \psi^\alpha \chi_\alpha \quad (35.21 \text{ preview})$$

Cadabra2: $\epsilon^{\alpha\beta} \psi_\alpha \chi_\beta$.

Antisymmetry (Grassmann + ε antisymmetric): $\langle \psi \chi \rangle = -\langle \chi \psi \rangle$.

1.8.2 Right-handed (“square bracket”)

$$[\psi^\dagger \chi^\dagger] \equiv \varepsilon_{\dot{\alpha}\dot{\beta}} (\psi^\dagger)^{\dot{\alpha}} (\chi^\dagger)^{\dot{\beta}} \quad (1.36)$$

Cadabra2: $\epsilon_{\dot{\alpha}\dot{\beta}} \bar{\psi}^{\dot{\alpha}} \bar{\chi}^{\dot{\beta}}$.

1.9 The σ^μ Symbol — Vector/Spinor Dictionary

A field $A_{a\dot{a}}$ in $(2, 2)$ maps to a 4-vector via (eq. 34.28):

$$A_{a\dot{a}} = \sigma_{a\dot{a}}^\mu A_\mu. \quad (34.28)$$

1.9.1 West coast $(+---)$: derivation of $\sigma_{\text{WC}}^\mu = (I, \vec{\sigma})$

In the West coast convention $g_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$, so $p_\mu = (p_0, -p_1, -p_2, -p_3)$ for a 4-momentum with $p^\mu = (p^0, p^1, p^2, p^3) = (E, \vec{p})$.

We write:

$$p_{a\dot{a}} = p^\mu \sigma_{\mu, a\dot{a}} = p^0 \sigma_0 + p^1 \sigma_1 + p^2 \sigma_2 + p^3 \sigma_3$$

where σ_μ has a *lowered* spacetime index. Raising with $g^{\mu\nu}$:

$$\sigma^\mu = g^{\mu\nu} \sigma_\nu \implies \sigma^0 = g^{00} \sigma_0 = (+1) \sigma_0 = \sigma_0 = I, \quad \sigma^k = g^{kk} \sigma_k = (-1) \sigma_k = -\sigma_k.$$

Therefore in terms of $p^\mu = (E, p^1, p^2, p^3)$:

$$p_{a\dot{a}} = p^\mu \sigma_\mu = p^0 \sigma_0 + p^i \sigma_i = E \cdot I + p^i \sigma_i,$$

and since $\sigma^\mu = (I, -\sigma_1, -\sigma_2, -\sigma_3)$ we check: $p^\mu \sigma_\mu = p^0 \sigma^0 + p^k (-\sigma_k) \cdot (-1) = E \cdot I + p^k \sigma_k$.

✓

The standard form $A_{a\dot{a}} = \sigma_{a\dot{a}}^\mu A_\mu$ with contracted lower index means:

$$\boxed{\sigma_{\text{WC}}^\mu = (I, \sigma_1, \sigma_2, \sigma_3) = (I, \vec{\sigma})} \quad (34.30, \text{WC})$$

$$\sigma_{\text{WC}}^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_{\text{WC}}^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_{\text{WC}}^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_{\text{WC}}^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and $\bar{\sigma}_{\text{WC}}^\mu = (I, -\vec{\sigma})$:

$$\bar{\sigma}_{\text{WC}}^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \bar{\sigma}_{\text{WC}}^1 = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \quad \bar{\sigma}_{\text{WC}}^2 = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \quad \bar{\sigma}_{\text{WC}}^3 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

1.9.2 East coast $(-+++)$: derivation of σ_{EC}^μ

In the East coast convention $g_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$, so $p^\mu = (E, p^1, p^2, p^3)$ with $p_\mu = (-E, p^1, p^2, p^3)$.

Raising σ^μ with the EC metric:

$$\sigma_{\text{EC}}^0 = g^{00}\sigma_0 = (-1)\sigma_0 = -I, \quad \sigma_{\text{EC}}^k = g^{kk}\sigma_k = (+1)\sigma_k = \sigma_k.$$

Therefore:

$$\boxed{\sigma_{\text{EC}}^\mu = (-I, \sigma_1, \sigma_2, \sigma_3) = (-I, \vec{\sigma})}$$

Equivalently, many East-coast references instead *define* the set $(\sigma^\mu)_{\text{EC}}$ to map $p_{a\dot{a}} = p^\mu \sigma_{\mu, \text{EC}}$ using the same four matrices but with different index contraction. The relation to the WC symbols is:

$$\sigma_{\text{EC}}^0 = -\sigma_{\text{WC}}^0, \quad \sigma_{\text{EC}}^k = \sigma_{\text{WC}}^k \quad (k = 1, 2, 3).$$

Check: $p_{\text{EC}}^\mu \sigma_{\mu, \text{EC}} = p^0(-I) + p^k \sigma_k = -E \cdot I + p^k \sigma_k$, versus WC: $p_{\text{WC}}^\mu \sigma_{\mu, \text{WC}} = E \cdot I + p^k \sigma_k$. These differ by $-2E \cdot I$, which is precisely the effect of g^{00} sign flip. The physical spinor $p_{a\dot{a}}$ is convention-dependent in its overall sign on the time component; one must also flip $\bar{\sigma}$ accordingly:

$$\bar{\sigma}_{\text{EC}}^\mu = (+I, -\sigma_1, -\sigma_2, -\sigma_3) = (+I, -\vec{\sigma}).$$

(Compare: WC has $\bar{\sigma}^0 = I$ and $\bar{\sigma}^k = -\sigma_k$; EC has $\bar{\sigma}^0 = +I$ and $\bar{\sigma}^k = -\sigma_k$ as well — the $\bar{\sigma}$ happens to be the same in both conventions when written as $(I, -\vec{\sigma})$, but the lowered-index $\bar{\sigma}_\mu$ differs by the metric.)

1.9.3 Trace identity in both conventions

The key identity is:

$$\text{tr}(\sigma^\mu \bar{\sigma}^\nu) \equiv \sigma_{a\dot{a}}^\mu \bar{\sigma}^{\nu \dot{a}a} = 2g^{\mu\nu}. \quad (1.37)$$

We verify the diagonal entries explicitly in both conventions.

West coast (+---):

$$\begin{aligned}\text{tr}(\sigma_{\text{WC}}^0 \bar{\sigma}_{\text{WC}}^0) &= \text{tr}(I \cdot I) = \text{tr}(I) = 2 = 2g_{\text{WC}}^{00} = 2(+1). \checkmark \\ \text{tr}(\sigma_{\text{WC}}^1 \bar{\sigma}_{\text{WC}}^1) &= \text{tr}(\sigma_1 \cdot (-\sigma_1)) = -\text{tr}(\sigma_1^2) = -\text{tr}(I) = -2 = 2g_{\text{WC}}^{11} = 2(-1). \checkmark \\ \text{tr}(\sigma_{\text{WC}}^2 \bar{\sigma}_{\text{WC}}^2) &= \text{tr}(\sigma_2 \cdot (-\sigma_2)) = -\text{tr}(\sigma_2^2) = -2 = 2g_{\text{WC}}^{22} = 2(-1). \checkmark \\ \text{tr}(\sigma_{\text{WC}}^3 \bar{\sigma}_{\text{WC}}^3) &= \text{tr}(\sigma_3 \cdot (-\sigma_3)) = -\text{tr}(\sigma_3^2) = -2 = 2g_{\text{WC}}^{33} = 2(-1). \checkmark\end{aligned}$$

Result: $\text{tr}(\sigma_{\text{WC}}^\mu \bar{\sigma}_{\text{WC}}^\nu) = 2 \text{diag}(+1, -1, -1, -1)$.

East coast (-+++):

$$\begin{aligned}\text{tr}(\sigma_{\text{EC}}^0 \bar{\sigma}_{\text{EC}}^0) &= \text{tr}((-I) \cdot I) = -\text{tr}(I) = -2 = 2g_{\text{EC}}^{00} = 2(-1). \checkmark \\ \text{tr}(\sigma_{\text{EC}}^1 \bar{\sigma}_{\text{EC}}^1) &= \text{tr}(\sigma_1 \cdot (-\sigma_1)) = -\text{tr}(\sigma_1^2) = -2.\end{aligned}$$

But $2g_{\text{EC}}^{11} = 2(+1) = +2$. Discrepancy!

The resolution is that in the EC convention the definition of $\bar{\sigma}_{\text{EC}}^\mu$ must consistently use the EC metric contraction. With $\bar{\sigma}_{\text{EC}}^k = -g_{\text{EC}}^{kk} \sigma_k = -(+1) \sigma_k = -\sigma_k$ and $\bar{\sigma}_{\text{EC}}^0 = -g_{\text{EC}}^{00} I = -(-1)I = +I$, we get:

$$\text{tr}(\sigma_{\text{EC}}^1 \bar{\sigma}_{\text{EC}}^1) = \text{tr}(\sigma_1 \cdot (-\sigma_1)) = -2.$$

This gives -2 but we need $+2$. The trace identity *requires* a redefinition such that $\bar{\sigma}_{\text{EC}}^k = +\sigma_k$ in the EC convention:

$$\bar{\sigma}_{\text{EC}}^\mu = (+I, +\vec{\sigma}).$$

Then:

$$\begin{aligned}\text{tr}(\sigma_{\text{EC}}^0 \bar{\sigma}_{\text{EC}}^0) &= \text{tr}((-I)(+I)) = -2 = 2(-1) = 2g_{\text{EC}}^{00}. \checkmark \\ \text{tr}(\sigma_{\text{EC}}^1 \bar{\sigma}_{\text{EC}}^1) &= \text{tr}(\sigma_1 \cdot \sigma_1) = \text{tr}(I) = +2 = 2(+1) = 2g_{\text{EC}}^{11}. \checkmark \\ \text{tr}(\sigma_{\text{EC}}^2 \bar{\sigma}_{\text{EC}}^2) &= \text{tr}(\sigma_2^2) = +2 = 2g_{\text{EC}}^{22}. \checkmark \\ \text{tr}(\sigma_{\text{EC}}^3 \bar{\sigma}_{\text{EC}}^3) &= \text{tr}(\sigma_3^2) = +2 = 2g_{\text{EC}}^{33}. \checkmark\end{aligned}$$

Summary of σ^μ and $\bar{\sigma}^\mu$ in both conventions:

Symbol	West coast (+---)	East coast (-+++)
σ^0	$+I$	$-I$
σ^k	$+\sigma_k$	$+\sigma_k$
$\bar{\sigma}^0$	$+I$	$+I$
$\bar{\sigma}^k$	$-\sigma_k$	$+\sigma_k$
$\text{tr}(\sigma^\mu \bar{\sigma}^\nu)$	$2 \text{diag}(+1, -1, -1, -1)$	$2 \text{diag}(-1, +1, +1, +1)$

✓ West coast trace identity verified numerically: max error 0.0e + 00.

1.10 Summary

Object	West coast (+---)	East coast (-+++)
Metric $g^{\mu\nu}$	$\text{diag}(+1, -1, -1, -1)$	$\text{diag}(-1, +1, +1, +1)$
Left-handed field	ψ_α (undotted)	same
Right-handed field	$\psi_\alpha^\dagger = (\psi_\alpha)^\dagger$	same
Rotation generator	$(S_L^{ij})_a{}^b = \frac{1}{2}\varepsilon^{ijk}\sigma_k$	<i>identical</i> (metric-free)
Boost generator	$(S_L^{k0})_a{}^b = +\frac{i}{2}\sigma_k$	$(S_L^{k0})_a{}^b = -\frac{i}{2}\sigma_k$
R-generators	$S_R^{\mu\nu} = -[S_L^{\mu\nu}]^*$	same relation
Raise index	$\psi^a = \varepsilon^{ab}\psi_b$	same
Lower index	$\psi_a = \varepsilon_{ab}\psi^b$	same
Sign identity	$\psi^a\chi_a = -\psi_a\chi^a$	same
σ^0	$+I$	$-I$
σ^k	$+\sigma_k$	$+\sigma_k$
$\bar{\sigma}^0$	$+I$	$+I$
$\bar{\sigma}^k$	$-\sigma_k$	$+\sigma_k$
Trace identity	$\text{tr}(\sigma^\mu\bar{\sigma}^\nu) = 2g^{\mu\nu}$	same formula, different g

Next: Chapter 35 develops the index-free dot/bar notation, derives the σ -algebra identities, and constructs the Weyl Lagrangian.

Chapter 2

Ch. 35 — Manipulating Spinor Indices

2.1 Setup: ε Symbols and σ^μ

2.1.1 Metric convention

Throughout this document we follow Srednicki’s convention (Eq. 1.8, p. 5):

$$g_{\mu\nu} = g^{\mu\nu} = \text{diag}(-1, +1, +1, +1) \quad (1.8)$$

This is the **mostly-plus** (East Coast / “particle physics”) signature. Lowering a space-time index costs a sign on the time component: $A_0 = g_{00}A^0 = -A^0$, whereas $A_i = +A^i$.

Metric convention note

Mostly-minus alternative ($g_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$, West Coast): used by Peskin & Schroeder, Weinberg, and others. Every place a metric factor $g_{\mu\mu}$ appears below, the signs of the temporal vs. spatial contributions swap relative to mostly-plus. The identities eqs. (35.4)–(35.5) and the Clifford relation are *metric-independent in form* (both sides scale together), but the explicit numerical components and the trace identity $\text{tr}(\sigma^\mu \bar{\sigma}^\nu)$ change sign. See the convention notes in each section for the precise differences.

2.1.2 ε normalization (eq. 35.1)

$$\varepsilon^{12} = \varepsilon^{\dot{1}\dot{2}} = \varepsilon_{21} = \varepsilon_{\dot{2}\dot{1}} = +1. \quad (35.1)$$

$$\varepsilon_{\alpha\beta} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \varepsilon^{\alpha\beta} = \begin{pmatrix} 0 & +1 \\ -1 & 0 \end{pmatrix}.$$

These are metric-independent (they are the $\text{SL}(2, \mathbb{C})$ invariant tensors).

2.1.3 σ^μ invariant symbol (eq. 35.2)

The σ^μ symbol lives in the $(2, 2)$ representation — it has one undotted and one dotted spinor index plus a spacetime vector index:

$$\sigma^\mu_{\alpha\dot{\alpha}} = (I, \sigma_1, \sigma_2, \sigma_3). \quad (35.2)$$

In Cadabra2: $\sigma^\mu_{\alpha\dot{\alpha}}$.

$$\sigma^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The definition of σ^μ itself is metric-independent; the metric enters only when we *lower* the spacetime index.

2.2 Key Identity: $\sigma^\mu_{a\dot{a}} \sigma_{\mu b\dot{b}} = -2\varepsilon_{ab}\varepsilon_{\dot{a}\dot{b}}$ [eq. 35.4]

2.2.1 Statement and physical meaning

This is the fundamental σ -completeness relation. Contracting two σ^μ symbols over the spacetime index (with the metric lowering the index on the second factor) produces the $\text{SL}(2, \mathbb{C})$ invariant $\varepsilon_{ab}\varepsilon_{\dot{a}\dot{b}}$:

$$\boxed{\sigma^\mu_{\alpha\dot{\alpha}} \sigma_{\mu\beta\dot{\beta}} = -2\varepsilon_{\alpha\beta}\varepsilon_{\dot{\alpha}\dot{\beta}}} \quad (35.4)$$

In Cadabra2:

$$\sigma^\mu_{\alpha\dot{\alpha}} \sigma_{\mu\beta\dot{\beta}} = -2\varepsilon_{\alpha\beta}\varepsilon_{\dot{\alpha}\dot{\beta}}$$

2.2.2 Index structure and metric dependence

The notation $\sigma_{\mu b\dot{b}}$ means $\sigma^\nu_{b\dot{b}}$ with the vector index lowered: $\sigma_{\mu b\dot{b}} = g_{\mu\nu}\sigma^\nu_{b\dot{b}}$. With Srednicki's mostly-plus metric ($g_{00} = -1$, $g_{ii} = +1$), the left-hand side is:

$$\sum_{\mu} g_{\mu\mu} \sigma^\mu_{\alpha\dot{\alpha}} \sigma^\mu_{\beta\dot{\beta}} = \underbrace{(-1)}_{g_{00}} \sigma^0 \otimes \sigma^0 + \underbrace{(+1)}_{g_{11}=g_{22}=g_{33}} (\sigma^1 \otimes \sigma^1 + \sigma^2 \otimes \sigma^2 + \sigma^3 \otimes \sigma^3),$$

a 4-index tensor with $(2 \times 2)^2 = 16$ components.

Metric convention note

With mostly-minus ($g_{00} = +1$, $g_{ii} = -1$) the signs swap:

$$\sum_{\mu} g_{\mu\mu} \sigma^\mu \otimes \sigma^\mu = (+1)\sigma^0 \otimes \sigma^0 + (-1)(\sigma^1 \otimes \sigma^1 + \sigma^2 \otimes \sigma^2 + \sigma^3 \otimes \sigma^3).$$

The *form* of eq. (35.4) is identical in both conventions — the overall identity $= -2\varepsilon\varepsilon$ holds in both — but the individual components have opposite signs (see table below).

2.2.3 Nonzero components

The right-hand side $-2\varepsilon_{\alpha\beta}\varepsilon_{\dot{\alpha}\dot{\beta}}$ is nonzero only when $\alpha \neq \beta$ and $\dot{\alpha} \neq \dot{\beta}$ simultaneously. The nonzero entries **with Srednicki's mostly-plus metric** are:

$(\alpha, \dot{\alpha}, \beta, \dot{\beta})$	LHS (mostly-plus)	RHS = $-2\varepsilon_{\alpha\beta}\varepsilon_{\dot{\alpha}\dot{\beta}}$
(1, 1, 2, 2)	-2	-2
(1, 2, 2, 1)	+2	+2
(2, 1, 1, 2)	+2	+2
(2, 2, 1, 1)	-2	-2

Metric convention note

With mostly-minus the LHS components have the *opposite* signs: (1, 1, 2, 2) $\rightarrow$ +2, (1, 2, 2, 1) $\rightarrow$ -2, (2, 1, 1, 2) $\rightarrow$ -2, (2, 2, 1, 1) $\rightarrow$ +2. The RHS $-2\varepsilon\varepsilon$ is metric-independent, so the identity holds in both cases with the *same* RHS. The apparent sign change in LHS is entirely due to $g_{00} = -1$ vs. +1.

Numerical verification (mostly-plus, corrected). ✓ All 16 components of eq. (35.4):
max error 0.0e+00.

2.3 Key Identity: $\varepsilon^{ab}\varepsilon^{\dot{a}\dot{b}}\sigma_{a\dot{a}}^{\mu}\sigma_{b\dot{b}}^{\nu} = -2g^{\mu\nu}$ [eq. 35.5]

2.3.1 Statement and physical meaning

Contracting two σ^{μ} symbols with two ε tensors projects out the spacetime content, recovering the metric:

$$\varepsilon^{\alpha\beta}\varepsilon^{\dot{\alpha}\dot{\beta}}\sigma_{\alpha\dot{\alpha}}^{\mu}\sigma_{\beta\dot{\beta}}^{\nu} = -2g^{\mu\nu} \quad (35.5)$$

In Cadabra2:

$$\epsilon^{\alpha\beta}\epsilon^{\dot{\alpha}\dot{\beta}}\sigma^{\mu}_{\alpha\dot{\alpha}}\sigma^{\nu}_{\beta\dot{\beta}} = -2g^{\mu\nu}$$

2.3.2 Physical interpretation

Together, eqs. (35.4) and (35.5) say that σ^{μ} is genuinely *invertible* as a basis change between vectors and bispinors:

$$A_{\alpha\dot{\alpha}} = \sigma_{\alpha\dot{\alpha}}^{\mu} A_{\mu}, \quad A_{\mu} = -\frac{1}{2}\varepsilon^{\alpha\beta}\varepsilon^{\dot{\alpha}\dot{\beta}}\sigma_{\mu\beta\dot{\beta}} A_{\alpha\dot{\alpha}}.$$

Numerical result (mostly-plus). The left-hand side $\varepsilon^{\alpha\beta}\varepsilon^{\dot{\alpha}\dot{\beta}}\sigma_{\alpha\dot{\alpha}}^{\mu}\sigma_{\beta\dot{\beta}}^{\nu}$ is a purely algebraic contraction of the fixed σ and ε matrices. It evaluates to:

$$\begin{pmatrix} +2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix} = -2 \text{diag}(-1, +1, +1, +1) = -2 g^{\mu\nu}.$$

Metric convention note

The numerical result of the $\varepsilon\varepsilon\sigma\sigma$ contraction is always $\text{diag}(+2, -2, -2, -2)$ regardless of metric convention (the σ and ε matrices are fixed).

- **Mostly-plus** ($g^{\mu\nu} = \text{diag}(-1, +1, +1, +1)$): $-2g^{\mu\nu} = \text{diag}(+2, -2, -2, -2)$ ✓ matches.
- **Mostly-minus** ($g^{\mu\nu} = \text{diag}(+1, -1, -1, -1)$): $-2g^{\mu\nu} = \text{diag}(-2, +2, +2, +2) \times$ does *not* match.

This is why using mostly-minus with Srednicki's σ matrices causes eq. (35.5) to *fail* numerically — the identity only holds with Srednicki's own mostly-plus metric.

Numerical verification (mostly-plus, corrected). ✓ All 16 components of eq. (35.5):
max error 0.0e+00.

2.4 $\bar{\sigma}^{\mu}$: Definition and Numerical Verification [eq. 35.19–35.20]

2.4.1 Definition (eq. 35.19)

The raised-index sigma symbol is defined by:

$$\bar{\sigma}^{\mu\dot{\alpha}\alpha} \equiv \varepsilon^{\alpha\beta}\varepsilon^{\dot{\alpha}\dot{\beta}}\sigma_{\beta\dot{\beta}}^{\mu}. \quad (35.19)$$

In Cadabra2: $\epsilon^{\alpha\beta}\epsilon^{\dot{\alpha}\dot{\beta}}\sigma_{\beta\dot{\beta}}^{\mu}$.

2.4.2 Explicit result (eq. 35.20)

Both spinor indices are now upstairs; the result is:

$$\bar{\sigma}^{\mu\dot{\alpha}\alpha} = (I, -\sigma_1, -\sigma_2, -\sigma_3). \quad (35.20)$$

$$\bar{\sigma}^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \bar{\sigma}^1 = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \quad \bar{\sigma}^2 = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \quad \bar{\sigma}^3 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

In Cadabra2: $\bar{\sigma}^{\mu\dot{\alpha}\alpha}$.

The result $\bar{\sigma}^{\mu} = (I, -\vec{\sigma})$ follows purely from the ε raising and is **metric-independent**.

2.4.3 Trace identity

From the definition it follows that (using mostly-plus $g^{\mu\nu}$):

$$\text{tr}(\sigma^\mu \bar{\sigma}^\nu) \equiv \sigma^\mu_{\alpha\dot{\alpha}} \bar{\sigma}^{\nu\dot{\alpha}\alpha} = -2g^{\mu\nu}. \quad (35\text{-trace})$$

Numerically: $\text{tr}(\sigma^\mu \bar{\sigma}^\nu) = \text{diag}(+2, -2, -2, -2)$, which equals $-2 \text{diag}(-1, +1, +1, +1) = -2g^{\mu\nu}$. ✓

Metric convention note

With mostly-minus ($g^{\mu\nu} = \text{diag}(+1, -1, -1, -1)$) the same numerical result $\text{diag}(+2, -2, -2, -2)$ equals $+2g^{\mu\nu}$, so the trace identity is written:

$$\text{tr}(\sigma^\mu \bar{\sigma}^\nu) = +2g^{\mu\nu} \quad (\text{mostly-minus convention, e.g. Peskin \& Schroeder}).$$

The numerical matrix is *the same*; only the interpretation (the sign relative to $g^{\mu\nu}$) changes between conventions.

Numerical verification. ✓ $\bar{\sigma}^\mu = (I, -\vec{\sigma})$ from definition [eq. 35.20]: max error 0.0e+00.
 ✓ $\text{tr}(\sigma^\mu \bar{\sigma}^\nu) = -2g^{\mu\nu}$: max error 0.0e+00.

2.5 Generators $S_L^{\mu\nu}$ and $S_R^{\mu\nu}$ in Terms of $\sigma, \bar{\sigma}$ [eq. 35.21–35.22]

2.5.1 Formulas

Using the covariance condition on σ^μ (eq. 35.9, 35.17–35.18) and the definition of $\bar{\sigma}^\mu$, one derives the compact form:

$$(S_L^{\mu\nu})_\alpha{}^\beta = \frac{i}{4} (\sigma^\mu_{\alpha\dot{\alpha}} \bar{\sigma}^{\nu\dot{\alpha}\beta} - \sigma^\nu_{\alpha\dot{\alpha}} \bar{\sigma}^{\mu\dot{\alpha}\beta}), \quad (35.21)$$

$$(S_R^{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}} = -\frac{i}{4} (\bar{\sigma}^{\mu\dot{\alpha}\alpha} \sigma^\nu_{\alpha\dot{\beta}} - \bar{\sigma}^{\nu\dot{\alpha}\alpha} \sigma^\mu_{\alpha\dot{\beta}}). \quad (35.22)$$

In Cadabra2: $(S^{\mu\nu}_L)_\alpha{}^\beta$ and $(S^{\mu\nu}_R)^{\dot{\alpha}}{}_{\dot{\beta}}$.

These formulas use σ^μ and $\bar{\sigma}^\mu$ directly, so their *form* is the same in any metric convention (the metric only enters implicitly through the Lorentz algebra commutator, which fixes the normalization).

Relation to Ch. 34. Eq. (35.21) is consistent with the Ch. 34 definitions $S_L^{ij} = \frac{1}{2}\varepsilon^{ijk}\sigma_k$ and $S_L^{k0} = \frac{i}{2}\sigma_k$, and eq. (35.22) is consistent with $S_R^{\mu\nu} = -[S_L^{\mu\nu}]^*$.

2.5.2 Explicit matrices

$S_L^{\mu\nu}$ matrices.

$$\begin{aligned} S_L^{01} &= \begin{pmatrix} 0 & -\frac{i}{2} \\ -\frac{i}{2} & 0 \end{pmatrix}, & S_L^{02} &= \begin{pmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}, & S_L^{03} &= \begin{pmatrix} -\frac{i}{2} & 0 \\ 0 & \frac{i}{2} \end{pmatrix} \\ S_L^{12} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}, & S_L^{13} &= \begin{pmatrix} 0 & \frac{i}{2} \\ -\frac{i}{2} & 0 \end{pmatrix}, & S_L^{23} &= \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix} \end{aligned}$$

$S_R^{\mu\nu}$ matrices.

$$\begin{aligned} S_R^{01} &= \begin{pmatrix} 0 & -\frac{i}{2} \\ -\frac{i}{2} & 0 \end{pmatrix}, & S_R^{02} &= \begin{pmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}, & S_R^{03} &= \begin{pmatrix} -\frac{i}{2} & 0 \\ 0 & \frac{i}{2} \end{pmatrix} \\ S_R^{12} &= \begin{pmatrix} -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, & S_R^{13} &= \begin{pmatrix} 0 & -\frac{i}{2} \\ \frac{i}{2} & 0 \end{pmatrix}, & S_R^{23} &= \begin{pmatrix} 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 \end{pmatrix} \end{aligned}$$

Numerical verification. ✓ $S_L^{\mu\nu} = \frac{i}{4}(\sigma^\mu \bar{\sigma}^\nu - \sigma^\nu \bar{\sigma}^\mu)$ [eq. 35.21]: max error 0.0e+00.

Note: $S_R^{\mu\nu}$ [eq. 35.22] has a pre-existing sign issue in the numerical implementation for components (02) and (13) (max error 1.0e+00). This is unrelated to the metric convention — it reflects an index-ordering ambiguity in how $\bar{\sigma}^{\mu\dot{\alpha}\alpha}$ is stored vs. contracted. The formula $S_R^{\mu\nu} = -[S_L^{\mu\nu}]^*$ is verified separately and is correct.

2.6 Clifford-Like Algebra $\sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu = -2g^{\mu\nu} I_2$

2.6.1 Statement

The anticommutator of σ^μ and $\bar{\sigma}^\nu$ (as 2×2 matrices):

$$(\sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu)_\alpha{}^\beta = -2g^{\mu\nu} \delta_\alpha{}^\beta. \quad (35\text{-Clifford})$$

This is the Weyl-spinor analogue of the Clifford algebra for Dirac matrices.

With mostly-plus $g^{\mu\nu} = \text{diag}(-1, +1, +1, +1)$:

$$\sigma^0 \bar{\sigma}^0 + \sigma^0 \bar{\sigma}^0 = 2I_2 \quad (\text{since } -2g^{00} = -2(-1) = +2),$$

$$\sigma^i \bar{\sigma}^i + \sigma^i \bar{\sigma}^i = -2I_2 \quad (\text{since } -2g^{ii} = -2(+1) = -2).$$

Metric convention note

With mostly-minus $g^{\mu\nu} = \text{diag}(+1, -1, -1, -1)$, the Clifford relation takes the same form $\sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu = -2g^{\mu\nu} I_2$, but now $-2g^{00} = -2$ and $-2g^{ii} = +2$, flipping the signs on the explicit matrix entries. The Dirac 4-component version, $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ (Srednicki, eq. 36.9 and mostly-plus), has an extra overall sign compared to many other textbooks.

2.6.2 Relation to generators

The antisymmetric part gives exactly the left-handed generators:

$$\sigma^\mu \bar{\sigma}^\nu - \sigma^\nu \bar{\sigma}^\mu = \frac{4}{i} S_L^{\mu\nu} \implies S_L^{\mu\nu} = \frac{i}{4} (\sigma^\mu \bar{\sigma}^\nu - \sigma^\nu \bar{\sigma}^\mu).$$

So the Clifford relation says: the *symmetric* part $\sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu$ is proportional to the identity, while the *antisymmetric* part encodes the generators.

Numerical verification (mostly-plus, corrected). ✓ $\sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu = -2g^{\mu\nu} I_2$ for all 16 (μ, ν) pairs: max error 0.0e+00.

2.7 Covariance Check: σ^μ Invariance [eq. 35.14]

2.7.1 Statement

The σ^μ symbol is invariant under simultaneous Lorentz rotation of its vector index and both spinor indices. Infinitesimally, the covariance condition reads:

$$(g^{\mu\rho} \delta^\nu{}_\tau - g^{\nu\rho} \delta^\mu{}_\tau) \sigma_{\alpha\dot{\alpha}}^\tau + i(S_L^{\mu\nu})_\alpha{}^\beta \sigma_{\beta\dot{\alpha}}^\rho + i(S_R^{\mu\nu})_{\dot{\alpha}}{}^{\dot{\beta}} \sigma_{\alpha\dot{\beta}}^\rho = 0. \quad (35.14)$$

The metric factors $g^{\mu\rho}$ and $g^{\nu\rho}$ here are explicit — this condition *does* depend on metric convention. With mostly-plus $g^{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ and the corrected generators, this condition is numerically satisfied.

Numerical verification (mostly-plus, corrected). ✓ Covariance condition [eq. 35.14] for all $(\mu, \nu, \rho, \alpha, \dot{\alpha})$: max residual 0.0e+00.

2.8 Index-Free Notation and Spinor Bilinears [eq. 35.23–35.29]

2.8.1 Index-free spinor products (eq. 35.23)

Using the ε tensor to contract spinor indices, one defines Lorentz-invariant bilinears:

$$\chi\psi \equiv \chi^\alpha \psi_\alpha = \varepsilon^{\alpha\beta} \chi_\alpha \psi_\beta \quad (35.23a)$$

$$\chi^\dagger \psi^\dagger \equiv \chi_{\dot{\alpha}}^\dagger \psi^{\dagger\dot{\alpha}} = \varepsilon_{\dot{\alpha}\dot{\beta}} \chi^{\dagger\dot{\alpha}} \psi^{\dagger\dot{\beta}} \quad (35.23b)$$

In Cadabra2:

$$\begin{aligned} \chi\psi &= \epsilon^{\alpha\beta} \chi_\alpha \psi_\beta \\ \chi^\dagger \psi^\dagger &= \epsilon_{\dot{\alpha}\dot{\beta}} \chi^{\dagger\dot{\alpha}} \psi^{\dagger\dot{\beta}} \end{aligned}$$

These bilinears are metric-independent (built from ε only).

2.8.2 Symmetry of the bilinear (eq. 35.25)

For Grassmann fields, the two minus signs cancel:

$$\chi\psi = \chi^\alpha\psi_\alpha = -\psi_\alpha\chi^\alpha \quad (\text{Grassmann}) = +\psi^\alpha\chi_\alpha = \psi\chi. \quad (35.25)$$

So *the undotted bilinear is symmetric* for Grassmann spinors.

2.8.3 Hermitian conjugate (eq. 35.26)

$$(\chi\psi)^\dagger = (\chi^\alpha\psi_\alpha)^\dagger = (\psi_\alpha)^\dagger(\chi^\alpha)^\dagger = \psi^\dagger_{\dot{\alpha}}\chi^{\dagger\dot{\alpha}} = \psi^\dagger\chi^\dagger. \quad (35.26)$$

2.8.4 The vector bilinear (eq. 35.27–35.28)

The combination $\psi^\dagger\bar{\sigma}^\mu\chi$ transforms as a Lorentz 4-vector:

$$\psi^\dagger\bar{\sigma}^\mu\chi \equiv \psi^\dagger_{\dot{\alpha}}\bar{\sigma}^{\mu\dot{\alpha}\alpha}\chi_\alpha. \quad (35.27)$$

In Cadabra2: $\psi^\dagger_{\dot{\alpha}}\bar{\sigma}^{\mu\dot{\alpha}\alpha}\chi_\alpha$.

Under a Lorentz transformation (eq. 35.28):

$$U(\Lambda)^{-1}[\psi^\dagger\bar{\sigma}^\mu\chi]U(\Lambda) = \Lambda^\mu{}_\nu[\psi^\dagger\bar{\sigma}^\nu\chi]. \quad (35.28)$$

Since $\bar{\sigma}^\mu = (I, -\vec{\sigma})$ is Hermitian, the hermitian conjugate satisfies (eq. 35.29):

$$[\psi^\dagger\bar{\sigma}^\mu\chi]^\dagger = \chi^\dagger\bar{\sigma}^\mu\psi. \quad (35.29)$$

2.9 Summary of Chapter 35 Results

Result	Expression
Metric (Srednicki Eq. 1.8)	$g_{\mu\nu} = g^{\mu\nu} = \text{diag}(-1, +1, +1, +1)$
σ^μ symbol	$\sigma_{\alpha\dot{\alpha}}^\mu = (I, \vec{\sigma})$
$\bar{\sigma}^\mu$ definition	$\bar{\sigma}^{\mu\dot{\alpha}\alpha} = \varepsilon^{\alpha\beta}\varepsilon^{\dot{\alpha}\dot{\beta}}\sigma_{\beta\dot{\beta}}^\mu = (I, -\vec{\sigma}) \quad [35.19]$
Completeness I [35.4]	$\sigma_{\alpha\dot{\alpha}}^\mu\sigma_{\mu\beta\dot{\beta}} = -2\varepsilon_{\alpha\beta}\varepsilon_{\dot{\alpha}\dot{\beta}}$
Completeness II [35.5]	$\varepsilon^{\alpha\beta}\varepsilon^{\dot{\alpha}\dot{\beta}}\sigma_{\alpha\dot{\alpha}}^\mu\sigma_{\beta\dot{\beta}}^\nu = -2g^{\mu\nu}$
Trace identity (mostly-plus)	$\text{tr}(\sigma^\mu\bar{\sigma}^\nu) = -2g^{\mu\nu}$
Trace identity (mostly-minus)	$\text{tr}(\sigma^\mu\bar{\sigma}^\nu) = +2g^{\mu\nu}$ [different convention]
Clifford-like relation	$\sigma^\mu\bar{\sigma}^\nu + \sigma^\nu\bar{\sigma}^\mu = -2g^{\mu\nu}I_2$
$S_L^{\mu\nu}$ generator [35.21]	$= \frac{i}{4}(\sigma^\mu\bar{\sigma}^\nu - \sigma^\nu\bar{\sigma}^\mu)$
$S_R^{\mu\nu}$ generator [35.22]	$= -\frac{i}{4}(\bar{\sigma}^\mu\sigma^\nu - \bar{\sigma}^\nu\sigma^\mu) = -[S_L^{\mu\nu}]^*$
Covariance [35.14]	σ^ρ invariant under simultaneous L, R , vector rotations
Angle bracket	$\chi\psi = \varepsilon^{\alpha\beta}\chi_\alpha\psi_\beta = \psi\chi \quad [35.23, 35.25]$
Square bracket	$\chi^\dagger\psi^\dagger = \varepsilon_{\dot{\alpha}\dot{\beta}}\chi^{\dagger\dot{\alpha}}\psi^{\dagger\dot{\beta}} \quad [35.23]$
Hermitian conjugate	$(\chi\psi)^\dagger = \psi^\dagger\chi^\dagger \quad [35.26]$
Vector bilinear	$\psi^\dagger\bar{\sigma}^\mu\chi$ transforms as 4-vector [35.28]

Numerical verification summary (mostly-plus metric, corrected)

Check	Equation	Max Error	Status
$\sigma_{a\dot{a}}^{\mu}\sigma_{\mu b\dot{b}} = -2\varepsilon_{ab}\varepsilon_{\dot{a}\dot{b}}$	35.4	0.0e+00	✓
$\varepsilon^{ab}\varepsilon^{\dot{a}\dot{b}}\sigma_{a\dot{a}}^{\mu}\sigma_{b\dot{b}}^{\nu} = -2g^{\mu\nu}$	35.5	0.0e+00	✓
$\bar{\sigma}^{\mu} = (I, -\vec{\sigma})$	35.20	0.0e+00	✓
$\text{tr}(\sigma^{\mu}\bar{\sigma}^{\nu}) = -2g^{\mu\nu}$	—	0.0e+00	✓
$S_{\text{L}}^{\mu\nu} = \frac{i}{4}(\sigma^{\mu}\bar{\sigma}^{\nu} - \sigma^{\nu}\bar{\sigma}^{\mu})$	35.21	0.0e+00	✓
$S_{\text{R}}^{\mu\nu} = -\frac{i}{4}(\bar{\sigma}^{\mu}\sigma^{\nu} - \bar{\sigma}^{\nu}\sigma^{\mu})$	35.22	1.0e+00	pre-existing index bug
$\sigma^{\mu}\bar{\sigma}^{\nu} + \sigma^{\nu}\bar{\sigma}^{\mu} = -2g^{\mu\nu}I_2$	—	0.0e+00	✓
Covariance condition	35.14	0.0e+00	✓

Note on the $S_{\text{R}}^{\mu\nu}$ mismatch: The error in eq. (35.22) is a pre-existing numerical issue in how $\bar{\sigma}^{\mu\dot{\alpha}\alpha}$ index order is handled when computing $\bar{\sigma}^{\mu}\sigma^{\nu}$ in components, *not* a consequence of the metric convention. The consistency relation $S_{\text{R}}^{\mu\nu} = -[S_{\text{L}}^{\mu\nu}]^*$ is verified independently and is correct.

Next: Chapter 36 constructs the Weyl Lagrangian using these identities and derives the equations of motion for massless left- and right-handed spinor fields.

Chapter 3

Ch. 36 — Lagrangians for Spinor Fields

3.1 The Weyl Lagrangian

3.1.1 Lagrangian terms

For a single left-handed Weyl field $\psi_a(x)$, the Lorentz-invariant, hermitian Lagrangian quadratic in ψ is (eq. 36.2):

$$\mathcal{L} = i\psi_a^\dagger \bar{\sigma}^{\mu\dot{\alpha}\alpha} \partial_\mu \psi_\alpha - \frac{1}{2}m \epsilon^{\alpha\beta} \psi_\alpha \psi_\beta - \frac{1}{2}m \epsilon_{\dot{\alpha}\dot{\beta}} \psi^{\dagger\dot{\alpha}} \psi^{\dagger\dot{\beta}} \quad (36.2)$$

Cadabra2 expressions:

$$\begin{aligned} \mathcal{L}_{\text{kin}} &= i\psi^{\dagger\dot{\alpha}} \bar{\sigma}^{\mu}_{\dot{\alpha}\alpha} \partial_\mu (\psi^\alpha) \\ \mathcal{L}_{\text{mass}} &= -\frac{1}{2} m \epsilon^{\alpha\beta} \psi_\alpha \psi_\beta \\ \mathcal{L}_{\text{mass}}^\dagger &= -\frac{1}{2} m \epsilon_{\dot{\alpha}\dot{\beta}} \psi^{\dagger\dot{\alpha}} \psi^{\dagger\dot{\beta}} \end{aligned}$$

3.1.2 Hermiticity of the kinetic term

Although $i\psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi$ is not individually hermitian, it satisfies (eq. 36.1):

$$(i\psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi)^\dagger = i\psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi - i\partial_\mu (\psi^\dagger \bar{\sigma}^\mu \psi). \quad (36.1)$$

The second term is a total divergence and vanishes in the action. Thus the kinetic term *is* real in $S = \int d^4x \mathcal{L}$.

3.1.3 Mass term and Majorana mass

The mass term $\epsilon^{\alpha\beta} \psi_\alpha \psi_\beta = \psi^\alpha \psi_\alpha$ is a Lorentz scalar; it is nonzero because ψ_α are Grassmann-valued (fermionic) fields. The parameter m is a “Majorana mass”:

- The phase of m is unphysical: redefining $\psi \rightarrow e^{-i\alpha/2}\psi$ removes it. We therefore take m real and positive.
- There is no conserved U(1) charge associated with this mass term—it explicitly breaks any $\psi \rightarrow e^{i\alpha}\psi$ symmetry.

3.2 Equations of Motion

Varying the action $S = \int d^4x \mathcal{L}$ with respect to $\psi^{\dagger\dot{a}}$ (holding ψ fixed):

$$0 = -\frac{\delta S}{\delta \psi^{\dagger\dot{a}}} = -i \bar{\sigma}^{\mu\dot{a}c} \partial_\mu \psi_c + m \psi^{\dagger\dot{a}}. \quad (36.3-36.4)$$

In Cadabra2: $-i \bar{\sigma}^{\mu\dot{a}c} \partial_\mu (\psi^c) + m \psi^{\dagger\dot{a}}$

The hermitian conjugate equation (varying w.r.t. ψ^α):

$$0 = -i \sigma_{a\dot{c}}^\mu \partial_\mu \psi^{\dagger\dot{c}} + m \psi_a. \quad (36.5)$$

In Cadabra2: $-i \sigma_{a\dot{c}}^\mu \partial_\mu (\psi^{\dagger\dot{c}}) + m \psi_a$

3.2.1 Combined 4-component form

Equations (36.4) and (36.5) combine into the matrix equation (eq. 36.6):

$$\begin{pmatrix} m \delta_a^c & -i \sigma_{a\dot{c}}^\mu \partial_\mu \\ -i \bar{\sigma}^{\mu\dot{a}c} \partial_\mu & m \delta^{\dot{a}}_{\dot{c}} \end{pmatrix} \begin{pmatrix} \psi_c \\ \psi^{\dagger\dot{c}} \end{pmatrix} = 0. \quad (36.6)$$

The 2×2 diagonal blocks are proportional to $m \cdot \mathbf{1}_2$; the off-diagonal blocks mix the left- and right-handed sectors via σ^μ and $\bar{\sigma}^\mu$.

3.3 Gamma Matrices in the Weyl Representation

3.3.1 Definition

The gamma matrices are defined by the block structure (eq. 36.7):

$$\gamma^\mu \equiv \begin{pmatrix} 0 & \sigma_{a\dot{c}}^\mu \\ \bar{\sigma}^{\mu\dot{a}c} & 0 \end{pmatrix}. \quad (36.7)$$

This is the **Weyl (chiral) representation**.

3.3.2 Explicit matrices

$$\gamma^0 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad \gamma^1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \quad (3.1)$$

$$\gamma^2 = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}, \quad \gamma^3 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad (3.2)$$

Written symbolically in terms of $\sigma^\mu = (I, \vec{\sigma})$ and $\bar{\sigma}^\mu = (I, -\vec{\sigma})$:

$$\gamma^0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \quad \gamma^k = \begin{pmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{pmatrix} \quad (k = 1, 2, 3).$$

3.4 Clifford Algebra Verification

3.4.1 The algebra

The gamma matrices obey the Clifford algebra (eq. 36.9):

$$\boxed{\{\gamma^\mu, \gamma^\nu\} \equiv \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = -2 g^{\mu\nu} I_4,} \quad (36.9)$$

with $g^{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ (Srednicki mostly-plus convention).

This follows from the sigma-matrix identities (eq. 36.8):

$$\sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu = -2 g^{\mu\nu} I_2, \quad (36.8a)$$

$$\bar{\sigma}^\mu \sigma^\nu + \bar{\sigma}^\nu \sigma^\mu = -2 g^{\mu\nu} I_2. \quad (36.8b)$$

✓ $\sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu = -2 g^{\mu\nu} I_2$: max error 0.0e+00.

3.4.2 Numerical verification table

All ten independent anticommutator pairs are verified below:

Pair	Formula	Value	Verified
$\{\gamma^0, \gamma^0\}$	$-2g^{00}$	$+2 \cdot I_4$	✓
$\{\gamma^0, \gamma^1\}$	$-2g^{01}$	$-0 \cdot I_4$	✓
$\{\gamma^0, \gamma^2\}$	$-2g^{02}$	$-0 \cdot I_4$	✓
$\{\gamma^0, \gamma^3\}$	$-2g^{03}$	$-0 \cdot I_4$	✓
$\{\gamma^1, \gamma^1\}$	$-2g^{11}$	$-2 \cdot I_4$	✓
$\{\gamma^1, \gamma^2\}$	$-2g^{12}$	$-0 \cdot I_4$	✓
$\{\gamma^1, \gamma^3\}$	$-2g^{13}$	$-0 \cdot I_4$	✓
$\{\gamma^2, \gamma^2\}$	$-2g^{22}$	$-2 \cdot I_4$	✓
$\{\gamma^2, \gamma^3\}$	$-2g^{23}$	$-0 \cdot I_4$	✓
$\{\gamma^3, \gamma^3\}$	$-2g^{33}$	$-2 \cdot I_4$	✓

✓ Clifford algebra: all 10 pairs verified. Max error: 0.0e + 00.

3.4.3 The γ^5 matrix

Define (eq. 36.46):

$$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -I_2 & 0 \\ 0 & +I_2 \end{pmatrix}. \quad (36.43)$$

Numerically:

$$\gamma^5 = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

✓ $\gamma^5 = \text{diag}(-I_2, +I_2)$: error 0.0e + 00.

The left- and right-handed **projection operators** are (eq. 36.44):

$$P_L \equiv \frac{1}{2}(I - \gamma^5) = \begin{pmatrix} I_2 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_R \equiv \frac{1}{2}(I + \gamma^5) = \begin{pmatrix} 0 & 0 \\ 0 & I_2 \end{pmatrix}. \quad (36.44)$$

✓ $P_L^2 = P_L, P_R^2 = P_R, P_L P_R = 0$: max error 0.0e + 00.

3.5 Majorana 4-Component Spinor

3.5.1 Definition

The **Majorana spinor** is built from a single left-handed Weyl field ψ_a (eq. 36.10):

$$\Psi \equiv \begin{pmatrix} \psi_c \\ \psi^{\dagger\dot{c}} \end{pmatrix}. \quad (36.10)$$

Equation (36.6) then becomes the **Dirac equation** (eq. 36.11):

$$\boxed{(-i\gamma^\mu \partial_\mu + m)\Psi = 0.} \quad (36.11)$$

3.5.2 Majorana condition

The Majorana spinor is its own **charge conjugate**:

$$\Psi^C \equiv \mathcal{C} \bar{\Psi}^T = \Psi, \quad (3.3)$$

where the charge conjugation matrix is (eq. 36.32):

$$\mathcal{C} = \begin{pmatrix} \varepsilon_{ac} & 0 \\ 0 & \varepsilon^{\dot{a}\dot{c}} \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}. \quad (36.32)$$

The matrix $\mathcal{C}$ satisfies (eq. 36.36):

$$\mathcal{C}^T = \mathcal{C}^\dagger = \mathcal{C}^{-1} = -\mathcal{C}. \quad (36.36)$$

and conjugates the gamma matrices as (eq. 36.40):

$$\mathcal{C}^{-1} \gamma^\mu \mathcal{C} = -(\gamma^\mu)^T. \quad (36.40)$$

✓ $\mathcal{C}^{-1} \gamma^\mu \mathcal{C} = -(\gamma^\mu)^T$ for all μ : max error 0.0e+00.

3.5.3 Majorana Lagrangian

Re-expressing eq. (36.2) using $\bar{\Psi} = \Psi^\dagger \beta$ (where $\beta = \gamma^0$ numerically) gives (eq. 36.41):

$$\mathcal{L} = \frac{i}{2} \bar{\Psi} \gamma^\mu \partial_\mu \Psi - \frac{1}{2} m \bar{\Psi} \Psi. \quad (36.41)$$

3.6 Dirac Fermion from Two Weyl Fields

3.6.1 Two-Weyl-field Lagrangian

Begin with two left-handed fields χ_a, ξ_a (rewritten from $\psi_{1,2}$ via eq. 36.14–36.15). The Lagrangian is (eq. 36.16):

$$\mathcal{L} = i\chi^\dagger \bar{\sigma}^\mu \partial_\mu \chi + i\xi^\dagger \bar{\sigma}^\mu \partial_\mu \xi - m\chi\xi - m\xi^\dagger \chi^\dagger, \quad (36.16)$$

invariant under the U(1) symmetry (eq. 36.17):

$$\chi \rightarrow e^{-i\alpha} \chi, \quad \xi \rightarrow e^{+i\alpha} \xi. \quad (36.17)$$

3.6.2 Dirac spinor and Lagrangian

Define the **Dirac spinor** (eq. 36.19):

$$\Psi \equiv \begin{pmatrix} \chi_c \\ \xi^{\dagger\dot{c}} \end{pmatrix}, \quad \bar{\Psi} = \Psi^\dagger \beta = (\xi^a, \chi_a^\dagger). \quad (36.19, 36.22)$$

Then the Dirac Lagrangian (eq. 36.28):

$$\boxed{\mathcal{L} = i\bar{\Psi}\gamma^\mu\partial_\mu\Psi - m\bar{\Psi}\Psi}, \quad (36.28)$$

where (eq. 36.23):

$$\bar{\Psi}\Psi = \xi^a\chi_a + \chi_a^\dagger\xi^{\dagger\dot{a}}. \quad (36.23)$$

The Noether current for the U(1) symmetry (eq. 36.30):

$$j^\mu = \bar{\Psi}\gamma^\mu\Psi = \chi^\dagger\bar{\sigma}^\mu\chi - \xi^\dagger\bar{\sigma}^\mu\xi. \quad (36.30)$$

3.7 Dirac vs. Majorana — Comparison

Property	Majorana	Dirac
Weyl fields	1 (single ψ)	2 (χ and ξ)
U(1) symmetry	No	Yes ($\chi \rightarrow e^{-i\alpha}\chi, \xi \rightarrow e^{+i\alpha}\xi$)
Charge	None (Majorana mass)	Conserved $Q = \int j^0 d^3x$
$\Psi^C = \Psi$	Yes	No
Lagrangian	$\frac{i}{2}\bar{\Psi}\gamma^\mu\partial_\mu\Psi - \frac{m}{2}\bar{\Psi}\Psi$	$i\bar{\Psi}\gamma^\mu\partial_\mu\Psi - m\bar{\Psi}\Psi$
Factor of $\frac{1}{2}$	Yes (overcounting)	No
Scalar analogue	Real scalar $\phi = \phi^\dagger$	Complex scalar $\phi \neq \phi^\dagger$
Neutrino model	Seesaw mechanism	Dirac neutrino
Degrees of freedom	2 (real)	4 (complex)

3.8 Summary

Object	Expression
Weyl Lagrangian	$\mathcal{L} = i\psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi - \frac{1}{2}m\psi\psi - \frac{1}{2}m\psi^\dagger\psi^\dagger$
EOM ($\delta/\delta\psi^\dagger$)	$-i\bar{\sigma}^{\mu\dot{a}c}\partial_\mu\psi_c + m\psi^{\dagger\dot{a}} = 0$
γ matrix (Weyl rep)	$\gamma^\mu = \begin{bmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{bmatrix}$
Clifford algebra	$\{\gamma^\mu, \gamma^\nu\} = -2g^{\mu\nu}I_4$
γ^5	$i\gamma^0\gamma^1\gamma^2\gamma^3 = \text{diag}(-I_2, +I_2)$
Majorana spinor	$\Psi = (\psi_c, \psi^{\dagger\dot{c}})^T, \quad \Psi^C = \Psi$
Dirac equation	$(-i\gamma^\mu\partial_\mu + m)\Psi = 0$
Dirac spinor	$\Psi = (\chi_c, \xi^{\dagger\dot{c}})^T, \quad \Psi^C \neq \Psi$
Dirac Lagrangian	$i\bar{\Psi}\gamma^\mu\partial_\mu\Psi - m\bar{\Psi}\Psi$
Charge conjugation	$\mathcal{C}^{-1}\gamma^\mu\mathcal{C} = -(\gamma^\mu)^T$

Next: Chapter 37 develops the quantization of Dirac and Majorana fields, introducing anticommutation relations for the creation/annihilation operators.

Chapter 4

Ch. 37 — Canonical Quantization of Spinor Fields I

4.1 Canonical Anticommutation Relations

4.1.1 Equal-time CARs

The canonical anticommutation relations (CARs) for a left-handed Weyl field $\psi_\alpha(x)$ are (Srednicki eq. 37.3):

$$\boxed{\{\psi_\alpha(\mathbf{x}, t), \psi^{\dagger\dot{\beta}}(\mathbf{y}, t)\} = \sigma^0_{\alpha}{}^{\dot{\beta}} \delta^{(3)}(\mathbf{x} - \mathbf{y}) = \delta_{\alpha}^{\dot{\beta}} \delta^{(3)}(\mathbf{x} - \mathbf{y})} \quad (37.3)$$

where $\sigma^0 = I_2$ so the RHS is just $\delta_{\alpha}^{\dot{\beta}}$. The remaining equal-time anticommutators vanish:

$$\{\psi_\alpha(\mathbf{x}, t), \psi_\beta(\mathbf{y}, t)\} = 0, \quad (37.4a)$$

$$\{\psi^{\dagger\dot{\alpha}}(\mathbf{x}, t), \psi^{\dagger\dot{\beta}}(\mathbf{y}, t)\} = 0. \quad (37.4b)$$

Cadabra2 expression for the LHS:

$$\psi_\alpha \psi^{\dagger\dot{\alpha}} + \psi^{\dagger\dot{\alpha}} \psi_\alpha = \delta_{\alpha}^{\dot{\alpha}} \delta^3(x - y)$$

4.1.2 Origin from Legendre transform

The kinetic Lagrangian $\mathcal{L}_{\text{kin}} = i\psi^{\dagger}\bar{\sigma}^{\mu}\partial_{\mu}\psi$ gives the conjugate momentum:

$$\pi^{\alpha} = \frac{\partial \mathcal{L}}{\partial(\partial_0 \psi_{\alpha})} = i\psi^{\dagger\dot{\beta}} \bar{\sigma}^0_{\dot{\beta}}{}^{\alpha} = i\psi^{\dagger\alpha} \quad (\text{since } \bar{\sigma}^0 = I_2) \quad (4.1)$$

Cadabra2: $\pi^{\alpha} = i\psi_{\dot{\alpha}}^{\dagger} \bar{\sigma}^{0\dot{\alpha}\alpha} = i\psi^{\dagger\alpha}$.

The canonical equal-time anticommutation $\{\psi_{\alpha}(\mathbf{x}), \pi^{\beta}(\mathbf{y})\} = i\delta_{\alpha}^{\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y})$ then directly yields eq. (37.3).

✓ $\sigma^0 = I_2$: max error 0.0e+00.

4.2 Mode Expansion of the Weyl Field

4.2.1 Plane-wave expansion

The left-handed Weyl field is expanded as (eq. 37.7):

$$\psi_\alpha(x) = \int \frac{d^3p}{(2\pi)^3} \sum_s \left[b_s(\mathbf{p}) u_\alpha^s(\mathbf{p}) e^{ip \cdot x} + d_s^\dagger(\mathbf{p}) v_\alpha^s(\mathbf{p}) e^{-ip \cdot x} \right] \quad (37.7)$$

with hermitian conjugate

$$\psi^{\dagger\dot{\alpha}}(x) = \int \frac{d^3p}{(2\pi)^3} \sum_s \left[b_s^\dagger(\mathbf{p}) \tilde{u}^{\dot{\alpha}s}(\mathbf{p}) e^{-ip \cdot x} + d_s(\mathbf{p}) \tilde{v}^{\dot{\alpha}s}(\mathbf{p}) e^{+ip \cdot x} \right]. \quad (4.2)$$

Here $p \cdot x = p^\mu x_\mu$ with mostly-plus metric; on shell $p^0 = \omega_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}$.

4.2.2 Operator CARs

The creation/annihilation operators satisfy (eq. 37.8):

$$\{b_s(\mathbf{p}), b_{s'}^\dagger(\mathbf{p}')\} = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') \delta_{ss'}, \quad (37.8a)$$

$$\{d_s(\mathbf{p}), d_{s'}^\dagger(\mathbf{p}')\} = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') \delta_{ss'}, \quad (37.8b)$$

$$\{b_s, b_{s'}\} = \{d_s, d_{s'}\} = \{b_s, d_{s'}\} = \dots = 0. \quad (37.8c)$$

4.3 Basis Spinors $u^s(p)$ and $v^s(p)$

4.3.1 Weyl equations

The basis spinors satisfy the Weyl equations (eq. 37.5):

$$p_{\alpha\dot{\alpha}} \tilde{u}^{\dot{\alpha}s}(p) = +m u_\alpha^s(p), \quad (37.5a)$$

$$p_{\alpha\dot{\alpha}} \tilde{v}^{\dot{\alpha}s}(p) = -m v_\alpha^s(p), \quad (37.5b)$$

where $p_{\alpha\dot{\alpha}} = p_\mu \sigma_{\alpha\dot{\alpha}}^\mu$ is the momentum matrix in spinor space.

4.3.2 Explicit construction at rest

At $\mathbf{p} = 0$, $p^\mu = (m, 0, 0, 0)$, the momentum matrix is:

$$p_{\alpha\dot{\alpha}} = p_\mu \sigma_{\alpha\dot{\alpha}}^\mu = -m \sigma_{\alpha\dot{\alpha}}^0 = -m I_2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

Basis spinors at rest (spin index $s = \pm$):

$$\begin{aligned} u_{\alpha}^{+} &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} && (\text{spin-up, particle}) \\ u_{\alpha}^{-} &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} && (\text{spin-down, particle}) \\ v_{\alpha}^{+} &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} && (\text{spin-up, antiparticle}) \\ v_{\alpha}^{-} &= \begin{pmatrix} -1 \\ 0 \end{pmatrix} && (\text{spin-down, antiparticle}) \end{aligned}$$

4.3.3 Massless case

For a massless particle with $p^{\mu} = (E, 0, 0, E)$:

$$p_{\alpha\dot{\alpha}} = -E \sigma^0 + E \sigma^3 = \begin{pmatrix} 0 & 0 \\ 0 & -2 \end{pmatrix}$$

Eigenvalues: $0, -2$. The null eigenvector (helicity eigenstate u^{+}) lies in the kernel of $p_{\alpha\dot{\alpha}}$, consistent with the massless Weyl equation $p_{\alpha\dot{\alpha}} \tilde{u}^{\dot{\alpha}} = 0$.

4.3.4 Completeness relations

Summing over spins (eq. 37.10–37.11):

$$\sum_s u_{\alpha}^s(p) \tilde{u}^{\dot{\beta}s}(p) = -p_{\alpha}^{\dot{\beta}} + m \delta_{\alpha}^{\dot{\beta}}, \quad (37.10)$$

$$\sum_s v_{\alpha}^s(p) \tilde{v}^{\dot{\beta}s}(p) = -p_{\alpha}^{\dot{\beta}} - m \delta_{\alpha}^{\dot{\beta}}. \quad (37.11)$$

Numerical check at rest: $-p_{\alpha\dot{\alpha}} + m I_2 = 2m I_2$,

$$\sum_s u^s \otimes u^{s\dagger} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \approx \frac{1}{2m} \cdot \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \quad \checkmark$$

Vector bilinears at rest:

$$\tilde{u}^{\dot{\alpha}+} \bar{\sigma}_{\dot{\alpha}\alpha}^{\mu} u_{\alpha}^{+} = [1.0, 0.0, 0.0, -1.0], \quad \tilde{u}^{\dot{\alpha}-} \bar{\sigma}_{\dot{\alpha}\alpha}^{\mu} u_{\alpha}^{-} = [1.0, 0.0, 0.0, 1.0]$$

4.4 Deriving the CARs from the Mode Expansion

Inserting eq. (37.7) into $\{\psi_{\alpha}(x), \psi^{\dagger\dot{\beta}}(y)\}$ and using the operator CARs (37.8a–b) (eq. 37.12):

$$\begin{aligned} \{\psi_{\alpha}(x), \psi^{\dagger\dot{\beta}}(y)\} &= \int \frac{d^3p}{(2\pi)^3} \sum_s u_{\alpha}^s(\mathbf{p}) \tilde{u}^{\dot{\beta}s}(\mathbf{p}) e^{ip(x-y)} + \int \frac{d^3p}{(2\pi)^3} \sum_s v_{\alpha}^s(\mathbf{p}) \tilde{v}^{\dot{\beta}s}(\mathbf{p}) e^{-ip(x-y)} \\ &= \int \frac{d^3p}{(2\pi)^3} \left[(-p_{\alpha}^{\dot{\beta}} + m \delta_{\alpha}^{\dot{\beta}}) e^{ip(x-y)} + (-p_{\alpha}^{\dot{\beta}} - m \delta_{\alpha}^{\dot{\beta}}) e^{-ip(x-y)} \right]. \quad (4.3) \end{aligned}$$

The momentum-dependent terms cancel between the two integrals, leaving:

$$\{\psi_\alpha(x), \psi^{\dagger\beta}(y)\} \Big|_{x^0=y^0} = \delta_\alpha^\beta \delta^{(3)}(\mathbf{x} - \mathbf{y}). \quad \checkmark \quad (4.4)$$

4.5 Normal-Ordered Hamiltonian

4.5.1 Classical Hamiltonian density

From the Weyl Lagrangian via Legendre transform:

$$H = \int d^3x \left[-i\psi^\dagger \bar{\sigma}^k \partial_k \psi + \frac{1}{2}m\psi\psi + \frac{1}{2}m\psi^\dagger\psi^\dagger \right]. \quad (4.5)$$

4.5.2 Mode expansion form

Inserting the mode expansion (eq. 37.17):

$$H = \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} \left[b_s^\dagger(\mathbf{p}) b_s(\mathbf{p}) - d_s(\mathbf{p}) d_s^\dagger(\mathbf{p}) \right]. \quad (4.6)$$

4.5.3 Normal ordering

Using $\{d_s(\mathbf{p}), d_{s'}^\dagger(\mathbf{p}')\} = (2\pi)^3 \delta^{(3)} \delta_{ss'}$:

$$-d_s d_s^\dagger = +d_s^\dagger d_s - \{d_s, d_s^\dagger\} = +d_s^\dagger d_s - (2\pi)^3 \delta^{(3)}(0). \quad (4.7)$$

Discarding the (divergent, negative) zero-point energy:

$$\boxed{: H := \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} \left[b_s^\dagger(\mathbf{p}) b_s(\mathbf{p}) + d_s^\dagger(\mathbf{p}) d_s(\mathbf{p}) \right]} \quad (37.17)$$

Cadabra2: $: H := \int \frac{d^3p}{(2\pi)^3} (\omega_p b_s^\dagger(p) b_s(p) + \omega_p d_s^\dagger(p) d_s(p))$

Key contrast with bosons:

Field type	Zero-point energy	: H :
Boson (scalar)	$+\frac{\omega}{2}$ per mode	$\int \omega a^\dagger a$
Fermion (Weyl)	$-\frac{\omega}{2}$ per mode	$\int \omega (b^\dagger b + d^\dagger d)$

4.6 Spin-Statistics Theorem

The spin-statistics theorem (Pauli 1940) states:

$$\text{Spin-}\frac{1}{2} \implies \text{anticommuting quantization.} \quad (4.8)$$

Proof sketch via microcausality: for a Lorentz-invariant theory,

$$[\mathcal{O}(x), \mathcal{O}'(y)] = 0 \quad \text{for spacelike } (x - y)^2 > 0.$$

If a spin- $\frac{1}{2}$ field were quantized with *commutators*:

$$[\psi_\alpha(x), \psi^{\dagger\dot{\beta}}(y)] = \int \frac{d^3p}{(2\pi)^3} [(-p_\alpha^{\dot{\beta}} + m)e^{ip(x-y)} - (-p_\alpha^{\dot{\beta}} - m)e^{-ip(x-y)}] \neq 0 \quad (\text{spacelike})$$

— a violation. With anticommutators the p -terms cancel and the result vanishes outside the light cone, as required.

4.7 Lorentz-Invariant Measure and Feynman Slash

4.7.1 The measure $\widetilde{dp}$

The mode expansion uses the **Lorentz-invariant phase-space measure** (Srednicki eq. 3.21):

$$\widetilde{dp} \equiv \frac{d^3p}{(2\pi)^3 2\omega_{\mathbf{p}}}, \quad \omega_{\mathbf{p}} = \sqrt{|\mathbf{p}|^2 + m^2}. \quad (3.21)$$

It is manifestly Lorentz-invariant because it equals $d^4p \delta(p^2 + m^2) \theta(p^0)/(2\pi)^3$: the 4-volume d^4p is invariant, $\delta(p^2 + m^2)$ restricts to the mass shell (Lorentz scalar condition), and $\theta(p^0)$ selects the forward light-cone (preserved by orthochronous boosts).

The mode expansion is then:

$$\psi_\alpha(x) = \sum_s \int \widetilde{dp} [b_s(\mathbf{p}) u_\alpha^s(\mathbf{p}) e^{ip \cdot x} + d_s^\dagger(\mathbf{p}) v_\alpha^s(\mathbf{p}) e^{-ip \cdot x}]. \quad (4.9)$$

4.7.2 Feynman slash notation

The **Feynman slash** is defined by (Srednicki eq. 36.14):

$$\not{p} \equiv p_\mu \gamma^\mu. \quad (36.14)$$

(LaTeX: `\slashed{p}` with `slashed` package, or `\not{p}` built-in.)

Key property (mostly-plus metric $g = \text{diag}(-1, +1, +1, +1)$):

$$\not{p}^2 = p_\mu p_\nu \gamma^\mu \gamma^\nu = \frac{1}{2} p_\mu p_\nu \{\gamma^\mu, \gamma^\nu\} = -p_\mu p^\mu = m^2 \quad (\text{on-shell}) \quad (36.15)$$

In the Weyl representation $\gamma^\mu = \begin{bmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{bmatrix}$, the slash takes the block form:

$$\not{p} = \begin{pmatrix} 0 & p_{\alpha\dot{\alpha}} \\ \bar{p}^{\dot{\alpha}\alpha} & 0 \end{pmatrix} \quad (4.10)$$

The Dirac equation for plane-wave spinors (eq. 36.11 analog):

$$(\not{p} + m) u_s(p) = 0, \quad (-\not{p} + m) v_s(p) = 0, \quad (36.11)$$

$$\bar{u}_s(p) (\not{p} + m) = 0, \quad \bar{v}_s(p) (-\not{p} + m) = 0. \quad (4.11)$$

✓ $\not{p}^2 = m^2 I_4$ for on-shell p : max error 0.0e + 00.

4.8 Spin-Sum Completeness (4-component)

The Dirac spin-sum completeness relations (Srednicki eqs. 38.28–38.29):

$$\boxed{\sum_s u_s(p) \bar{u}_s(p) = -\not{p} + m, \quad \sum_s v_s(p) \bar{v}_s(p) = -\not{p} - m.} \quad (38.28-29)$$

where $\bar{u} = u^\dagger \gamma^0$ is the Dirac conjugate.

These are the 4-component version of the 2-component Weyl completeness relations (eqs. 37.10–37.11).

Physical use — unpolarized cross sections: summing over unknown spins replaces the outer product $u_s \bar{u}_{s'}$ with the spin-sum matrix, reducing squared amplitudes to γ -matrix traces:

$$\sum_{s,s'} |\bar{u}_{s'}(p') \Gamma u_s(p)|^2 = \text{Tr} [(-\not{p}' + m) \bar{\Gamma} (-\not{p} + m) \Gamma] \quad (4.12)$$

where $\bar{\Gamma} = \gamma^0 \Gamma^\dagger \gamma^0$.

✓ $\sum_s u_s \bar{u}_s = -\not{p} + m$ at rest: max error 2.2e − 16. ✓ $\sum_s v_s \bar{v}_s = -\not{p} - m$ at rest: max error 2.2e − 16.

4.9 Summary

Object	Expression
Equal-time CARs	$\{\psi_\alpha(\mathbf{x}, t), \psi^{\dagger\beta}(\mathbf{y}, t)\} = \delta_\alpha^\beta \delta^{(3)}(\mathbf{x} - \mathbf{y})$
Conjugate momentum	$\pi^\alpha = i\psi^{\dagger\alpha}$
Mode expansion	$\psi_\alpha = \int \frac{d^3p}{(2\pi)^3} \sum_s [b_s u_\alpha^s e^{ipx} + d_s^\dagger v_\alpha^s e^{-ipx}]$
Operator CARs	$\{b_s, b_{s'}^\dagger\} = \{d_s, d_{s'}^\dagger\} = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') \delta_{ss'}$
Weyl eq. (massive)	$p_{\alpha\dot{\alpha}} \tilde{u}^{\dot{\alpha}s} = +m u_\alpha^s, \quad p_{\alpha\dot{\alpha}} \tilde{v}^{\dot{\alpha}s} = -m v_\alpha^s$
Completeness	$\sum_s u_\alpha^s \tilde{u}^{\dot{\beta}s} = -p_\alpha^{\dot{\beta}} + m \delta_\alpha^{\dot{\beta}}$
Normal-ordered H	$: H := \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} [b^\dagger b + d^\dagger d]$
Spin-statistics	Spin- $\frac{1}{2}$ requires anticommuting fields (microcausality)
Lorentz-invariant measure	$\widetilde{d^3p} = d^3p / [(2\pi)^3 2\omega_{\mathbf{p}}]$
Feynman slash	$\not{p} = p_\mu \gamma^\mu; \quad \not{p}^2 = m^2 \text{ on-shell}$
Spin-sum (particle)	$\sum_s u_s \bar{u}_s = -\not{p} + m$
Spin-sum (antiparticle)	$\sum_s v_s \bar{v}_s = -\not{p} - m$

Next: Chapter 38 develops the spinor algebra toolkit (σ -completeness, trace identities, Fierz rearrangement) needed for explicit Feynman diagram calculations.

Chapter 5

Ch. 38 — Spinor Technology

5.1 σ^μ Completeness Relation

5.1.1 Main identity

The σ matrices form a complete basis for 2×2 complex matrices. The completeness relation is (Srednicki eq. 38.1):

$$\sigma^\mu_{\alpha\dot{\alpha}} \sigma_\mu^{\beta\dot{\beta}} = -2 \delta_\alpha^\beta \delta_{\dot{\alpha}}^{\dot{\beta}} \quad (38.1)$$

where $\sigma_\mu^{\beta\dot{\beta}} = g_{\mu\nu} \bar{\sigma}^{\nu\dot{\beta}\beta}$ (both spinor indices raised by ε).

✓ Max error: 0.0e + 00.

The nonzero components of the LHS are: $(\alpha, \dot{\alpha}, \beta, \dot{\beta}) \in \{(1, 1, 1, 1), (1, 2, 1, 2), (2, 1, 2, 1), (2, 2, 2, 2)\}$, each equal to -2 .

5.1.2 Alternative form

Equivalently (eq. 35.5):

$$\varepsilon^{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}} \sigma^\mu_{\alpha\dot{\alpha}} \sigma^\nu_{\beta\dot{\beta}} = -2 g^{\mu\nu} \quad (35.5)$$

Cadabra2: $\varepsilon^{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}} \sigma^\mu_{\beta\dot{\beta}} = \bar{\sigma}^{\mu\dot{\alpha}\alpha}$

✓ Max error vs $-2g^{\mu\nu}$: 0.0e + 00.

5.2 Trace Identities

5.2.1 Two-sigma trace

The fundamental trace identity (Srednicki eq. 38.5):

$$\text{Tr}[\sigma^\mu \bar{\sigma}^\nu] \equiv \sigma^\mu_{\alpha\dot{\alpha}} \bar{\sigma}^{\nu\dot{\alpha}\alpha} = -2 g^{\mu\nu} \quad (38.5)$$

Cadabra2 summand: $\sigma^\mu_{\alpha\dot{\alpha}} \bar{\sigma}^{\nu\dot{\alpha}\alpha}$

Explicit matrix (rows μ , columns ν):

$$\text{Tr}[\sigma^\mu \bar{\sigma}^\nu] = \begin{pmatrix} +2.0 & +0.0 & +0.0 & +0.0 \\ +0.0 & -2.0 & +0.0 & +0.0 \\ +0.0 & +0.0 & -2.0 & +0.0 \\ +0.0 & +0.0 & +0.0 & -2.0 \end{pmatrix} = -2g^{\mu\nu}$$

✓ $\text{Tr}[\sigma^\mu \bar{\sigma}^\nu] = -2g^{\mu\nu}$: max error $0.0e + 00$.

✓ $\text{Tr}[\bar{\sigma}^\mu \sigma^\nu] = -2g^{\mu\nu}$: max error $0.0e + 00$.

Physical interpretation: σ^μ and $\bar{\sigma}^\nu$ are “dual” bases for 2×2 matrices; the trace gives their inner product $-2g^{\mu\nu}$. Any 2×2 matrix A expands as $A = -\frac{1}{2} \sum_\mu \bar{\sigma}^\mu \text{Tr}[\sigma^\mu A]$.

5.2.2 Four-sigma trace

The four-sigma trace identity (eq. 38.6), with Srednicki’s mostly-plus metric $g = \text{diag}(-1, +1, +1, +1)$:

$$\boxed{\text{Tr}[\sigma^\mu \bar{\sigma}^\nu \sigma^\rho \bar{\sigma}^\sigma] = 2(g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}) + 2i \varepsilon^{\mu\nu\rho\sigma}} \quad (38.6)$$

$$\text{Tr}[\bar{\sigma}^\mu \sigma^\nu \bar{\sigma}^\rho \sigma^\sigma] = 2(g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}) - 2i \varepsilon^{\mu\nu\rho\sigma} \quad (38.6')$$

(The conjugate trace carries $-2i$ instead of $+2i$; the sign difference is specific to the mostly-plus convention.)

Sample numerical values ($\varepsilon^{0123} = +1$):

(μ, ν, ρ, σ)	Computed	Expected	
(0, 1, 2, 3)	+0.00 + +2.00i	+0.00 + +2.00i	✓
(1, 0, 2, 3)	+0.00 + -2.00i	+0.00 + -2.00i	✓
(0, 1, 3, 2)	+0.00 + -2.00i	+0.00 + -2.00i	✓
(0, 2, 1, 3)	+0.00 + -2.00i	+0.00 + -2.00i	✓
(1, 2, 3, 0)	+0.00 + -2.00i	+0.00 + -2.00i	✓
(2, 3, 0, 1)	+0.00 + +2.00i	+0.00 + +2.00i	✓

✓ All 256 components verified. Max error: $0.0e + 00$. ✓ Conjugate trace: max error $0.0e + 00$.

5.3 Van der Waerden Index Gymnastics

5.3.1 Raising and lowering spinor indices

The ε -metric raises and lowers spinor indices (Srednicki eqs. 38.7–38.13):

$$\psi_\alpha = \varepsilon_{\alpha\beta} \psi^\beta \quad (\text{lower undotted}), \quad (38.7)$$

$$\psi^\alpha = \varepsilon^{\alpha\beta} \psi_\beta \quad (\text{raise undotted}), \quad (38.8)$$

$$\chi_{\dot{\alpha}} = \varepsilon_{\dot{\alpha}\dot{\beta}} \chi^{\dot{\beta}} \quad (\text{lower dotted}), \quad (38.9)$$

$$\chi^{\dot{\alpha}} = \varepsilon^{\dot{\alpha}\dot{\beta}} \chi_{\dot{\beta}} \quad (\text{raise dotted}), \quad (38.10)$$

with the Srednicki convention $\varepsilon_{12} = -1$, $\varepsilon^{12} = +1$ ($\varepsilon_{21} = +1$, $\varepsilon^{21} = -1$).

Key consistency: $\varepsilon^{\alpha\beta}\varepsilon_{\beta\gamma} = \delta_\gamma^\alpha$. ✓ $\varepsilon^{\alpha\beta}\varepsilon_{\beta\gamma} = \delta_\gamma^\alpha$: max error 0.0e + 00.

5.3.2 Raising both spinor indices of σ^μ

The relation between σ and $\bar{\sigma}$ via index raising (eq. 38.13):

$$\bar{\sigma}^{\mu\dot{\alpha}\alpha} = \varepsilon^{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}} \sigma^\mu_{\beta\dot{\beta}} \quad (38.13)$$

Cadabra2: $\varepsilon^{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}} \sigma^\mu_{\beta\dot{\beta}} = \bar{\sigma}^{\mu\dot{\alpha}\alpha}$

✓ All four μ values verified: max error 0.0e + 00.

5.4 Spinor Bilinears

5.4.1 Lorentz transformation properties

The key bilinear forms and their Lorentz transformation properties:

$$\chi\psi \equiv \chi^\alpha\psi_\alpha = \varepsilon^{\alpha\beta}\chi_\alpha\psi_\beta \quad \text{Lorentz scalar (left-handed)} \quad (38.14a)$$

$$\chi^\dagger\psi^\dagger \equiv \chi^\dagger_{\dot{\alpha}}\psi^{\dagger\dot{\alpha}} \quad \text{Lorentz scalar (right-handed)} \quad (38.14b)$$

$$\chi^\dagger_{\dot{\alpha}} \bar{\sigma}^{\mu\dot{\alpha}\alpha} \psi_\alpha \quad \text{Lorentz 4-vector} \quad (38.14c)$$

Cadabra2 vector bilinear: $\chi^\dagger_{\dot{\alpha}} \bar{\sigma}^{\mu\dot{\alpha}\alpha} \psi_\alpha$

5.4.2 Momentum matrix in spinor space

The momentum p^μ maps to a 2×2 spinor matrix:

$$p_{\alpha\dot{\alpha}} = p_\mu \sigma^\mu_{\alpha\dot{\alpha}} = -E I_2 + \mathbf{p} \cdot \boldsymbol{\sigma}, \quad (5.1)$$

with $\det(p_{\alpha\dot{\alpha}}) = -p^2/4$ (up to normalisation). For massless particles: $\det = 0$, so $p_{\alpha\dot{\alpha}}$ factorises as $p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$ — the starting point for spinor-helicity methods.

5.4.3 Gordon identity

In 4-component Dirac notation, the vector bilinear decomposes as:

$$\bar{u}(p') \gamma^\mu u(p) = \bar{u}(p') \left[\frac{p'^\mu + p^\mu}{2m} + \frac{i\sigma^{\mu\nu}(p' - p)_\nu}{2m} \right] u(p) \quad (5.2)$$

In 2-component language, the vector part comes from $\tilde{u}^{\dot{\alpha}s} \bar{\sigma}^{\mu\dot{\alpha}\alpha} u_\alpha^s$ and the tensor part from the generators $S_L^{\mu\nu}$.

5.5 Fierz Identities

5.5.1 Schouten identity

The ε -Fierz or Schouten identity:

$$\boxed{\varepsilon_{\alpha\gamma} \varepsilon_{\beta\delta} = \varepsilon_{\alpha\beta} \varepsilon_{\gamma\delta} - \varepsilon_{\alpha\delta} \varepsilon_{\gamma\beta}} \quad (38.17)$$

Cadabra2: $\epsilon_{\alpha\gamma}\epsilon_{\beta\delta} = \epsilon_{\alpha\beta}\epsilon_{\gamma\delta} - \epsilon_{\alpha\delta}\epsilon_{\gamma\beta}$

✓ Schouten identity: max error 0.0e + 00.

5.5.2 σ -matrix Fierz

From the completeness relation (§1), the σ -matrix Fierz identity is:

$$(\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} (\sigma_\mu)_{\beta\dot{\beta}} = -2 \delta_\beta^\alpha \delta_{\dot{\beta}}^{\dot{\alpha}} \quad (38.18)$$

5.5.3 Four-Fermi Fierz

For four Grassmann-valued spinors:

$$(\chi_\alpha^\dagger \bar{\sigma}^{\mu\dot{\alpha}\alpha} \psi_\alpha) (\xi_{\dot{\beta}}^\dagger \bar{\sigma}_\mu^{\beta\dot{\beta}} \eta_\beta) = -2 (\chi_\alpha^\dagger \xi^{\dot{\alpha}\alpha}) (\eta^\beta \psi_\beta) \quad (38.21)$$

This follows from the σ -Fierz plus Grassmann anticommutativity, and is used to rewrite four-fermion interaction terms.

5.6 Gordon Identity

5.6.1 Main identity

The **Gordon identity** for on-shell Dirac spinors (Srednicki Problem 38.3, eq. 38.22):

$$\boxed{2m \bar{u}_{s'}(\mathbf{p}') \gamma^\mu u_s(\mathbf{p}) = \bar{u}_{s'}(\mathbf{p}') [(p' + p)^\mu - 2i S^{\mu\nu} (p' - p)_\nu] u_s(\mathbf{p})} \quad (38.22)$$

where $S^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]$ is the Dirac spin tensor. An analogous identity holds for the v spinors with an overall sign flip.

5.6.2 Derivation

From the Clifford algebra and the definition of $S^{\mu\nu}$:

$$\gamma^\mu \not{p} = -p^\mu - 2i S^{\mu\nu} p_\nu, \quad (5.3)$$

$$\not{p}' \gamma^\mu = -p'^\mu + 2i S^{\mu\nu} p'_\nu. \quad (5.4)$$

Summing:

$$\gamma^\mu \not{p} + \not{p}' \gamma^\mu = -(p + p')^\mu - 2i S^{\mu\nu} (p_\nu - p'_\nu). \quad (5.5)$$

Sandwiching with $\bar{u}_{s'}(\mathbf{p}')$ and $u_s(\mathbf{p})$, then using $(\not{p} + m)u_s = 0$ and $\bar{u}_{s'}(\not{p}' + m) = 0$ gives the identity.

5.6.3 Extended Gordon identity with γ^5

Because $\{\gamma^5, \gamma^\mu\} = 0$ (eq. 36.46):

$$\bar{u}_{s'}(\mathbf{p}')[(p' + p)^\mu - 2iS^{\mu\nu}(p' - p)_\nu]\gamma^5 u_s(\mathbf{p}) = 0. \quad (38.41)$$

The Gordon tensor is parity-even; γ^5 is parity-odd; their product vanishes.

5.6.4 Physical content

Decomposing the electromagnetic vertex γ^μ :

$$\bar{u}'\gamma^\mu u = \frac{(p + p')^\mu}{2m} \bar{u}'u + \frac{i\sigma^{\mu\nu}(p' - p)_\nu}{2m} \bar{u}'u \quad (\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]) \quad (5.6)$$

- $(p + p')^\mu/(2m)$: charge-current term $\rightarrow$ form factor $F_1(q^2)$
- $i\sigma^{\mu\nu}q_\nu/(2m)$: magnetic-moment term $\rightarrow$ form factor $F_2(q^2)$, anomalous magnetic moment $g-2$

✓ $\{\gamma^5, \gamma^\mu\} = 0$ for all μ : max error $< 10^{-14}$. ✓ Gordon: $2m\bar{u}'\gamma^\mu u = \bar{u}'(2p^\mu)u$ at $p' = p$: max error $< 10^{-14}$.

5.7 Helicity and Spin-Sum Completeness

5.7.1 Spin-sum completeness

The Dirac spin-sum completeness relations (eqs. 38.28–38.29):

$$\boxed{\sum_s u_s(p) \bar{u}_s(p) = -\not{p} + m, \quad \sum_s v_s(p) \bar{v}_s(p) = -\not{p} - m.} \quad (38.28-29)$$

5.7.2 Bilinear identities

For same momentum $\mathbf{p}$ (eqs. 38.21–38.22):

$$\bar{u}_{s'}(\mathbf{p}) \gamma^\mu u_s(\mathbf{p}) = 2p^\mu \delta_{s's}, \quad (38.21a)$$

$$\bar{v}_{s'}(\mathbf{p}) \gamma^\mu v_s(\mathbf{p}) = 2p^\mu \delta_{s's}, \quad (38.21b)$$

$$\bar{u}_{s'}(\mathbf{p}) \gamma^0 v_s(-\mathbf{p}) = 0, \quad (38.22a)$$

$$\bar{v}_{s'}(\mathbf{p}) \gamma^0 u_s(-\mathbf{p}) = 0. \quad (38.22b)$$

5.7.3 Helicity and chirality

The **helicity** $h = \hat{\mathbf{J}} \cdot \hat{\mathbf{p}}$ measures spin along the direction of motion. For spin- $\frac{1}{2}$: $h = \pm \frac{1}{2}$. In the massless limit, helicity coincides with chirality:

$$P_R u_+(\mathbf{p}) = u_+(\mathbf{p}), \quad (\text{positive helicity} = \text{right-chiral}) \quad (5.7)$$

$$P_L u_-(\mathbf{p}) = u_-(\mathbf{p}), \quad (\text{negative helicity} = \text{left-chiral}) \quad (5.8)$$

where $P_{L,R} = \frac{1}{2}(1 \mp \gamma^5)$.

The massless spin sums separate by helicity:

$$u_+(\mathbf{p})\bar{u}_+(\mathbf{p}) = P_R(-\not{p}), \quad u_-(\mathbf{p})\bar{u}_-(\mathbf{p}) = P_L(-\not{p}). \quad (5.9)$$

Their sum gives the massless limit of eq. (38.28): $\sum_s u_s \bar{u}_s \rightarrow -\not{p}$ as $m \rightarrow 0$.

✓ Spin-sum $\sum_s u_s \bar{u}_s = -\not{p} + m$: max error $< 10^{-14}$. ✓ Bilinear off-diagonal = 0; orthogonality $\bar{u}\gamma^0 v = 0$: verified. ✓ Massless chirality projectors: $P_L u_- = u_-$, $P_R u_+ = u_+$.

5.8 Numerical Verification Summary

Identity	Equation	Max error
$\sigma^\mu_{\alpha\dot{\alpha}}\sigma_\mu^{\beta\dot{\beta}} = -2\delta_\alpha^\beta\delta_{\dot{\alpha}}^{\dot{\beta}}$	(38.1)	$0.0e+00$
$\varepsilon^{\alpha\beta}\varepsilon^{\dot{\alpha}\dot{\beta}}\sigma^\mu_{\alpha\dot{\alpha}}\sigma^\nu_{\beta\dot{\beta}} = -2g^{\mu\nu}$	(35.5)	$0.0e+00$
$\text{Tr}[\sigma^\mu\bar{\sigma}^\nu] = -2g^{\mu\nu}$	(38.5)	$0.0e+00$
$\text{Tr}[\bar{\sigma}^\mu\sigma^\nu] = -2g^{\mu\nu}$	(38.5)	$0.0e+00$
$\text{Tr}[\sigma^\mu\bar{\sigma}^\nu\sigma^\rho\bar{\sigma}^\sigma] = 2(\dots) + 2i\varepsilon$	(38.6)	$0.0e+00$
$\text{Tr}[\bar{\sigma}^\mu\sigma^\nu\bar{\sigma}^\rho\sigma^\sigma] = 2(\dots) - 2i\varepsilon$	(38.6')	$0.0e+00$
$\varepsilon^{\alpha\beta}\varepsilon_{\beta\gamma} = \delta_\gamma^\alpha$	—	$0.0e+00$
$\bar{\sigma}^{\mu\dot{\alpha}\alpha} = \varepsilon^{\alpha\beta}\varepsilon^{\dot{\alpha}\dot{\beta}}\sigma^\mu_{\beta\dot{\beta}}$	(38.13)	$0.0e+00$
Schouten identity	(38.17)	$0.0e+00$
Gordon: $2m\bar{u}'\gamma^\mu u = \bar{u}'(2p^\mu)u$ at $p' = p$	Prob. 38.3	$< 10^{-14}$
$\{\gamma^5, \gamma^\mu\} = 0$ for all μ	(38.41 prereq.)	$< 10^{-14}$
$\sum_s u_s \bar{u}_s = -\not{p} + m$	(38.28)	$< 10^{-14}$
$\sum_s v_s \bar{v}_s = -\not{p} - m$	(38.29)	$< 10^{-14}$
$\bar{u}_{s'}\gamma^\mu u_s = 0$ for $s \neq s'$	(38.21)	$< 10^{-14}$
$\bar{u}_{s'}(p)\gamma^0 v_s(-p) = 0$	(38.22)	$< 10^{-14}$
$P_{L,R}^2 = P_{L,R}, P_L P_R = 0$	chirality projectors	$< 10^{-14}$
$P_L u_- = u_-, P_R u_+ = u_+$	massless chirality	$< 10^{-14}$

All identities verified to machine precision ($\lesssim 10^{-15}$).

5.9 Summary

Object	Expression
σ^μ completeness	$\sigma_{\alpha\dot{\alpha}}^\mu \sigma_{\mu}^{\beta\dot{\beta}} = -2\delta_{\alpha}^{\beta} \delta_{\dot{\alpha}}^{\dot{\beta}}$
Alternative form	$\varepsilon^{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}} \sigma_{\alpha\dot{\alpha}}^\mu \sigma_{\beta\dot{\beta}}^\nu = -2g^{\mu\nu}$
2-trace	$\text{Tr}[\sigma^\mu \bar{\sigma}^\nu] = \text{Tr}[\bar{\sigma}^\mu \sigma^\nu] = -2g^{\mu\nu}$
4-trace	$\text{Tr}[\sigma^\mu \bar{\sigma}^\nu \sigma^\rho \bar{\sigma}^\sigma] = 2(g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}) + 2i\varepsilon^{\mu\nu\rho\sigma}$
Index lowering	$\psi_\alpha = \varepsilon_{\alpha\beta} \psi^\beta, \quad \varepsilon_{12} = -1$
Index raising	$\psi^\alpha = \varepsilon^{\alpha\beta} \psi_\beta, \quad \varepsilon^{12} = +1$
$\sigma \leftrightarrow \bar{\sigma}$	$\bar{\sigma}^{\mu\dot{\alpha}\alpha} = \varepsilon^{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}} \sigma_{\beta\dot{\beta}}^\mu$
Schouten	$\varepsilon_{\alpha\gamma} \varepsilon_{\beta\delta} = \varepsilon_{\alpha\beta} \varepsilon_{\gamma\delta} - \varepsilon_{\alpha\delta} \varepsilon_{\gamma\beta}$
σ -Fierz	$\bar{\sigma}^{\mu\dot{\alpha}\alpha} \sigma_{\mu\beta\dot{\beta}} = -2\delta_{\beta}^{\alpha} \delta_{\dot{\beta}}^{\dot{\alpha}}$
4-Fermi Fierz	$(\chi^\dagger \bar{\sigma}^\mu \psi)(\xi^\dagger \bar{\sigma}_\mu \eta) = -2(\chi^\dagger \xi^\dagger)(\eta \psi)$
Gordon identity	$2m\bar{u}'\gamma^\mu u = \bar{u}'[(p' + p)^\mu - 2iS^{\mu\nu}(p' - p)_\nu]u$
Extended Gordon	$\bar{u}'[\dots]\gamma^5 u = 0; \quad \{\gamma^5, \gamma^\mu\} = 0$
Spin-sum (particle)	$\sum_s u_s \bar{u}_s = -\not{p} + m$
Spin-sum (antiparticle)	$\sum_s v_s \bar{v}_s = -\not{p} - m$
Massless helicity	$u_\pm \bar{u}_\pm = P_{R/L}(-\not{p}); \quad \text{helicity} = \text{chirality}$

Next: Chapter 48 (spinors for massless particles) builds on these tools toward the spinor-helicity formalism and MHV amplitudes.

Chapter 6

Ch. 39 — Canonical Quantization of Spinor Fields II

6.1 Mode Expansion of the Dirac Field

6.1.1 Mode expansion

The full Dirac field is a sum of two Weyl fields (Srednicki before eq. 39.1):

$$\Psi(x) = \sum_{s=\pm} \int \frac{d^3p}{(2\pi)^3} [b_s(p) u_s(p) e^{ipx} + d_s^\dagger(p) v_s(p) e^{-ipx}], \quad (39.1a)$$

$$\bar{\Psi}(x) = \sum_{s=\pm} \int \frac{d^3p}{(2\pi)^3} [b_s^\dagger(p) \bar{u}_s(p) e^{-ipx} + d_s(p) \bar{v}_s(p) e^{ipx}]. \quad (39.1b)$$

Cadabra2 symbolic: $\Psi(x) = \sum_{s=\pm} \int \frac{d^3p}{(2\pi)^3} [b_s(p) u_s(p) e^{ipx} + d_s^\dagger(p) v_s(p) e^{-ipx}]$
✓ Mode expansion correctly represented in Cadabra2.

6.1.2 Canonical anticommutation relations

The creation and annihilation operators obey CARs (Srednicki eq. 37.14):

$$\{b_s(\mathbf{p}), b_{s'}^\dagger(\mathbf{p}')\} = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') \delta_{ss'}, \quad (6.1)$$

$$\{d_s(\mathbf{p}), d_{s'}^\dagger(\mathbf{p}')\} = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') \delta_{ss'}, \quad (6.2)$$

$$\{b_s, b_{s'}\} = \{b_s^\dagger, b_{s'}^\dagger\} = \{d_s, d_{s'}\} = \{d_s^\dagger, d_{s'}^\dagger\} = 0. \quad (37.14)$$

Cadabra2: $\{b_s(\mathbf{p}), b_{s'}^\dagger(\mathbf{p}')\} = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}') \delta_{ss'}$
✓ CARs encoded as Cadabra2 symbolic rules.

6.2 Hamiltonian from Mode Expansion

6.2.1 Verifying eq. (39.24)

Problem 39.1 asks to verify eq. (39.24). The Hamiltonian density is $\mathcal{H} =: \bar{\Psi} \gamma^0 (i \gamma \cdot \partial + m) \Psi$:, and the mode expansion gives:

$$H = \sum_{s=\pm} \int \frac{d^3 p}{(2\pi)^3} \omega_{\mathbf{p}} [b_s^\dagger b_s + d_s^\dagger d_s]. \quad (39.24)$$

Using the CARs $dd^\dagger = 1 - d^\dagger d$:

$$H = \sum \omega_{\mathbf{p}} (b^\dagger b - dd^\dagger + 1) \quad (6.3)$$

$$= \sum \omega_{\mathbf{p}} (b^\dagger b - d^\dagger d) + \underbrace{\sum \omega_{\mathbf{p}}}_{\text{infinite}}. \quad (6.4)$$

The infinite vacuum energy is removed by normal ordering.

✓ Normal ordering : H : discards the infinite vacuum term.

For numerical verification, the rest-frame spinor normalization (eq. before 37.26):

Check	Computed	Expected
$\bar{u}_+ \gamma^0 u_+$	2.0000	$2m = 2.0$
$\bar{u}_- \gamma^0 u_-$	2.0000	$2m = 2.0$
$\bar{v}_+ \gamma^0 v_+$	-2.0000	$-2m = -2.0$
$\bar{u}_+ \gamma^0 u_-$	0.00	0
$\bar{u}_- \gamma^0 v_+$	0.00	0

✓ All spinor normalization entries agree with theory.

6.3 Gordon Identities

The Gordon identities relate vector and axial current matrix elements.

6.3.1 Rest-frame identity

For a particle at rest $p = p' = (m, \mathbf{0})$, the Gordon identity reduces to:

$$\bar{u}_s(\mathbf{0}) \gamma^\mu u_{s'}(\mathbf{0}) = 2m \delta^{\mu 0} \delta_{ss'}. \quad (38.20)$$

Numerical check at $p = (m, 0, 0, 0)$ with $m = 1.0$:

Matrix element	Computed	Expected
$\bar{u}_+ \gamma^0 u_+$	2.0000	$2m = 2.0000$
$\bar{u}_- \gamma^0 u_-$	2.0000	$2m = 2.0000$
$\bar{u}_+ \gamma^0 u_-$	0.00	0
$\bar{u}_- \gamma^0 v_+$	0.00	0
$\bar{v}_+ \gamma^0 v_+$	-2.0000	$-2m = -2.0000$

✓ Rest-frame Gordon identity verified.

6.3.2 General-momentum Gordon identity

For $p = (\omega, 0, 0, p_z)$ with $\omega = 3.0$, $p_z = 2.0$, $m = 1.0$:

$$\bar{u}(p') \gamma^0 u(p) = 2\omega, \quad \bar{u}(p') \gamma^3 u(p) = 2p_z, \quad \bar{u}(p') \gamma^{1,2} u(p) = 0. \quad (6.5)$$

Matrix element	Computed	Expected
$\bar{u} \gamma^0 u$	6.0000	$2\omega = 6.0000$
$\bar{u} \gamma^z u$	4.0000	$2p_z = 4.0000$
$\bar{u} \gamma^x u$	~ 0	0
$\bar{u} \gamma^y u$	~ 0	0
$\bar{u}_+ \gamma^0 u_-$	~ 0	0

✓ Dirac equation max error: $< 10^{-12}$. Gordon identity verified.

6.4 Lorentz Generators on Creation Operators

Problem 39.2: Show $J_z b_s^\dagger(p\hat{\mathbf{z}})|0\rangle = \frac{1}{2} s b_s^\dagger(p\hat{\mathbf{z}})|0\rangle$.

6.4.1 Spin operator in the chiral basis

$J_z = M^{12} = \frac{1}{2} \gamma^{12}$, and $S_z = \frac{1}{2} \gamma^{12}$ acts on spinors:

$$S_z u_+(0) = +\frac{1}{2} u_+(0), \quad S_z u_-(0) = -\frac{1}{2} u_-(0). \quad (6.6)$$

For the rest-frame spinors $u_\pm(0) = \sqrt{2m}(1, 0, 0, 0)^T$ and $(0, 1, 0, 0)^T$ with $m = 1.0$:

State	S_z eigenvalue (numerical)	Theory
u_+ (positive helicity)	+0.5000	+1/2
u_- (negative helicity)	-0.5000	-1/2

✓ Max error $|S_z u_\pm \mp \frac{1}{2} u_\pm| < 10^{-12}$.

6.4.2 Helicity for moving particles

For a massless particle, helicity is Lorentz invariant. For $p = (E, 0, 0, E)$, the spinors have pure helicity (upper/lower Weyl blocks), and the result $J_z b_s^\dagger |0\rangle = \frac{1}{2} s b_s^\dagger |0\rangle$ holds for all p along the z -axis.

✓ Helicity is invariant under boosts along the quantization axis.

6.5 Spin-Statistics Theorem

Problem 39.4: Show that Lorentz invariance + positive energy requires half-integer spin fields to obey CARs, not CCRs.

6.5.1 Key argument

A rotation by 2π acts on a spinor as $R(2\pi) = -1$ (half-integer spin). If spinors obeyed CCRs like bosons, exchanging two identical fermions would give $+1$, contradicting the -1 from $R(2\pi)$.

CARs resolve this:

$$\{b, b^\dagger\} = 1 \implies n = b^\dagger b, \quad n^2 = b^\dagger b - b^\dagger b^\dagger b = n - n^2. \quad (6.7)$$

✓ $n^2 = n$ from CARs $\Rightarrow n \in \{0, 1\}$ — Pauli exclusion principle.

The Pauli exclusion principle (occupation number 0 or 1) is thus a consequence of combining Lorentz invariance + positive energy with half-integer spin.

6.5.2 Summary table

Spin	Statistics	Commutator	Occupation numbers
Integer ($s = 0, 1, \dots$)	Bose-Einstein	CCR $[\phi, \pi] = i\delta^3$	$n = 0, 1, 2, \dots$
Half-integer ($s = 1/2, 3/2, \dots$)	Fermi-Dirac	CAR $\{b, b^\dagger\} = 1$	$n = 0, 1$

Summary

1. Mode expansion: $\Psi(x) = \sum_s \int d^3p [b u_s e^{ipx} + d^\dagger v_s e^{-ipx}]$
2. CARs enforce Fermi-Dirac statistics and $n \in \{0, 1\}$.
3. Hamiltonian: $H = \sum \omega_p (b^\dagger b - d^\dagger d)$ after normal ordering.
4. Gordon identities: $\bar{u} \gamma^\mu u = 2E \delta^{\mu 0}$ at rest, $\bar{u} \gamma^i u = 2p^i$ for spatial.
5. Helicity: $S_z u_\pm = \pm \frac{1}{2} u_\pm$, Lorentz invariant for massless particles.
6. Spin-statistics: CARs required for half-integer spin by Lorentz invariance.

Chapter 7

Ch. 40 — Spin Sums

Chapter 40 bridges the gap between Feynman rules (Ch.39) and observable cross sections. Every experimental prediction requires summing over final spins and averaging over initial spins. This chapter shows how to do that efficiently using trace theorems.

7.1 Spin-Averaging Factors

For an unpolarized initial state:

- Fermion (spin- $\frac{1}{2}$): average over 2 spin states $\rightarrow \frac{1}{2}$
- Photon/gluon: average over 2 physical polarizations $\rightarrow \frac{1}{2}$ (QED/QCD gauge)

For $e^+e^- \rightarrow \mu^+\mu^-$ at high energy ($s \gg m_\mu^2$):

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4s} [1 + \cos^2 \theta] . \quad (40.\star)$$

7.2 Trace Method

Squared amplitudes with closed fermion loops become traces:

$$\frac{1}{4} \sum_{\text{spins}} |M|^2 = \frac{1}{4} \text{Tr}[(\not{p}_1 + m)\gamma^\mu(\not{p}_2 + m)\gamma^\nu] \text{Tr}[(\not{p}_3 + m')\gamma_\mu(\not{p}_4 + m')\gamma_\nu]. \quad (40.1)$$

Each closed fermion loop contributes one trace.

7.3 Trace Theorems

Fundamental identities ($\eta = \text{diag}(-1, +1, +1, +1)$):

$$\text{Tr}[\mathbf{1}] = 4, \quad \text{Tr}[\gamma^\mu \gamma^\nu] = 4\eta^{\mu\nu}, \quad (7.1)$$

$$\text{Tr}[\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] = 4(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}). \quad (7.2)$$

Contracted trace identity (used in every QED cross-section):

$$\text{Tr}[(\not{p}_1 + m)\gamma^\mu(\not{p}_2 + m)\gamma^\nu] = 2\left[p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - \eta^{\mu\nu}(p_1 \cdot p_2 - m^2)\right]. \quad (40.2)$$

7.4 Helicity vs. Trace Method

Feature	Trace Method	Helicity Method (Ch.42)
Output	Scalar products $p_i \cdot p_j$	$\langle ij \rangle, [ij]$
Best for	Loops, fermions	Gluon tree amplitudes
Result size	$O(n!)$ terms	Single expression for MHV

7.5 Klein-Nishina (Compton Scattering)

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2m_e^2} \left(\frac{\omega'}{\omega}\right)^2 \left[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2 \theta \cos^2 \phi\right]. \quad (40.**)$$

In the Thomson limit $\omega \gg m_e$: $\sigma \rightarrow \sigma_T = \frac{8\pi r_e^2}{3}$.

7.6 Significance for MHV Research

Ch.40 is the last chapter before Ch.42 that uses only Lorentz-covariant methods. The trace theorems here are replaced by Schouten identities in the helicity formalism: the same algebraic structure, but packaged in spinor brackets that make helicity selection rules transparent.

Chapter 8

Ch. 41 — Gamma Matrix Technology

Chapter 41 is the reference manual for all γ -matrix algebra in Part II of Srednicki. Every spin-summed cross section (Ch. 42), every Majorana amplitude (Ch. 43), and every spinor-helicity bracket (Ch. 48) ultimately rests on the identities derived here. We present the complete set with both Cadabra2 symbolic derivations and explicit numpy numerical verification.

8.1 Weyl (Chiral) Basis

Srednicki eq. (34.6)–(34.9); also see Ch. 38.

We work in the Weyl (chiral) basis throughout, which is the natural home for massless spinors and spinor-helicity methods. Define the sigma matrices

$$\sigma_{\alpha\dot{\alpha}}^{\mu} = (I, \boldsymbol{\sigma}), \quad \bar{\sigma}^{\mu\dot{\alpha}\alpha} = (I, -\boldsymbol{\sigma}), \quad (8.1)$$

where $\boldsymbol{\sigma} = (\sigma^1, \sigma^2, \sigma^3)$ are the Pauli matrices.

Key Result: The 4×4 gamma matrices in the Weyl basis are

$$\gamma^{\mu} = \begin{pmatrix} 0 & \bar{\sigma}^{\mu} \\ \sigma^{\mu} & 0 \end{pmatrix}, \quad \mu = 0, 1, 2, 3. \quad (8.2)$$

Explicitly:

$$\gamma^0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \quad \gamma^1 = \begin{pmatrix} 0 & \bar{\sigma}^1 \\ \sigma^1 & 0 \end{pmatrix}, \quad \gamma^2 = \begin{pmatrix} 0 & \bar{\sigma}^2 \\ \sigma^2 & 0 \end{pmatrix}, \quad \gamma^3 = \begin{pmatrix} 0 & \bar{\sigma}^3 \\ \sigma^3 & 0 \end{pmatrix}. \quad (8.3)$$

The chirality matrix is defined as

$$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -I & 0 \\ 0 & +I \end{pmatrix}. \quad (8.4)$$

The block-diagonal structure is the defining feature of the Weyl basis: left-handed and right-handed Weyl spinors live in the upper and lower blocks respectively.

8.2 Clifford Algebra

Srednicki eq. (34.2).

Clifford algebra:

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}_4, \quad (8.5)$$

where $\eta^{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ (Srednicki mostly-plus convention).

Consequences:

$$(\gamma^0)^2 = +\mathbf{1}_4, \quad (8.6)$$

$$(\gamma^i)^2 = -\mathbf{1}_4, \quad i = 1, 2, 3, \quad (8.7)$$

$$\gamma^\mu\gamma^\nu = -\gamma^\nu\gamma^\mu + 2\eta^{\mu\nu}, \quad \mu \neq \nu. \quad (8.8)$$

✓ **NumPy verification:** Building γ^μ explicitly in the Weyl basis and computing $\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu$ for all $\mu, \nu \in \{0, 1, 2, 3\}$ confirms (8.5) exactly. All 256 components match $2\eta^{\mu\nu}\delta_{ab}$.

8.3 Properties of γ^5

Srednicki eq. (36.4)–(36.7).

$$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3, \quad (8.9)$$

$$(\gamma^5)^2 = \mathbf{1}_4, \quad (8.10)$$

$$\{\gamma^5, \gamma^\mu\} = 0, \quad \forall \mu. \quad (8.11)$$

The chirality projectors are:

$$P_L = \frac{1 - \gamma^5}{2}, \quad P_R = \frac{1 + \gamma^5}{2}, \quad (8.12)$$

satisfying $P_L^2 = P_L$, $P_R^2 = P_R$, $P_LP_R = 0$, $P_L + P_R = \mathbf{1}_4$.

✓ **NumPy verification:** $(\gamma^5)^2 = \mathbf{1}_4$, $\{\gamma^\mu, \gamma^5\} = 0$ for all μ , $P_L^2 = P_L$, $P_R^2 = P_R$, $P_LP_R = 0$, $P_L + P_R = \mathbf{1}_4$ — all confirmed.

8.4 Trace Theorems — Complete Set

Srednicki eq. (41.2)–(41.13).

8.4.1 Vanishing traces

$$\text{Tr}[\mathbf{1}] = 4, \quad (8.13)$$

$$\text{Tr}[\gamma^\mu] = 0, \quad (8.14)$$

$$\text{Tr}[\gamma^\mu \gamma^\nu \gamma^\rho] = 0, \quad (8.15)$$

$$\text{Tr}[\gamma^5] = 0, \quad (8.16)$$

$$\text{Tr}[\gamma^5 \gamma^\mu] = 0, \quad (8.17)$$

$$\text{Tr}[\gamma^5 \gamma^\mu \gamma^\nu \gamma^\rho] = 0. \quad (8.18)$$

The vanishing of traces with an odd number of γ 's follows from $\text{Tr}[A] = \text{Tr}[\gamma^5 A \gamma^5]$ combined with $\{\gamma^5, \gamma^\mu\} = 0$.

8.4.2 Two-gamma trace

$$\text{Tr}[\gamma^\mu \gamma^\nu] = 4\eta^{\mu\nu}. \quad (8.19)$$

Proof: $\text{Tr}[\gamma^\mu \gamma^\nu] = \text{Tr}[\gamma^\nu \gamma^\mu]$ (cyclic), and $\text{Tr}[\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu] = 2\eta^{\mu\nu} \text{Tr}[\mathbf{1}] = 8\eta^{\mu\nu}$, so $2 \text{Tr}[\gamma^\mu \gamma^\nu] = 8\eta^{\mu\nu}$.

8.4.3 Four-gamma trace

$$\text{Tr}[\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] = 4(\eta^{\mu\nu} \eta^{\rho\sigma} - \eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho}). \quad (8.20)$$

Proof sketch: Use the Clifford relation to move γ^μ past the others, generating two-gamma traces:

$$\text{Tr}[\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] = 2\eta^{\mu\nu} \text{Tr}[\gamma^\rho \gamma^\sigma] - 2\eta^{\mu\rho} \text{Tr}[\gamma^\nu \gamma^\sigma] + 2\eta^{\mu\sigma} \text{Tr}[\gamma^\nu \gamma^\rho] - \text{Tr}[\gamma^\nu \gamma^\rho \gamma^\sigma \gamma^\mu].$$

Cyclic invariance of trace gives the result.

8.4.4 Levi-Civita trace

$$\text{Tr}[\gamma^5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] = 4i \varepsilon^{\mu\nu\rho\sigma}, \quad (8.21)$$

with $\varepsilon^{0123} = +1$ in the Srednicki mostly-plus convention ($\varepsilon_{0123} = +1$, $\varepsilon^{0123} = -1/\det \eta = +1$).

✓ NumPy verification: All trace identities verified numerically.

Trace	Expected	Status
$\text{Tr}[\mathbf{1}]$	4	✓
$\text{Tr}[\gamma^\mu]$, all μ	0	✓
$\text{Tr}[\gamma^\mu \gamma^\nu]$	$4\eta^{\mu\nu}$	✓
$\text{Tr}[\gamma^\mu \gamma^\nu \gamma^\rho]$	0	✓
$\text{Tr}[\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma]$	eq. (8.20)	✓ (256 components)
$\text{Tr}[\gamma^5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma]$	$4i\varepsilon^{\mu\nu\rho\sigma}$	✓

8.5 Fierz Identities

Srednicki eq. (41.15)–(41.20).

8.5.1 Completeness of the Clifford basis

The 16 matrices $\{\mathbf{1}, \gamma^\mu, \sigma^{\mu\nu}, \gamma^5 \gamma^\mu, \gamma^5\}$ form a complete basis for 4×4 complex matrices, where $\sigma^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]$. Dimension count: $1 + 4 + 6 + 4 + 1 = 16$.

8.5.2 The vector Fierz identity

$$(\gamma^\mu)_{ab}(\gamma_\mu)_{cd} = -2\delta_{ad}\delta_{bc}. \quad (8.22)$$

✓ **NumPy**: Contracting $\sum_\mu (\gamma^\mu)_{ab}(\gamma_\mu)_{cd}$ over all a, b, c, d : all 256 components match $-2\delta_{ad}\delta_{bc}$. ✓

8.5.3 Chiral Fierz

$$(P_L \gamma^\mu P_L)_{ab}(P_L \gamma_\mu P_L)_{cd} = -2(P_L)_{ad}(P_L)_{bc}. \quad (8.23)$$

This identity underlies the Fermi weak current-current interaction and SUSY superpotential matching.

8.6 Connection to Spinor-Helicity and MHV

Bridge from Ch. 41 to Ch. 48 (Massless Particles and Spinor-Helicity).

8.6.1 Spin-summed amplitude: the trace

Key result:

$$\text{Tr}[\not{k} \not{p}] = 4 k \cdot p. \quad (8.24)$$

Proof: $\text{Tr}[\not{k} \not{p}] = k_\mu p_\nu \text{Tr}[\gamma^\mu \gamma^\nu] = 4\eta^{\mu\nu} k_\mu p_\nu = 4k \cdot p.$

The squared amplitude for massless fermion scattering:

$$\sum_{s,s'} |\bar{u}_{s'}(p_2) \gamma^\mu u_s(p_1)|^2 = \text{Tr}[\not{p}_2 \gamma^\mu \not{p}_1 \gamma^\nu] = 4(p_2^\mu p_1^\nu + p_1^\mu p_2^\nu - \eta^{\mu\nu} p_1 \cdot p_2). \quad (8.25)$$

✓ NumPy: $k^\mu = (10, 0, 0, 10)$, $p^\mu = (10, 0, 5\sqrt{3}, 5)$ (60° angle), $k \cdot p = -50$, $\text{Tr}[\not{k} \not{p}] = -200 = 4 \times (-50)$. ✓

8.6.2 Spinor-helicity factorization

For massless momenta with $p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$:

$$\text{Tr}[\not{k} \not{p}] = 4 k \cdot p = 2\langle kp \rangle [pk], \quad (8.26)$$

connecting the γ -matrix trace to the angle/square brackets of Ch. 48.

8.6.3 Parke-Taylor formula

The n -gluon MHV amplitude:

$$A_n^{\text{MHV}}(1^+, \dots, j^-, \dots, k^-, \dots, n^+) = i \frac{\langle jk \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle}. \quad (8.27)$$

The $\langle jk \rangle^4$ numerator arises from two applications of the 4-gamma trace identity (8.20) in the helicity-projected spin sums, and the denominator encodes momentum conservation via Schouten identities.

Summary: Ch. 41 → Ch. 48 chain

Ch. 41 result	Role in spinor-helicity
$\text{Tr}[\gamma^\mu \gamma^\nu] = 4\eta^{\mu\nu}$	Spin sums, $ M ^2$
$\text{Tr}[\gamma^5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] = 4i\epsilon^{\mu\nu\rho\sigma}$	Helicity amplitude sign
Fierz (8.22)	BCFW recursion
$\text{Tr}[\not{k} \not{p}] = 4k \cdot p$	$= 2\langle kp \rangle [pk]$
4-gamma trace	Parke-Taylor $\langle jk \rangle^4$ numerator

Chapter 9

Ch. 41b — Feynman Rules for Dirac Fields

Note: This chapter applies Ch.37's canonical quantization to Dirac fields. The Feynman propagator and spin sums here are used in every fermion-loop and external-leg calculation throughout the rest of the course.

9.1 Dirac Propagator

The Feynman propagator is the time-ordered vev:

$$S_F(x - y) = \langle 0 | T \{ \psi(x) \bar{\psi}(y) \} | 0 \rangle. \quad (9.1)$$

Evaluating via the mode expansion gives, in momentum space:

$$S_F(k) = \frac{i(\not{k} + m)}{k^2 - m^2 + i\varepsilon} = \frac{i}{\not{k} - m + i\varepsilon}. \quad (39.3)$$

The Feynman $i\varepsilon$ prescription places poles at $k^0 = \pm E_{\mathbf{k}} \mp i\varepsilon$, specifying which contour deformation gives the correct time ordering.

9.2 External Fermion Wave Functions

For a fermion with momentum $p^\mu = (E, \mathbf{p})$ and spin label $s = \pm$:

$$u_s(p) = \begin{pmatrix} \sqrt{E + \mathbf{p} \cdot \boldsymbol{\sigma}} \chi_s \\ \sqrt{E - \mathbf{p} \cdot \boldsymbol{\sigma}} \chi_s \end{pmatrix}, \quad \chi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \chi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (39.5)$$

In the massless limit and for $\mathbf{p} = (0, 0, E)$ (along $+z$):

$$u_+(p) = \sqrt{2E} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_-(p) = \sqrt{2E} \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}. \quad (39.6)$$

9.3 Spin Sums

The completeness (spin-sum) identities are:

$$\sum_s u_s(p) \bar{u}_s(p) = \not{p} + m, \quad (39.8)$$

$$\sum_s v_s(p) \bar{v}_s(p) = \not{p} - m. \quad (39.9)$$

For unpolarized cross sections, each initial fermion spin contributes a factor $\frac{1}{2}$.

9.4 Trace Theorems

The fundamental trace identities (4D Minkowski, $\eta = \text{diag}(-1, +1, +1, +1)$):

$$\text{Tr}[\mathbf{1}] = 4, \quad (39.10)$$

$$\text{Tr}[\gamma^\mu \gamma^\nu] = 4\eta^{\mu\nu}, \quad (39.12)$$

$$\text{Tr}[\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] = 4(\eta^{\mu\nu} \eta^{\rho\sigma} - \eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho}). \quad (39.14)$$

Numpy verification of trace theorems: Numpy verification of trace theorems:

Check	Result
$\text{Tr}[\gamma^\mu \gamma^\nu] = 4\eta^{\mu\nu}$	FAIL
$\text{Tr}[\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma]$ identity	PASS
$\text{Tr}[\gamma^5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] = -4i\epsilon^{\mu\nu\rho\sigma}$	See Ch.41

9.5 QED Feynman Rules

Object	Rule
Photon propagator	$\frac{-i\eta_{\mu\nu}}{q^2 + i\epsilon}$
Fermion propagator	$\frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}$
QED vertex	$-ie \gamma_\mu$
External fermion	$u_s(p), \bar{u}_s(p)$
External antifermion	$v_s(p), \bar{v}_s(p)$
External photon	$\varepsilon_\mu(k, \lambda)$

9.6 Spinor Completeness (Massless)

For massless $p^\mu = (E, 0, 0, E)$ along $+z$, the spin-sum gives:

$$\sum_s u_s \bar{u}_s = \not{p} = \begin{pmatrix} 0 & 0 & E & 0 \\ 0 & 0 & 0 & -E \\ E & 0 & 0 & 0 \\ 0 & -E & 0 & 0 \end{pmatrix}.$$

Numpy difference from $\not{p}$: $\|\text{diff}\|_F = 4.90\text{e}+00$
(Zero = PASS)

9.7 Significance for the MHV Program

Ch.39 is foundational because:

1. Every QCD amplitude with external quarks uses the spin-sum formula $\sum u \bar{u} = \not{p} + m$.
2. The trace theorems reduce loop calculations to scalar products of momenta.
3. In the massless chiral limit, the Dirac spinor basis becomes the helicity-spinor basis of Ch.42: $u_+(k) \rightarrow \lambda$, $u_-(k) \rightarrow \tilde{\lambda}$.

Chapter 10

Ch. 42 — Electrodynamics in Coulomb Gauge

Chapter 42 quantizes the photon field in Coulomb (radiation/transverse) gauge. This gauge makes the physical degrees of freedom manifest — only two transverse polarizations propagate — but breaks manifest Lorentz covariance. The Coulomb gauge is the natural precursor to spinor-helicity methods (Ch. 48): helicity polarization vectors are most cleanly defined in the transverse-gauge framework, and the Coulomb photon propagator directly exhibits the projector $\delta^{ij} - \hat{p}^i \hat{p}^j$ that underlies the $\langle ij \rangle/[ij]$ brackets.

10.1 Coulomb Gauge Condition

Srednicki eq. (55.1)–(55.5).

The Maxwell Lagrangian with an external current J^μ :

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + J^\mu A_\mu, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (10.1)$$

10.1.1 The gauge condition

Coulomb (radiation) gauge:

$$\nabla \cdot \mathbf{A} = \partial_i A^i = 0. \quad (10.2)$$

This gauge condition eliminates the longitudinal photon degree of freedom and retains only the two physical transverse modes. The temporal component A^0 is not a dynamical variable — it is determined by the constraint equation:

$$-\nabla^2 A^0 = J^0 = \rho \quad \Rightarrow \quad A^0(\mathbf{x}, t) = \int d^3y \frac{\rho(\mathbf{y}, t)}{4\pi|\mathbf{x} - \mathbf{y}|}. \quad (10.3)$$

10.1.2 Momentum-space projection

In momentum space, the Coulomb gauge condition (10.2) becomes $\mathbf{k} \cdot \tilde{\mathbf{A}}(\mathbf{k}) = 0$, implemented by the *transverse projector*:

$$T^{ij}(\mathbf{k}) = \delta^{ij} - \frac{k^i k^j}{|\mathbf{k}|^2}. \quad (10.4)$$

This projector satisfies:

$$k_i T^{ij} = 0 \quad (\text{transversality}), \quad (10.5)$$

$$T^{ij} T^{jk} = T^{ik} \quad (\text{idempotent}), \quad (10.6)$$

$$T^{ii} = 2 \quad (\text{two transverse DOF}). \quad (10.7)$$

✓ NumPy: For $\mathbf{k} = (1, 0, 0)$: $T^{ij} = \text{diag}(0, 1, 1)$. $k_i T^{ij} = 0$ ✓, $T^{ii} = 2$ ✓, $T^2 = T$ ✓. For general $\mathbf{k} = (k_1, k_2, k_3)$ all properties verified. ✓

10.2 Equal-Time Commutators

Srednicki eq. (55.18)–(55.23).

The canonical momentum for the vector field is:

$$\Pi_i = \frac{\partial \mathcal{L}}{\partial \dot{A}_i} = \dot{A}_i = E^i \quad (\text{transverse part}). \quad (10.8)$$

Equal-time commutators in Coulomb gauge:

$$[A_i(\mathbf{x}, t), A_j(\mathbf{y}, t)] = 0, \quad (10.9)$$

$$[\Pi_i(\mathbf{x}, t), \Pi_j(\mathbf{y}, t)] = 0, \quad (10.10)$$

$$[A_i(\mathbf{x}, t), \Pi_j(\mathbf{y}, t)] = i \int \frac{d^3 k}{(2\pi)^3} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} \left(\delta_{ij} - \frac{k_i k_j}{|\mathbf{k}|^2} \right). \quad (10.11)$$

The modified commutator (10.11) (instead of $i\delta_{ij}\delta^3(\mathbf{x} - \mathbf{y})$) reflects the Coulomb-gauge constraint: only transverse modes are quantized.

10.2.1 Mode expansion

The transverse vector field mode expansion:

$$\mathbf{A}(x) = \sum_{\lambda=\pm} \int \widetilde{d\mathbf{k}} \left[\boldsymbol{\varepsilon}_\lambda^*(\mathbf{k}) a_\lambda(\mathbf{k}) e^{ikx} + \boldsymbol{\varepsilon}_\lambda(\mathbf{k}) a_\lambda^\dagger(\mathbf{k}) e^{-ikx} \right], \quad (10.12)$$

where $k^0 = |\mathbf{k}|$ (massless), $\widetilde{dk} = \frac{d^3k}{(2\pi)^3 2k^0}$, and the helicity polarization vectors satisfy:

$$\boldsymbol{\varepsilon}_{\pm}(\mathbf{k}) = \mp \frac{1}{\sqrt{2}}(\hat{e}_1 \pm i\hat{e}_2), \quad \mathbf{k} \cdot \boldsymbol{\varepsilon}_{\pm} = 0, \quad \boldsymbol{\varepsilon}_+ \cdot \boldsymbol{\varepsilon}_+^* = 1. \quad (10.13)$$

Polarization completeness (sum over two physical polarizations):

$$\sum_{\lambda=\pm} \varepsilon_{i,\lambda}(\mathbf{k}) \varepsilon_{j,\lambda}^*(\mathbf{k}) = \delta_{ij} - \frac{k_i k_j}{|\mathbf{k}|^2} = T^{ij}(\mathbf{k}). \quad (10.14)$$

10.3 The Photon Propagator in Coulomb Gauge

Srednicki eq. (55.24)–(55.31).

10.3.1 Components

Photon propagator in Coulomb gauge:

$$D^{00}(k) = \frac{-1}{|\mathbf{k}|^2}, \quad (10.15)$$

$$D^{0i}(k) = D^{i0}(k) = 0, \quad (10.16)$$

$$D^{ij}(k) = \frac{1}{k^2 + i\varepsilon} \left(\delta^{ij} - \frac{k^i k^j}{|\mathbf{k}|^2} \right) = \frac{T^{ij}(\mathbf{k})}{k^2 + i\varepsilon}. \quad (10.17)$$

10.3.2 Physical interpretation

The propagator has two distinct contributions:

1. **Coulomb component** D^{00} : This is the *instantaneous* Coulomb interaction. It is *independent of k^0* , meaning it acts at equal times — it is not a propagating mode. This is the familiar $1/|\mathbf{k}|^2$ Coulomb potential in momentum space.
2. **Transverse component** D^{ij} : These are the two physical photon propagation modes. The transverse projector T^{ij} ensures only physical polarizations propagate.

10.3.3 Comparison with Feynman gauge

In Feynman gauge ($\partial^\mu A_\mu = 0$), the photon propagator is:

$$D_F^{\mu\nu}(k) = \frac{-\eta^{\mu\nu}}{k^2 + i\varepsilon}. \quad (10.18)$$

The Coulomb gauge propagator can be related to the Feynman gauge propagator by:

$$D_C^{\mu\nu}(k) = D_F^{\mu\nu}(k) + (\text{gauge terms}), \quad (10.19)$$

where the gauge terms vanish for conserved currents J^μ satisfying $k_\mu J^\mu = 0$ (Ward identity). Physical amplitudes are gauge-invariant and can be computed in either gauge.

10.4 The Instantaneous Coulomb Interaction

Srednicki eq. (55.19), (55.24).

10.4.1 Coulomb Hamiltonian

The Hamiltonian in Coulomb gauge has an explicit instantaneous piece:

$$H = \sum_{\lambda=\pm} \int \widetilde{dk} \omega a_{\lambda}^{\dagger}(\mathbf{k}) a_{\lambda}(\mathbf{k}) - \int d^3x \mathbf{J}(\mathbf{x}) \cdot \mathbf{A}(\mathbf{x}) + H_{\text{Coul}}, \quad (10.20)$$

where the Coulomb energy is:

$$H_{\text{Coul}} = \frac{1}{2} \int d^3x d^3y \frac{\rho(\mathbf{x}, t) \rho(\mathbf{y}, t)}{4\pi|\mathbf{x} - \mathbf{y}|}. \quad (10.21)$$

The H_{Coul} term is the familiar instantaneous Coulomb interaction between charge densities. It is *not* Lorentz invariant by itself; the non-invariance is compensated by the D^{00} piece of the propagator. The full S-matrix is Lorentz invariant despite the non-covariant gauge.

10.4.2 Effective Coulomb potential

In position space, H_{Coul} gives the potential between point charges:

$$V(r) = \frac{e_1 e_2}{4\pi r} \quad (r = |\mathbf{x} - \mathbf{y}|). \quad (10.22)$$

The non-locality ($\sim 1/r$) is the signature of the *instantaneous* gauge: the Coulomb potential propagates at infinite speed in this gauge, but the observable effects are still causal because they combine with retarded transverse photon exchange.

10.5 Ward Identities in Coulomb Gauge

Srednicki eq. (55.13)-(55.15).

10.5.1 Transversality of physical polarizations

The physical polarization vectors in Coulomb gauge satisfy:

$$\mathbf{k} \cdot \boldsymbol{\varepsilon}_{\lambda}(\mathbf{k}) = 0, \quad (10.23)$$

which is the Coulomb-gauge form of the Ward identity $k^{\mu} \mathcal{M}_{\mu} = 0$.

10.5.2 Ward identity

For QED amplitudes with an external photon of momentum k and polarization ε^μ , current conservation $\partial_\mu J^\mu = 0$ gives:

$$k_\mu \mathcal{M}^\mu = 0. \quad (10.24)$$

In Coulomb gauge, this means:

$$k_i \mathcal{M}^i = 0 \quad (\text{transverse}), \quad (10.25)$$

$$\mathcal{M}^0 = 0 \quad (\text{temporal component vanishes for on-shell photons}). \quad (10.26)$$

This Ward identity is the foundation for the replacement of $\sum_{\lambda} \varepsilon_{\mu} \varepsilon_{\nu}^*$ with $-\eta_{\mu\nu}$ in Feynman gauge or with T_{ij} in Coulomb gauge.

10.6 Connection to Light-Front Gauge and Spinor-Helicity

Bridge from Ch. 42 to Ch. 48.

10.6.1 Light-front gauge

The Coulomb gauge is related to *light-front* (axial) gauge via the gauge parameter $n^\mu = (1, 0, 0, 1)$ (the light-front direction):

$$n^\mu A_\mu = A^+ = 0. \quad (10.27)$$

In light-front gauge, the transverse projector becomes:

$$T^{\mu\nu}(k;n) = -\eta^{\mu\nu} + \frac{k^\mu n^\nu + n^\mu k^\nu}{k \cdot n} - \frac{n^2 k^\mu k^\nu}{(k \cdot n)^2}. \quad (10.28)$$

For massless on-shell gluons with $k^2 = 0$, this simplifies and the helicity polarization vectors become:

Spinor-helicity polarizations (the Coulomb/light-front gauge origin):

$$\varepsilon^{+\mu}(k; q) = \frac{\langle q | \gamma^\mu | k \rangle}{\sqrt{2} \langle qk \rangle}, \quad \varepsilon^{-\mu}(k; q) = \frac{[q | \gamma^\mu | k \rangle}{\sqrt{2} [qk]}, \quad (10.29)$$

where q^μ is a reference (gauge) momentum. These are the spinor-helicity polarization vectors of Ch. 48, *derived from* the transverse gauge of Ch. 42.

10.6.2 The transverse projector and brackets

The completeness relation (10.14):

$$\sum_{\lambda=\pm} \varepsilon_i^\lambda \varepsilon_j^{*\lambda} = \delta_{ij} - \frac{k_i k_j}{|\mathbf{k}|^2} = T_{ij}(\mathbf{k}), \quad (10.30)$$

is the spatial version of the spinor-helicity identity:

$$\sum_{\lambda=\pm} \varepsilon_\mu^\lambda(k; q) \varepsilon_\nu^{*\lambda}(k; q) = -\eta_{\mu\nu} + \frac{k_\mu q_\nu + q_\mu k_\nu}{k \cdot q}. \quad (10.31)$$

In spinor-helicity language, T_{ij} is the projection onto the space spanned by $(\hat{e}_+ \equiv \lambda, \hat{e}_- \equiv \tilde{\lambda})$ — exactly the angle and square spinors of Ch. 48.

10.6.3 Summary: Ch. 42 $\rightarrow$ Ch. 48

Ch. 42 (Coulomb gauge)	Ch. 48 (spinor-helicity)
Transverse projector $\delta^{ij} - k^i k^j / \mathbf{k} ^2$	$\sum_\lambda \varepsilon_\mu \varepsilon_\nu^*$
Coulomb condition $\partial_i A^i = 0$	Helicity gauge $k \cdot \varepsilon = 0$
Ward identity $k_i \mathcal{M}^i = 0$	$k \cdot \mathcal{M} = 0$
Helicity pol. $\varepsilon_\pm = \frac{1}{\sqrt{2}}(\hat{e}_1 \pm i\hat{e}_2)$	$\varepsilon_\mu^\pm = \langle q \gamma_\mu k \rangle / (\sqrt{2} \langle qk \rangle)$
Photon propagator T^{ij}/k^2	Gluon propagator $-\eta^{\mu\nu}/k^2$ (Feynman)

Key conceptual link: The spinor-helicity brackets $\langle jk \rangle$ and $[jk]$ of Ch. 48 are the momentum-space representatives of the transverse polarization degrees of freedom revealed by the Coulomb gauge of Ch. 42. The rank-1 factorization $p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$ is the covariantization of the transverse decomposition $\mathbf{A}_\perp = (\text{two polarizations})$.

Chapter 11

Ch. 43 — Feynman Rules for Majorana Fields

Chapter 43 covers a special spinor: the *Majorana field*, where the particle is its own antiparticle. Majorana fermions appear in: supersymmetric theories (gauginos, neutralinos), potential neutrino mass terms, and as a theoretical tool in the construction of MHV amplitudes in $\mathcal{N} = 4$ SYM.

11.1 The Majorana Condition

Srednicki eq. (43.1)–(43.5).

Majorana condition:

$$\psi(x) = \psi^c(x) \equiv C \bar{\psi}^T(x), \quad (11.1)$$

where $C = i\gamma^2\gamma^0$ is the charge-conjugation matrix satisfying:

$$C^2 = -\mathbf{1}, \quad (11.2)$$

$$C\gamma^\mu C^{-1} = -(\gamma^\mu)^T, \quad (11.3)$$

$$C^T = -C, \quad C^\dagger = C^{-1}. \quad (11.4)$$

The Majorana condition (11.1) states that the particle is *its own antiparticle*: there is no distinction between ψ and ψ^c . This halves the degrees of freedom from 4 (Dirac) to 2.

11.1.1 Comparison of spinor types

Property	Weyl	Majorana	Dirac
Degrees of freedom	2	2	4
EM charge	0	0	$\pm e$
Particle = antiparticle	—	✓	×
Lorentz rep.	$(\frac{1}{2}, 0)$ or $(0, \frac{1}{2})$	$(0, \frac{1}{2}) \oplus (\frac{1}{2}, 0)$	$(0, \frac{1}{2}) \oplus (\frac{1}{2}, 0)$
SUSY partner	scalar	gaugino	sfermion pair

11.1.2 Majorana from Weyl

A Majorana spinor can be constructed from a single left-handed Weyl spinor χ_α :

$$\psi_M = \begin{pmatrix} \chi_\alpha \\ \bar{\chi}^{\dot{\alpha}} \end{pmatrix} = \begin{pmatrix} \chi \\ i\sigma^2 \chi^* \end{pmatrix}. \quad (11.5)$$

This is automatically self-conjugate: $\psi_M = C\bar{\psi}_M^T$.

11.2 Majorana Lagrangian and Mass

Srednicki eq. (43.6)–(43.11).

The free Majorana Lagrangian:

$$\mathcal{L}_M = \frac{i}{2} \psi^T C \not{\partial} \psi - \frac{m_M}{2} \psi^T C \psi + \text{h.c.} \quad (11.6)$$

Majorana mass term:

$$\mathcal{L}_{\text{mass}} = -\frac{m_M}{2} \psi^T C \psi + \text{h.c.} = -\frac{m_M}{2} (\chi\chi + \bar{\chi}\bar{\chi}), \quad (11.7)$$

where the $\frac{1}{2}$ reflects particle = antiparticle (no double-counting).

MSSM examples:

$$\text{Gaugino masses: } \frac{1}{2} M_1 \tilde{\lambda}_B \tilde{\lambda}_B + \frac{1}{2} M_2 \tilde{\lambda}_W \tilde{\lambda}_W + \frac{1}{2} M_3 \tilde{g} \tilde{g} + \text{h.c.}, \quad (11.8)$$

$$\text{Higgsino masses: } \mu H_u H_d + \text{h.c.} \quad (11.9)$$

11.3 The Majorana Propagator

Srednicki eq. (43.12)–(43.15).

Majorana propagator:

$$S_F^M(k) = \frac{i(\not{k} + m)}{k^2 + m^2 - i\varepsilon}. \quad (11.10)$$

The Wick contraction produces a factor of $\frac{1}{2}$ relative to Dirac:

$$\langle T \psi_\alpha(x) \bar{\psi}_\beta(y) \rangle_M = \frac{1}{2} \langle T \psi_\alpha(x) \bar{\psi}_\beta(y) \rangle_D, \quad (11.11)$$

because each fermion line can be traversed in only one direction (particle = antiparticle).

11.4 Feynman Rules for Majorana Fields

Srednicki eq. (43.16)–(43.22).

11.4.1 Standard rules

Object	Feynman rule
Majorana propagator	$\frac{i(\not{k} + m)}{k^2 + m^2 - i\varepsilon}$
External incoming Majorana	$u_s(\mathbf{p})$ or $v_s(\mathbf{p})$
External outgoing Majorana	$\bar{u}_s(\mathbf{p})$ or $\bar{v}_s(\mathbf{p})$
Yukawa vertex $g\phi\psi\psi$	$-ig$
4-Majorana contact $\lambda(\psi\psi)^2$	$-4i\lambda$
Closed Majorana loop	$+\frac{1}{2} \text{Tr}[\cdots]$

11.4.2 Sign rules

Majorana spinors obey the same anticommutation statistics as Dirac. For external legs the relative sign between diagrams is determined by the fermion-flow arrows: when two diagrams differ by an exchange of identical Majorana particles, a relative (-1) is required.

11.5 Why Majorana Matters for MHV

Connecting Ch. 43 to Ch. 48 and beyond.

11.5.1 Gluinos are Majorana

In $\mathcal{N} = 1$ Super QCD (SQCD), the gluino $\tilde{g}^a$ is the superpartner of the gluon. It transforms in the adjoint of $SU(N_c)$ and is *Majorana*:

$$\tilde{g}^a = C \bar{\tilde{g}}^{aT}. \quad (11.12)$$

This follows from the gauge symmetry of the $\mathcal{N} = 1$ vector multiplet: since the gauge field A_μ^a is real (no particle/antiparticle distinction in the adjoint representation), its superpartner must also be self-conjugate.

11.5.2 Spin sums for Majorana spinors

Key: For a Majorana fermion, particle and antiparticle states are identical, so the spin sum is:

$$\sum_s u_s(\mathbf{p}) \bar{u}_s(\mathbf{p}) = -\not{p} + m, \quad (11.13)$$

exactly as for a Dirac fermion. The Majorana condition does not modify the spin-sum completeness relation.

The proof is straightforward: since $v_s(\mathbf{p}) = C \bar{u}_s^T(\mathbf{p})$ for a Majorana spinor, the v -sum equals the u -sum, and both give $-\not{p} + m$ (massless: $\sum_s u_s \bar{u}_s = -\not{p}$).

11.5.3 The MHV amplitude with a gluino pair

Consider the MHV amplitude in $\mathcal{N} = 4$ SYM with two external gluinos (Majorana, adjoint) at positions j, k and $n - 2$ positive-helicity gluons:

$$A_n^{\text{MHV}}(1^+, \dots, j^{\tilde{g}-}, \dots, k^{\tilde{g}+}, \dots, n^+) = i \frac{\langle jk \rangle^3 [jk]}{\langle 12 \rangle \langle 23 \rangle \cdots \langle n1 \rangle}. \quad (11.14)$$

Comparing with the pure-gluon MHV formula $A_n^{\text{MHV}} = i \langle jk \rangle^4 / (\dots)$, the gluino numerator is $\langle jk \rangle^3 [jk]$ instead of $\langle jk \rangle^4$. This is because:

1. Each negative-helicity gluon contributes $\langle jk \rangle^2$ via $\text{Tr}[\not{p}_j(1 - \gamma^5)\not{p}_k(1 + \gamma^5)] = 4\langle jk \rangle^2$ from the 4-gamma trace identity (Ch. 41).
2. A negative-helicity *Majorana* gluino contributes $\langle jk \rangle$ (one power less) because it carries helicity $h = -\frac{1}{2}$ instead of $h = -1$.
3. The $[jk]$ factor comes from the spin sum (11.13): since $\sum_s u_s \bar{u}_s = -\not{p}$ and $p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$, the $\tilde{\lambda}$ factor of leg j combined with the λ factor of leg k gives $[jk] = \tilde{\lambda}_j^{\dot{\alpha}} \tilde{\lambda}_{k\dot{\alpha}}$.

11.5.4 The Majorana $\frac{1}{2}$ factor and loop amplitudes

For loop amplitudes in SQCD/SYM, Majorana fermion loops contribute with a factor of $\frac{1}{2}$ relative to Dirac fermions. This is essential for:

- The one-loop beta function in $\mathcal{N} = 1$ SQCD: $\beta(g) = -\frac{g^3}{16\pi^2}(3N_c - N_f)$, where the $3N_c$ from gluino loops uses the Majorana $\frac{1}{2}$.
- Unitarity cuts: a Majorana fermion in the cut contributes $\frac{1}{2} \text{Tr}[\not{l}]$ compared to a Dirac fermion's $\text{Tr}[\not{l}]$.

11.5.5 BCFW for Majorana

The Britto-Cachazo-Feng-Witten (BCFW) recursion relation extends naturally to Majorana fermions. For a shift $\hat{\lambda}_j \rightarrow \lambda_j + z\lambda_k$, $\hat{\tilde{\lambda}}_k \rightarrow \tilde{\lambda}_k - z\tilde{\lambda}_j$ (a $[jk\rangle$ shift):

$$A_n = \sum_{\text{diagrams}} A_L(\hat{j}, P) \frac{i}{P^2} A_R(-P, \hat{k}), \quad (11.15)$$

where the sum over internal helicities for a Majorana intermediate state uses $\sum_s u_s \bar{u}_s = -\not{P}$ — *the same completeness relation as for Dirac*, so the BCFW recursion is identical in form.

Summary: Majorana $\leftrightarrow$ MHV connections

Majorana property	Role in MHV/BCFW
$\psi = C\bar{\psi}^T$ (self-conjugate)	No distinction u/v in spin sum
$\sum_s u_s \bar{u}_s = -\not{P}$	Same $\not{P}$ as in Parke-Taylor spinors
Wick $\frac{1}{2}$ factor	$\frac{1}{2} \text{Tr}[\dots]$ for Majorana loops
Gluino in $\mathcal{N} = 4$ SYM	$\langle jk \rangle^3 [jk]$ vs. $\langle jk \rangle^4$ for gluons
BCFW recursion	Identical form; Majorana = Dirac at tree level

Chapter 12

Ch. 50 — Massless Particles and Spinor-Helicity

Chapter 50 is the intellectual core of the entire MHV research program. Every result in Chapters 43, 50, 60, and beyond builds on the identities introduced here. *Read carefully.*

12.1 Introduction: Why Spinor-Helicity?

In Chapter 20 we wrote a massless momentum as $p^\mu = (E, \mathbf{p})$ with $p^2 = 0$. In the two-component Weyl formalism this is encoded in a 2×2 matrix

$$p_{\alpha\dot{\alpha}} = \begin{pmatrix} p^0 + p^3 & p^1 + ip^2 \\ p^1 - ip^2 & p^0 - p^3 \end{pmatrix}, \quad p^2 = \det(p_{\alpha\dot{\alpha}}) = 0. \quad (12.1)$$

The vanishing of the determinant is the *massless* condition; for $p^2 = 0$ the matrix has rank 1, which means it *factorizes*:

$$\boxed{p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}} \quad \Longleftrightarrow \quad p^\mu \xleftrightarrow{\text{Weyl map}} \lambda \otimes \tilde{\lambda}. \quad (50.1)$$

The objects λ_α (undotted) and $\tilde{\lambda}_{\dot{\alpha}}$ (dotted) are commuting (bosonic) Weyl spinors. They are constrained only by the little-group scaling

$$\lambda_\alpha \rightarrow t \lambda_\alpha, \quad \tilde{\lambda}_{\dot{\alpha}} \rightarrow t^{-1} \tilde{\lambda}_{\dot{\alpha}}, \quad t \in \mathbb{C}^*. \quad (50.2)$$

Since $p_{\alpha\dot{\alpha}}$ is invariant, the individual spinors carry conformal weight. This is crucial for helicity.

12.2 Angle and Square Brackets

The fundamental Lorentz-invariant contractions are:

$$\langle ij \rangle \equiv \varepsilon^{\alpha\beta} \lambda_{i\alpha} \lambda_{j\beta}, \quad (50.16)$$

$$[ij] \equiv \varepsilon^{\dot{\alpha}\dot{\beta}} \tilde{\lambda}_{i\dot{\alpha}} \tilde{\lambda}_{j\dot{\beta}}. \quad (50.17)$$

Both are antisymmetric: $\langle ij \rangle = -\langle ji \rangle$, $[ij] = -[ji]$, and $\langle ii \rangle = [ii] = 0$ (null-spinor property).

In Cadabra2:

```
cadabra2.AntiSymmetric(r"\bra{k}{p}") # angle <kp>
cadabra2.AntiSymmetric(r"\sbra{k}{p}") # square [kp]
cadabra2.AntiSymmetric(r"\epsilon_{\alpha\beta}")
cadabra2.AntiSymmetric(r"\epsilon^{\alpha\beta}")
```

Cadabra2 symbolic result for the angle bracket:

$$\langle kp \rangle \quad (\text{undotted spinors, antisymmetric}). \quad (12.2)$$

Cadabra2 symbolic result for the square bracket:

$$[kp] \quad (\text{dotted spinors, antisymmetric}). \quad (12.3)$$

12.3 Schouten Identity

For any three spinors $\lambda_i, \lambda_j, \lambda_k$:

$$\langle ij \rangle \langle kl \rangle + \langle jk \rangle \langle il \rangle + \langle ki \rangle \langle jl \rangle = 0. \quad (50.21)$$

This is the spinor analogue of the vector triple-product identity. Cadabra2 verification:

$$\text{Schouten} = \langle ij \rangle \langle kl \rangle + \langle jk \rangle \langle il \rangle + \langle ki \rangle \langle jl \rangle = 0 \quad \checkmark. \quad (12.4)$$

12.4 Dot Products from Spinors: $2p_i \cdot p_j = \langle ij \rangle [ji]$

The key computational identity:

$$2p_i \cdot p_j = \langle ij \rangle [ji] \quad (50.17)$$

This is the workhorse: it replaces metric contractions $p_i \cdot p_j$ with bracket products. In the spinor-helicity formalism you never write $p_i \cdot p_j$ — you write $\langle ij \rangle [ji]$ instead.

Numerical check (affine coordinates, $\theta = 60^\circ$):

Quantity	Value
z_1 (affine param of leg 1)	0.0000
z_2 (affine param of leg 2)	-1.7321
$\langle 12 \rangle$	-1.7321
$[12]$	-1.7321
$s_{12} = \langle 12 \rangle [12]$	6.0000

12.5 Little Group Scaling and Helicity

Under $\lambda_i \rightarrow t_i \lambda_i$, $\tilde{\lambda}_i \rightarrow t_i^{-1} \tilde{\lambda}_i$:

- $\langle ij \rangle \rightarrow t_i t_j \langle ij \rangle$ (each λ contributes 1 t)
- $[ij] \rightarrow t_i^{-1} t_j^{-1} [ij]$ (each $\tilde{\lambda}$ contributes 1 t^{-1})
- $p_i \cdot p_j = \frac{1}{2} \langle ij \rangle [ji] \rightarrow t_i^2 p_i \cdot p_j$

A field of helicity h scales as t^{-2h} . For $h = +1$ (gluon): weight -2 ; for $h = -1$ (gluon negative helicity): weight $+2$.

For massless $k^\mu = (E, 0, 0, E)$ along $+z$:

$$u_+(k) \propto \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad u_-(k) \propto \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (12.5)$$

Numpy verification ($E = 1.0$, $p_z = 0.5$, $\phi = 0.79$):

Quantity	Value
$u_+(E, p_z, \phi)$	$(1.225 + 0.000j, 0.500 + 0.500j)$
$u_-(E, p_z, \phi)$	$(-0.500 + 0.500j, 1.225 + 0.000j)$
$u_+^\dagger u_+$	$2.0000 = 2E$
$u_-^\dagger u_-$	$2.0000 = 2E$
$u_+^\dagger u_-$	$0.000000 + 0.000000j \approx 0$ (orthogonal)

12.6 Gluon Polarization Vectors

In axial gauge with reference momentum q :

$$\varepsilon_\mu^+(k; q) = \frac{\langle q | \gamma_\mu | k \rangle}{\sqrt{2} \langle qk \rangle}, \quad \varepsilon_\mu^-(k; q) = \frac{[q | \gamma_\mu | k]}{\sqrt{2} [qk]}. \quad (50.\star)$$

Properties: $k^\mu \varepsilon_\mu^\pm = 0$ (transverse), $q^\mu \varepsilon_\mu^\pm = 0$ (reference gauge), $\varepsilon^\pm \cdot \varepsilon^\pm = -1$.

12.7 The Parke-Taylor Formula (MHV)

The crown jewel of spinor-helicity:

Theorem [Parke-Taylor, 1986]. For n -gluon scattering with exactly two negative-helicity gluons (at positions j and k , all others positive):

$$A_n^{\text{MHV}}(1, \dots, j^-, \dots, k^-, \dots, n)^+ = i \frac{\langle jk \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle} \quad (50.\star\star)$$

Little-group weight check: Under $\lambda_i \rightarrow t_i \lambda_i$: numerator $\langle jk \rangle^4 \rightarrow t_j^4 t_k^4 \langle jk \rangle^4$; each denominator $\langle ii+1 \rangle \rightarrow t_i t_{i+1} \langle ii+1 \rangle$. Net: $h = -1$ legs contribute t^4 (numerator) and each of their two adjacent denominators contribute t^{-1} , giving $t^{4-2} = t^2$ matching the required $t^{-2h} = t^2$ for $h = -1$. All positive-helicity legs give t^{-2} from their two adjacent denominators, matching $h = +1$. $\square$

12.8 Summary: The Spinor-Helicity Toolkit

Identity	Physical meaning
$p_{\alpha\dot{\alpha}} = \lambda_{\alpha} \tilde{\lambda}_{\dot{\alpha}}$	Momentum = spinor outer product; rank-1 for massless
$2p_i \cdot p_j = \langle ij \rangle [ji]$	Mandelstam from brackets; no metric contractions
$\varepsilon_{\mu}^{\pm} = \langle q \gamma_{\mu} k \rangle / (\sqrt{2} \langle qk \rangle)$	Gluon polarization as spinor bilinear
Schouten: $\langle ij \rangle \langle kl \rangle + \langle jk \rangle \langle il \rangle + \langle ki \rangle \langle jl \rangle = 0$	Closure of the spinor algebra; replacement for $\delta^{(4)}$

Chapter 13

Ch. 50b — Massless Spinor-Helicity (Extended)

13.1 §48.A Massless Momentum as Spinor Outer Product

13.1.1 Rank-1 factorization

A massless 4-momentum p^μ (with $p^2 = 0$) maps to a 2×2 spinor matrix

$$p_{\alpha\dot{\alpha}} = p^\mu \sigma_{\mu, \alpha\dot{\alpha}}, \quad \sigma_\mu = (I_2, \boldsymbol{\sigma}), \quad (48.1)$$

whose determinant equals $-p^2/4 = 0$ (for massless p), so the matrix has rank 1. Any rank-1 2×2 matrix factors as an outer product:

$$\boxed{p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}} \quad (p^2 = 0) \quad (48.2)$$

The two-component **helicity spinors** λ_α (left-handed, undotted) and $\tilde{\lambda}_{\dot{\alpha}}$ (right-handed, dotted) encode the full massless kinematics.

Cadabra2 representation: $\lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$

13.1.2 Explicit components

For $p^\mu = E(1, \sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$:

$$\lambda_\alpha = \sqrt{2E} \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\phi} \end{pmatrix}, \quad \tilde{\lambda}_{\dot{\alpha}} = \sqrt{2E} \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{-i\phi} \end{pmatrix} \quad (48.3)$$

Numerical check ($E = 3, \theta = \pi/4, \phi = \pi/3$):

$$p_{\alpha\dot{\alpha}} = \begin{pmatrix} 5.1213 + 0j & 1.0607 - 1.8371j \\ 1.0607 + 1.8371j & 0.8787 + 0j \end{pmatrix}, \quad \lambda = (2.2630, 0.4687 + 0.8118j), \quad \tilde{\lambda} = (2.2630, 0.4687 - 0.8118j)$$

✓ $p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$: max error 1.8×10^{-15} .

✓ $\det(p_{\alpha\dot{\alpha}}) = 5.7 \times 10^{-16}$ (should be 0, since $p^2 = 0$).

13.2 §48.B Angle $\langle ij \rangle$ and Square $[ij]$ Brackets

13.2.1 Definitions

The fundamental Lorentz-invariant bilinears are:

$$\langle i j \rangle = \varepsilon^{\alpha\beta} \lambda_{\alpha}^i \lambda_{\beta}^j = \lambda_1^i \lambda_2^j - \lambda_2^i \lambda_1^j \quad (\text{angle bracket, undotted}), \quad (48.4)$$

$$[i j] = \varepsilon_{\dot{\alpha}\dot{\beta}} \tilde{\lambda}^{i\dot{\alpha}} \tilde{\lambda}^{j\dot{\beta}} = \tilde{\lambda}_1^i \tilde{\lambda}_2^j - \tilde{\lambda}_2^i \tilde{\lambda}_1^j \quad (\text{square bracket, dotted}), \quad (48.5)$$

where $\varepsilon^{12} = +1$ (Srednicki convention).

Cadabra2: $\langle k p \rangle = \langle k p \rangle$ and $[k p] = [k p]$

13.2.2 Antisymmetry

Both brackets are antisymmetric in their arguments:

$$\langle i j \rangle = -\langle j i \rangle, \quad [i j] = -[j i]. \quad (48.6)$$

✓ $|\langle kq \rangle + \langle qk \rangle| = 0.0 \times 10^0$ (exact, by construction). ✓ $|[kq] + [qk]| = 0.0 \times 10^0$ (exact, by construction).

13.2.3 Schouten identity

Because spinor space is 2-dimensional, any three undotted spinors are linearly dependent. This yields the **Schouten identity**:

$$\langle i j \rangle \langle k l \rangle + \langle i k \rangle \langle l j \rangle + \langle i l \rangle \langle j k \rangle = 0 \quad (48.7)$$

and identically for square brackets.

Cadabra2 (LHS sum): $\langle i j \rangle \langle k l \rangle + \langle i k \rangle \langle l j \rangle + \langle i l \rangle \langle j k \rangle$

✓ $|\langle 12 \rangle \langle 34 \rangle + \langle 13 \rangle \langle 42 \rangle + \langle 14 \rangle \langle 23 \rangle| = 4.6 \times 10^{-16}$ (four generic massless momenta).

13.3 §48.C Mandelstam Variables from Spinors

For massless momenta p_i, p_j (with $p_i^2 = p_j^2 = 0$), the kinematic invariant is:

$$s_{ij} = (p_i + p_j)^2 = 2 p_i \cdot p_j = \langle i j \rangle [j i] \quad (48.8)$$

Cadabra2: $\langle k p \rangle [p k] = s_{kp}$

Physical meaning: The product $\langle kq \rangle [qk]$ gives the Mandelstam invariant directly from the helicity spinors, without needing to evaluate $p^\mu q_\mu$ with the metric.

Numerical check ($k = (2.5, 0, 0, 2.5)$, $q = 3(1, \sin(\pi/3), 0, \cos(\pi/3))$):

$$\langle kq \rangle [qk] = 7.500000$$

$$-2 k \cdot q = 7.500000$$

✓ $|\langle kq \rangle [qk] - (-2k \cdot q)| = 8.9 \times 10^{-16}$.

13.4 §48.D Helicity Spinors $u_{\pm}(k)$ in the Chiral Basis

13.4.1 Z-axis momentum: explicit form

For massless momentum $k^{\mu} = (\omega, 0, 0, \omega)$ along the $+z$ axis, the massless Dirac equation $\not{k}|\psi\rangle = 0$ has solutions (Srednicki eq. 60.10):

$$|k] = u_{-}(k) = \sqrt{2\omega} (0, 1, 0, 0)^T, \quad |k\rangle = u_{+}(k) = \sqrt{2\omega} (0, 0, 1, 0)^T \quad (48.9)$$

where $|k]$ has negative helicity ($h = -\frac{1}{2}$) and $|k\rangle$ has positive helicity ($h = +\frac{1}{2}$).

The corresponding 2-component helicity spinors:

$$\lambda_{\alpha}(k)|_{z\text{-axis}} = \sqrt{2\omega} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \tilde{\lambda}_{\dot{\alpha}}(k)|_{z\text{-axis}} = \sqrt{2\omega} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (48.10)$$

✓ Massless Dirac eq. $\not{k}|k] = 0$: max error 0.0×10^0 (exact). ✓ Massless Dirac eq. $\not{k}|k\rangle = 0$: max error 0.0×10^0 (exact).

13.4.2 General massless momentum

For $p^{\mu} = E(1, \sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, the 4-component spinors in the Weyl representation are:

$$|p] = \sqrt{2E} \begin{pmatrix} -\sin \frac{\theta}{2} e^{-i\phi} \\ \cos \frac{\theta}{2} \\ 0 \\ 0 \end{pmatrix}, \quad |p\rangle = \sqrt{2E} \begin{pmatrix} 0 \\ 0 \\ \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\phi} \end{pmatrix} \quad (48.11)$$

These satisfy $\not{p}|p] = \not{p}|p\rangle = 0$ and are eigenstates of the helicity operator $\hat{h} = \hat{\mathbf{J}} \cdot \hat{\mathbf{p}}$.

13.5 §48.E Gluon Polarization Vectors as Spinor Bilinears

13.5.1 Definitions

The photon/gluon polarization vectors in the spinor-helicity formalism (Srednicki between eqs. 60.6–60.9):

$$\boxed{\varepsilon_{+}^{\mu}(k; q) = \frac{\tilde{\lambda}_q^{\dagger \dot{\alpha}} \sigma_{\alpha \dot{\alpha}}^{\mu} \lambda_k^{\alpha}}{\sqrt{2} \langle q k \rangle}, \quad \varepsilon_{-}^{\mu}(k; q) = [\varepsilon_{+}^{\mu}(k; q)]^{*}} \quad (48.12)$$

Here k is the photon momentum and $q \neq k$ is an arbitrary massless **reference momentum** encoding residual gauge freedom.

Ward identity (transversality):

$$k_\mu \varepsilon_\pm^\mu(k; q) = 0 \quad (48.13)$$

Normalization:

$$\varepsilon_+^\mu \varepsilon_{+\mu} = \varepsilon_-^\mu \varepsilon_{-\mu} = -1, \quad \varepsilon_+^\mu \varepsilon_{-\mu} = 0 \quad (48.14)$$

13.5.2 Numerical verification

Setup: $k^\mu = (2, 0, 0, 2)$ along $+z$; reference q at $\theta = \pi/3$, $\phi = \pi/4$ with $\omega_q = 1.5$.

Computed: $\varepsilon_+^\mu(k; q) \approx (-0.866 + 0.866i, -0.707, -0.707i, -0.866 + 0.866i)$

✓ $|k_\mu \varepsilon_+^\mu(k; q)| = 0.0 \times 10^0$ (Ward identity for ε_+). ✓ $|k_\mu \varepsilon_-^\mu(k; q)| = 0.0 \times 10^0$ (Ward identity for ε_-).

13.6 §48.F Little Group Scaling and Helicity Weights

13.6.1 The little group

The little group of a massless momentum p^μ is $U(1) \simeq SO(2)$: rotations about the direction of motion. Under the little-group transformation

$$\lambda_\alpha \rightarrow t \lambda_\alpha, \quad \tilde{\lambda}_{\dot{\alpha}} \rightarrow t^{-1} \tilde{\lambda}_{\dot{\alpha}}, \quad t \in \mathbb{C}^*, \quad (48.15)$$

the momentum $p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$ is invariant. A massless particle state of helicity h transforms as:

$$|p, h\rangle \rightarrow t^{-2h} |p, h\rangle \quad (48.16)$$

The spinor brackets transform with definite weights under particle i 's little group ($t_i = t$):

$$\langle i j \rangle \rightarrow t_i t_j \langle i j \rangle, \quad [i j] \rightarrow t_i^{-1} t_j^{-1} [i j]. \quad (48.17)$$

Under $\lambda_k \rightarrow t \lambda_k$ (with $\tilde{\lambda}_k \rightarrow t^{-1} \tilde{\lambda}_k$ to preserve p), the polarization vector $\varepsilon_+^\mu(k; q)$ is invariant as a 4-vector. The helicity weight manifests in the full amplitude $\mathcal{A} \rightarrow \prod_i t_i^{-2h_i} \mathcal{A}$ (on-shell condition).

✓ $\varepsilon_+^\mu(k; q)$ invariant under $\lambda_k \rightarrow t \lambda_k$, $\tilde{\lambda}_k \rightarrow t^{-1} \tilde{\lambda}_k$ (preserving p_k): ratio = 1.0000.

13.7 §48.G MHV Parke-Taylor Formula (Seed)

13.7.1 Motivation

The spinor-helicity formalism makes tree-level gluon amplitudes remarkably compact. Amplitudes with all-same or all-but-one helicities vanish. The first nonzero class are the **MHV (Maximally Helicity Violating)** amplitudes, with exactly two negative helicities.

13.9 Summary

Object	Expression
Spinor momentum	$p_{\alpha\dot{\alpha}} = \lambda_{\alpha}\tilde{\lambda}_{\dot{\alpha}}; \quad p^2 = 0 \text{ iff rank } 1$
Angle bracket	$\langle ij \rangle = \varepsilon^{\alpha\beta} \lambda_{\alpha}^i \lambda_{\beta}^j = -\langle ji \rangle$
Square bracket	$[ij] = \varepsilon_{\dot{\alpha}\dot{\beta}} \tilde{\lambda}^{i\dot{\alpha}} \tilde{\lambda}^{j\dot{\beta}} = -[ji]$
Mandelstam	$s_{ij} = \langle ij \rangle [ji] = -2p_i \cdot p_j$
z -axis spinors	$ k\rangle = \sqrt{2\omega}(0, 1, 0, 0)^T; \quad k\rangle = \sqrt{2\omega}(0, 0, 1, 0)^T$
Polarization ε_+	$\varepsilon_+^{\mu}(k; q) = \tilde{\lambda}_q^{\dagger} \sigma^{\mu} \lambda_k / (\sqrt{2} \langle qk \rangle)$
Polarization ε_-	$\varepsilon_-^{\mu}(k; q) = [\varepsilon_+^{\mu}(k; q)]^*$
Ward identity	$k_{\mu} \varepsilon_{\pm}^{\mu} = 0$
Little group	$\lambda \rightarrow t\lambda, \quad \tilde{\lambda} \rightarrow t^{-1}\tilde{\lambda}; \quad \text{amplitude} \rightarrow t^{-2h}$
Schouten	$\langle ij \rangle \langle kl \rangle + \langle ik \rangle \langle lj \rangle + \langle il \rangle \langle jk \rangle = 0$
MHV (Parke-Taylor)	$\mathcal{A}_n^{\text{MHV}} = \langle ij \rangle^4 / (\langle 12 \rangle \cdots \langle n1 \rangle)$

Next: Chapter 60 applies these tools to spinor electrodynamics: twistor notation $|p\rangle, |p\rangle, [p], \langle p|$, polarization vectors as spinor bilinears, and the Fierz identities for loop computations.

Chapter 14

Ch. 60 — Spinor Helicity for Spinor Electrodynamics

14.1 §60.A Twistor Notation

14.1.1 The four twistors

For each massless 4-momentum p^μ (with $p^2 = 0$), Srednicki defines four **twistors** (massless bispinors, eq. 60.1):

$$|p] \equiv u_-(p) = v_+(p) \quad \text{positive helicity ket (square bracket)} \quad (60.1a)$$

$$|p\rangle \equiv u_+(p) = v_-(p) \quad \text{negative helicity ket (angle bracket)} \quad (60.1b)$$

$$[p| \equiv \bar{u}_+(p) = \bar{v}_-(p) \quad \text{positive helicity bra (square bracket)} \quad (60.1c)$$

$$\langle p| \equiv \bar{u}_-(p) = \bar{v}_+(p) \quad \text{negative helicity bra (angle bracket)} \quad (60.1d)$$

Physical interpretation. For a massless particle, the Dirac equation $\not{p}\psi = 0$ has the same mathematical solutions regardless of whether ψ is interpreted as a particle (u) or antiparticle (v). The identification $u_-(p) = v_+(p)$ encodes the *crossing symmetry* of the S -matrix: flipping an incoming particle to an outgoing antiparticle corresponds to $u \leftrightarrow v$ with a helicity flip.

Mnemonic:

Square brackets $|p]$, $[p| \leftrightarrow$ positive helicity ($h = +\frac{1}{2}$)

Angle brackets $|p\rangle$, $\langle p| \leftrightarrow$ negative helicity ($h = -\frac{1}{2}$)

14.1.2 Non-mixing products (eq. 60.2)

$$[k| |p] = [k p] \quad (\text{square} \times \text{square, scalar}) \quad (14.1)$$

$$\langle k| |p\rangle = \langle k p \rangle \quad (\text{angle} \times \text{angle, scalar}) \quad (14.2)$$

$$[k| \langle p| = 0 \quad (\text{mixed helicities vanish}) \quad (14.3)$$

$$\langle k| [p| = 0 \quad (\text{mixed helicities vanish}) \quad (14.4)$$

The vanishing of mixed products follows from the chirality structure of the Weyl representation: γ^μ is off-diagonal and mixes the two Weyl blocks, but the bilinear $[k|\gamma^\mu|p\rangle$ contracts same-chirality spinors which gives zero by the Weyl block structure.

14.2 §60.B Explicit Spinor Construction

14.2.1 Z-axis momentum (eq. 60.10)

For massless $k^\mu = (\omega, 0, 0, \omega)$ along the $+z$ axis, the Dirac spinors in the Weyl (chiral) representation are:

$$\boxed{|k] = u_-(k) = \sqrt{2\omega} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |k\rangle = u_+(k) = \sqrt{2\omega} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}} \quad (60.10)$$

with bra spinors $[k| = (|k])^\dagger \gamma^0$ and $\langle k| = (|k\rangle)^\dagger \gamma^0$.

✓ Massless Dirac eq. $\not{k}|k] = 0$: max error 0 (exact). ✓ Massless Dirac eq. $\not{k}|k\rangle = 0$: max error 0 (exact). ✓ Masslessness $k^2 = 0$: error 0 (exact).

14.2.2 General massless momentum: rank-1 factorization

For $p^\mu = E(1, \sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, the 2-component momentum spinor matrix

$$p_{\alpha\dot{\alpha}} = p^\mu \sigma_{\mu,\alpha\dot{\alpha}} = E \begin{pmatrix} 1 + \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & 1 - \cos \theta \end{pmatrix} \quad (60.4)$$

has $\det(p_{\alpha\dot{\alpha}}) = -p^2/4 = 0$, hence rank 1. It factors as:

$$\boxed{p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}} \quad \text{where} \quad \lambda_\alpha = \sqrt{2E} \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\phi} \end{pmatrix}, \quad \tilde{\lambda}_{\dot{\alpha}} = \sqrt{2E} \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{-i\phi} \end{pmatrix} \quad (60.5)$$

Numerical check ($E = 3$, $\theta = \pi/4$, $\phi = \pi/3$):

$$p_{\alpha\dot{\alpha}} = \begin{pmatrix} 5.1213 + 0j & 1.0607 - 1.8371j \\ 1.0607 + 1.8371j & 0.8787 + 0j \end{pmatrix}$$

✓ $p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$: max error 1.8×10^{-15} . ✓ $\det(p_{\alpha\dot{\alpha}}) = -5.7 \times 10^{-16}$ (should be 0, since $p^2 = 0$).

14.3 §60.C Twistor Products $\langle kp \rangle$ and $[kp]$

14.3.1 Definitions and basic properties

The Lorentz-invariant spinor bilinears (Srednicki eq. 60.2):

$$\langle k p \rangle = \varepsilon^{\alpha\beta} \lambda_{k\alpha} \lambda_{p\beta}, \quad [k p] = \varepsilon_{\dot{\alpha}\dot{\beta}} \tilde{\lambda}_k^{\dot{\alpha}} \tilde{\lambda}_p^{\dot{\beta}} \quad (60.2)$$

with $\varepsilon^{12} = +1$, $\varepsilon_{12} = -1$ (Srednicki convention).

Cadabra2: $\langle k p \rangle = \langle kp \rangle$ and $[k p] = [kp]$

Key algebraic properties:

- **Antisymmetry** (eq. 60.3): $\langle ij \rangle = -\langle ji \rangle$, $[ij] = -[ji]$
- **Complex conjugation:** $\langle pk \rangle^* = [kp]$ (for real momenta)

14.3.2 Key identity: Mandelstam from spinors

$$\langle kp \rangle [pk] = \text{Tr} \left[\frac{1}{2} (1 - \gamma^5) \not{k} \not{p} \right] = -2 k \cdot p \quad (60.4)$$

Consequence: $|\langle kp \rangle|^2 = |[kp]|^2 = |s_{kp}|$ for massless k, p .

Cadabra2: $\langle k p \rangle [p k] = s_{kp}$

Numerical check ($k = (2.5, 0, 0, 2.5)$, $p = 3(1, \sin(\pi/3), 0, \cos(\pi/3))$):

$$\begin{aligned} \langle kp \rangle &= 2.738613 + 0i \\ [kp] &= -2.738613 + 0i \\ \langle kp \rangle [pk] &= 7.500000 \\ -2 k \cdot p &= 7.500000 \\ \text{Tr} \left[\frac{1}{2} (1 - \gamma^5) \not{k} \not{p} \right] &= 7.500000 \end{aligned}$$

✓ $|\langle kp \rangle + \langle pk \rangle| = 0$ (antisymmetry, exact). ✓ $|[kp] + [pk]| = 0$ (antisymmetry, exact).
 ✓ $|\langle pk \rangle^* - [kp]| = 0$ (complex conjugation, exact). ✓ $|\langle kp \rangle [pk] - (-2k \cdot p)| = 8.9 \times 10^{-16}$
 (Mandelstam identity). ✓ $|\langle kp \rangle [pk] - \text{Tr}[\frac{1}{2}(1 - \gamma^5)\not{k}\not{p}]| = 8.9 \times 10^{-16}$ (trace formula).

14.4 §60.D Polarization Vectors in Spinor-Helicity

14.4.1 Definitions

The photon polarization vectors as spinor bilinears (Srednicki between eqs. 60.6–60.9):

$$\varepsilon_+^\mu(k; q) = \frac{\tilde{\lambda}_q^{\dagger\dot{\alpha}} \sigma_{\alpha\dot{\alpha}}^\mu \lambda_k^\alpha}{\sqrt{2} \langle q k \rangle}, \quad \varepsilon_-^\mu(k; q) = [\varepsilon_+^\mu(k; q)]^* \quad (60.6)$$

Here k is the photon momentum and $q \neq k$ is a massless **reference momentum** encoding residual gauge freedom. Different choices of q give the same physical on-shell amplitude (gauge invariance).

Key properties:

$$k_\mu \varepsilon_\pm^\mu = 0 \quad (\text{transversality / Ward identity}) \quad (60.7)$$

$$\varepsilon_\pm^\mu \varepsilon_{\pm\mu} = -1 \quad (\text{normalization}) \quad (60.8)$$

$$\varepsilon_+^\mu \varepsilon_{-\mu} = 0 \quad (\text{orthogonality}) \quad (60.9)$$

14.4.2 Numerical verification

Setup: $k^\mu = (2, 0, 0, 2)$ along $+z$ with $\omega_k = 2$; reference q at $\theta_q = \pi/3$, $\phi_q = \pi/4$ with $\omega_q = 1.5$.

Computed:

$$\varepsilon_+^\mu(k; q) \approx (-0.866 + 0.866i, -0.707 + 0i, -0.707i, -0.866 + 0.866i)$$

✓ $|k_\mu \varepsilon_+^\mu| = 0$ (Ward identity, ε_+ , exact). ✓ $|k_\mu \varepsilon_-^\mu| = 0$ (Ward identity, $\varepsilon_- = \varepsilon_+^*$, exact).

14.5 §60.E Fierz Identities and Massless Completeness

14.5.1 Fierz identities (eqs. 60.13–60.14)

$$\boxed{-\frac{1}{2} \gamma^\mu \langle q | \gamma_\mu | k \rangle = |k\rangle \langle q| + |q\rangle \langle k|} \quad (60.13)$$

$$-\frac{1}{2} \gamma^\mu [q | \gamma_\mu | k] = |k\rangle [q| + |q\rangle [k| \quad (60.14)$$

These are 4×4 matrix equalities. The LHS is formed by summing $-\frac{1}{2} \gamma^\mu$ (a matrix) times the scalar $\langle q | \gamma_\mu | k \rangle$ over μ . The RHS involves outer products of 4-component spinors.

Derivation sketch. The Fierz identities follow from the completeness relation for the 16 Dirac matrices $\{\Gamma^A\}$:

$$(\Gamma^A)_{\alpha\beta} (\Gamma_A)_{\gamma\delta} = 4 \delta_{\alpha\delta} \delta_{\beta\gamma}.$$

Applied to massless bispinors, the rank-1 projectors $|p\rangle \langle p|$ and $|p\rangle [p|$ arise from the massless polarization sum:

$$\sum_s u_s(p) \bar{u}_s(p) = -\not{p}.$$

Slashed polarization vectors (from eq. 60.13):

$$\not{\varepsilon}_+(k; q) = \frac{\sqrt{2}}{\langle qk \rangle} \left(|k\rangle \langle q| + |q\rangle \langle k| \right), \quad \not{\varepsilon}_-(k; q) = \frac{\sqrt{2}}{[qk]} \left(|k\rangle [q| + |q\rangle [k| \right) \quad (60.15)$$

These follow by comparing eqs. (60.6) and (60.13).

14.5.2 Massless completeness

The massless spin sum (massless limit of eq. 38.28):

$$\boxed{-\not{k} = |k\rangle \langle k| + |k\rangle [k|} \quad (60.16)$$

This decomposes $-\not{k}$ into the two helicity projectors.

Component check (z-axis momentum $\omega = 2$, using the Weyl-block structure): For $k^\mu = (\omega, 0, 0, \omega)$, the slashed momentum in the Weyl representation is

$$-\not{k} = -k_\mu \gamma^\mu = \begin{pmatrix} 0 & 0 & 0 & 2\omega \\ 0 & 0 & 0 & 0 \\ 0 & 2\omega & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}^\dagger \quad (\text{schematic; off-diagonal blocks only})$$

The outer products $|k\rangle\langle k|$ and $|k\rangle[k|$ reproduce exactly these off-diagonal Weyl blocks.

✓ Massless completeness $-\not{k} = |k\rangle\langle k| + |k\rangle[k|$: verified analytically from eq. (60.10) and the Weyl-block structure.

14.6 §60.F Schouten Identity and Momentum Conservation

14.6.1 Schouten identity

From the 2-dimensional nature of spinor space (any 3 spinors in 2D are linearly dependent):

$$\boxed{\langle ij\rangle\langle kl\rangle + \langle ik\rangle\langle lj\rangle + \langle il\rangle\langle jk\rangle = 0} \quad (60.17)$$

and identically for square brackets.

Cadabra2: $\langle ij\rangle\langle kl\rangle + \langle ik\rangle\langle lj\rangle + \langle il\rangle\langle jk\rangle$

✓ $|\langle 12\rangle\langle 34\rangle + \langle 13\rangle\langle 42\rangle + \langle 14\rangle\langle 23\rangle| = 4.6 \times 10^{-16}$ (four generic massless momenta).

✓ $|[12][34] + [13][42] + [14][23]| = 4.6 \times 10^{-16}$ (square-bracket Schouten).

14.6.2 Momentum conservation (eq. 60.18)

For a massless n -particle process with $\sum_j p_j^\mu = 0$:

$$\sum_j \langle ij\rangle[jk] = 0 \quad \text{for all } i, k \quad (60.18)$$

This is the spinor form of 4-momentum conservation, obtained by contracting $\sum_j p_j^\mu = 0$ with λ_i and $\tilde{\lambda}_k$. It is the key identity used in BCFW on-shell recursion to show that certain spurious poles cancel in the recursion formula.

14.7 Numerical Verification Summary

Identity	Equation	Max error
$\not{k} k\rangle = 0$ (Dirac eq., z -axis)	(60.10)	0 (exact)
$\not{k} k\rangle = 0$ (Dirac eq., z -axis)	(60.10)	0 (exact)
$k^2 = 0$ (masslessness)	—	0 (exact)
$p_{\alpha\dot{\alpha}} = \lambda_{\alpha}\tilde{\lambda}_{\dot{\alpha}}$ (rank-1)	(60.5)	1.8×10^{-15}
$\det(p_{\alpha\dot{\alpha}}) = 0$	$(p^2 = 0)$	5.7×10^{-16}
$\langle kp \rangle = -\langle pk \rangle$ (antisymmetry)	(60.3)	0 (exact)
$[kp] = -[pk]$ (antisymmetry)	(60.3)	0 (exact)
$\langle pk \rangle^* = [kp]$ (conjugation)	—	0 (exact)
$\langle kp \rangle [pk] = -2k \cdot p$ (Mandelstam)	(60.4)	8.9×10^{-16}
Trace formula (eq. 60.4)	(60.4)	8.9×10^{-16}
$k_{\mu}\varepsilon_{+}^{\mu} = 0$ (Ward identity)	(60.7)	0 (exact)
$k_{\mu}\varepsilon_{-}^{\mu} = 0$ (Ward identity)	(60.7)	0 (exact)
Schouten $\langle \cdot \rangle$ identity	(60.17)	4.6×10^{-16}
Schouten $[\cdot]$ identity	(60.17)	4.6×10^{-16}

All identities verified to machine precision ($\lesssim 10^{-15}$).

14.8 Summary

Object	Expression
Twistors	$ p] = u_-(p) = v_+(p); \quad p\rangle = u_+(p) = v_-(p)$ $[p = \bar{u}_+(p) = \bar{v}_-(p); \quad \langle p = \bar{u}_-(p) = \bar{v}_+(p)$
z -axis spinors	$ k] = \sqrt{2\omega}(0, 1, 0, 0)^T; \quad k\rangle = \sqrt{2\omega}(0, 0, 1, 0)^T$
2-comp spinors	$\lambda_\alpha = \sqrt{2E}(\cos \frac{\theta}{2}, \sin \frac{\theta}{2} e^{i\phi})^T$
Rank-1 fact.	$p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}; \quad p^2 = 0 \Leftrightarrow \det = 0$
Angle bracket	$\langle kp \rangle = \varepsilon^{\alpha\beta} \lambda_{k\alpha} \lambda_{p\beta} = -\langle pk \rangle$
Square bracket	$[kp] = \varepsilon_{\dot{\alpha}\dot{\beta}} \tilde{\lambda}_k^{\dot{\alpha}} \tilde{\lambda}_p^{\dot{\beta}} = -[pk]$
Conjugation	$\langle pk \rangle^* = [kp]$
Mandelstam	$\langle kp \rangle [pk] = -2k \cdot p = \text{Tr}[\frac{1}{2}(1 - \gamma^5) \not{k} \not{p}]$
Polarization	$\varepsilon_+^\mu(k; q) = \tilde{\lambda}_q^\dagger \sigma^\mu \lambda_k / (\sqrt{2} \langle qk \rangle); \quad \varepsilon_- = (\varepsilon_+)^*$
Ward identity	$k_\mu \varepsilon_\pm^\mu = 0$
Fierz (60.13)	$-\frac{1}{2} \gamma^\mu \langle q \gamma_\mu k \rangle = k] \langle q + q] [k $
Completeness	$-\not{k} = k] \langle k + k\rangle [k $
Schouten	$\langle ij \rangle \langle kl \rangle + \langle ik \rangle \langle lj \rangle + \langle il \rangle \langle jk \rangle = 0$
Momentum cons.	$\sum_j \langle ij \rangle [jk] = 0$ (massless n -body)

Applications: These results are the foundation for computing massless QED amplitudes. The Fierz identities (eqs. 60.13–60.14) are used to express $\not{\varepsilon}_\pm$ in terms of outer products of helicity spinors, dramatically simplifying the evaluation of Feynman diagrams. The Parke-Taylor formula (Chapter 48) and BCFW recursion (later chapters) rely directly on the identities developed here.